Software Test Plan (STP)

Collaborative Study Material & Quiz Platform

Team 17 | Revision 1.2 (DRAFT) | 18-10-2026

STATUS: Revision 1.2 draft — pending team approval. No test has been executed; no result, approval, build, CI workflow, sprint video or implementation status is claimed.

Item	Detail
Project	Collaborative Study Material & Quiz Platform
Group	Group 17
Document	Software Test Plan, Revision 1.2 (draft). Baseline revised: STP Version 1.0 dated 03-10-2026 and Revision 1.1 (draft) dated 18-10-2026.
Authors	Group 17 project team: Jagadeesh (Member 1, PES1UG24CS347), Alifiya (Member 2, PES1UG25CD841), Pushkar (Member 3, PES1UG24CS350), Vedh (Member 4, [SRN not yet supplied]). Ownership follows SRS Revision 1.2 Appendix D (baseline workload decision).
Verified against	SRS Revision 1.2 (draft, 17-10-2026; cover re-issued 18-10-2026), read in full; SAD Revision 1.1 (draft, 17-10-2026) through its audit report only - the SAD file itself was not supplied to this revision; Project Guidelines; Deliverables Part-2. Deliverables Part-1 and the official STP template were not supplied (OD-02).
Project guide	Ashok Kumar Patil
Date	18-10-2026
Companion files	STP_Team17_Rev1_2_Traceability_and_Inventory.xlsx; Team17_Submission_Set_Final_Report.docx; Team17_Deviation_Log.docx; Team17_Part2_Backlog.csv

Revision history

Version	Date	Author	Change summary
1.0	03-10-2026	Project team	Merged plan: 97 test cases, 20 unit-test candidates, against SRS v1.0 and SAD v1.0 (as supplied).
1.1 (draft)	18-10-2026	Project team	Audit and revision against SRS Rev 1.2 IDs (via SAD Rev 1.1) and Deliverables Part-2: all 97 test IDs and 20 unit-test IDs retained; 63 test cases and 8 unit tests added; status codes/messages aligned with SAD s4.3-4.5; traceability extended to SO, BR, DC, UI, COM; schedule rebased to Part-2 sprints; open-decision register added. See the audit report.
1.2 (draft)	18-10-2026	Project team	Re-baselined against SRS Revision 1.2 (read in full) and the confirmed four-member team (Jagadeesh, Alifiya, Pushkar, Vedh): Member 4 added to the title block, 1.1, 2.1, section 3 owner columns, 5, 9, 15, Annex B/C/E/J and the workbook; owners of 57 test cases and 10 unit-test candidates reassigned by SRS Appendix D (owner = Appendix D primary owner of the first-listed requirement); stale "SRS not supplied" statements replaced and SO, BR, UI, COM and DC wording taken from the SRS; revision labels corrected (document row, footer, "New in Rev 1.1"); SO-01 to SO-06 linked to the tests of their SEC requirements (SRS 3.3.3.4); OD-01 recorded as decided by the team (SRN pending); TOD-01 narrowed to the unsupplied SAD file; TC-QS-04 / SEC-ACC-04 link verified in Annex C; Annex H.3 and I updated; collaboration tests for REQ-COL-01 to 04 moved to Sprint 1 to match the Part-2 backlog. No test case, unit candidate or ID added or removed (160 / 28). No test executed; no approval given.

Revision 1.2 baseline note: SRS-Team17_Rev1.2_Draft.docx (17-10-2026, draft; cover re-issued 18-10-2026) is the authority for requirements and for ownership (Appendix D); SAD_Team17_Rev1_1_Draft.docx (17-10-2026, draft) is the design/API source (known to this revision through its audit report only); Project Guidelines.docx and Deliverables Part-2 define submission/process expectations (Deliverables Part-1 was not supplied); STP Revision 1.1 and its workbook are the historical test baseline. The source documents remain drafts and conflicting/open design decisions are not silently resolved.

How to read the status words in this document

- **Readiness (per test):** R1 Ready - executable once a build is deployed (build status is unverified); R2 Environment-dependent (load, devices, backup, TLS host, timing); R3 Blocked - the expected result is not agreed (open decision named); R4 Manual / inspection.
- **Execution status:** every test is "Not executed" in this revision. "Passed" and "Failed" appear nowhere because no run exists.
- **Coverage status (per requirement):** Covered = at least one designed, unblocked test addresses all stated behaviours (design coverage, not a pass); Partially covered; Blocked pending clarification; Not directly testable; Not covered. Coverage never means a test has run.

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1. Introduction

1.1 Purpose

This Test Plan defines how the Collaborative Study Material & Quiz Platform (Group 17) is verified as a whole: Account Management, authentication, sessions and access control (Member 1 - Jagadeesh); Question Bank and Quiz Authoring (Member 2 - Alifiya); Performance Analytics, Collaboration, study groups and moderation (Member 3 - Pushkar); and Material Management and Sharing, File Storage, Quiz Execution, attempts, timing, submission and grading (Member 4 - Vedh), with the related security, performance, safety and quality requirements and the team-level requirements NFR-QUAL-02, NFR-QUAL-05 and NFR-QUAL-06. It states objectives, scope, strategy, environment, schedule, deliverables, responsibilities, risks and criteria, and it provides the test cases and end-to-end traceability. Ownership follows SRS Revision 1.2 Appendix D. In v1.0 this purpose was repeated three times (once per member); it is stated once.

1.2 Scope

In scope (IDs of SRS Rev 1.2, cross-checked against SAD Rev 1.1 s3.8 as reported in the SAD audit): REQ-ACC-01 to 10, REQ-MAT-01 to 11, REQ-QB-01 to 07, REQ-QZ-01 to 18, REQ-PA-01 to 06, REQ-COL-01 to 07 (59 functional requirements); SEC-ACC-01 to 06, SEC-MAT-01 to 04, SEC-COM-01, SEC-QZ-01 to 05 (16); NFR-PERF-01 to 06, NFR-SAFE-01 to 03, NFR-QUAL-01 to 06 (15); security objectives SO-01 to SO-06; business rules BR-01 to BR-08; design constraints DC-01 to DC-11 (where testable); the domain requirement (English messages); user-interface items UI-01 to UI-18 and communication interface COM-01 (SRS 3.1.3 also lists COM-02 to COM-10, which are not traced separately: TOD-12); and the use cases of SRS s3 as cited in SAD s4.2 (UC-ACC, UC-MAT, UC-QB, UC-QZ, UC-PA, UC-COL). v1.0 covered the first 90 of these 115 IDs; SO, BR and DC (25 IDs) and the interface items were not traced.

The unit under test is the platform described in SAD Rev 1.1 (React client, Node.js/Express API, MongoDB, file storage, Nginx). Tests are planned against the specification; this document does not state that any component has been implemented.

1.3 Source documents and order of authority

Source	Version / date	Supplied?	Used for / limitation
Project Guidelines	Supplied file	Yes - read in full	Proposal contents (title, group number, team profile with name and SRN, plan of work and product ownership with one qualitative property per member) and the single-PDF submission rule. It defines no STP template. Compliance check: see Annex I and the final report.
Deliverables Part-1	-	No	Not in the files supplied to this revision; cited only through SRS 1.5. Part-1 requirements are unverified (OD-02).
Deliverables Part-2	PDF supplied	Yes - read in full	Backlog (ETA 12-Oct-2026), Sprint 1 (19-23 Oct), Sprint 2 (26-30 Oct), GitHub Actions, PR review, sprint videos, deviations, documentation, freeze.
SRS	Revision 1.2, 17-10-2026 (draft)	Yes - read in full	Authoritative for requirement IDs (115 REQ/SEC/NFR/SO/BR/DC IDs plus UI-01 to UI-18 and COM-01 to COM-10), wording, field limits (Appendix B), messages (B.7), security objectives (3.3.3.1, 3.3.3.4), business rules (3.3.5), design constraints (3.4) and ownership (Appendix D). Limits and messages stated in the test cards were checked against Appendix B; SAD-derived status codes and error-code strings are not re-verified (TOD-01).
SAD	Revision 1.1 draft, 17-10-2026	No (not supplied to this revision)	Components, API endpoints, status/error codes, messages, data model, security controls and open decisions OD-01 to OD-14 were taken over from STP Rev 1.1 and the SAD audit report. The SAD file was not available for re-verification (TOD-01).
SAD Rev 1.1 Audit Report	17-10-2026	Yes	Open decisions OD-01 to OD-14 and known inconsistencies.

Source	Version / date	Supplied?	Used for / limitation
STP (baseline)	Version 1.0, 03-10-2026; Revision 1.1 draft, 18-10-2026	Version 1.0 file: no; Revision 1.1: via its audit report and workbook	Baseline preserved and improved. The v1.0 file itself was not supplied to this revision, so v1.0-to-current differences rest on the revision histories and audit reports (see Team17_Deviation_Log.docx).
Official STP template	-	No	Not supplied; the v1.0 structure (15 numbered sections, annexes) is preserved. Template conformance is unverified (OD-02).
Repository, CONTRIBUTING.md, build	-	No	No implementation, build or CI evidence exists in the supplied material (OD-10).

Where sources conflict, the conflict is recorded in the register (Annex F) with the affected tests and the decision needed; no source has been silently changed. Order of authority: Guidelines/Deliverables, then SRS Rev 1.2, then SAD Rev 1.1, then STP v1.0.

1.4 References

- Software Requirements Specification (SRS), Revision 1.2 (draft), 17-10-2026 (cover re-issued 18-10-2026), Team 17.
- Software Architecture and Design Specification (SAD), Revision 1.1 (draft), 17-10-2026.
- SAD Revision 1.1 Audit Report, 17-10-2026.
- Software Test Plan Version 1.0, 03-10-2026.
- SE Mini-Project Deliverables Part-2.
- Requirements Traceability Matrix: SRS Appendix C and Annex C of this STP.

1.5 Definitions

STP, SRS, SAD, RTM, TC (test case), UT (unit-test candidate), NFR, UC, SO, BR, DC, UI, COM, OD (open decision of the SAD, Appendix 5.4) and TOD (open decision introduced by this STP) are used as in Annex J. Platform roles (Student, Faculty, Admin) are users of the system; Members 1-4 are project team members. They are never interchangeable.

2. Test Items and Objectives

2.1 Test items

v1.0 listed test items per member; Rev 1.1 aligned them with SAD components (SAD s3.4) and Rev 1.2 assigns owners by SRS Appendix D (four members).

Feature area	Test items (SAD component / endpoint group)	Owner
Account Management	Auth & Session Service (registration, log-in, lock-out, sessions, logout); Account Service (profile); API Layer middleware (authentication, RBAC, validation, rate limiting); UI-01 to UI-03, UI-07; /auth/*, /account/profile	Member 1 - Jagadeesh
Material Management and Sharing	Material Service, AccessGrant, File Storage (also used for profile pictures, REQ-ACC-09); UI-04 to UI-06; /materials/*	Member 4 - Vedh
Question Bank and Quiz Authoring	Question Bank Service; Quiz Engine (authoring: create, edit, delete, publish, question copy); owner views of submissions and results (REQ-QZ-16, 17, 18); UI-08 to UI-10, UI-14; /questions, /quizzes (authoring)	Member 2 - Alifiya
Quiz Execution, Attempts and Grading	Quiz Engine (execution: Student quiz list, start, timer, answers, auto-submit); Grading Service; Student attempts and results (REQ-QZ-06 to 15); UI-11 to UI-13; /quizzes (Student list, start), /attempts	Member 4 - Vedh
Performance Analytics & Collaboration	Performance Analytics Service (personal, topic-wise, class); Study Group & Collaboration Service (groups, membership, Admin moderation); UI screens for performance, groups and moderation; /performance/*, /groups, /collaboration/content/:id	Member 3 - Pushkar

Feature area	Test items (SAD component / endpoint group)	Owner
Cross-cutting	Nginx TLS termination, central error handler, logging, Docker deployment, backups (owner of operation unassigned: OD-06); availability, modularity, traceability (NFR-QUAL-02/05/06)	Member 1 coordinates security and cross-module NFRs (SRS App. D.4); Member 3 coordinates availability, modularity, traceability

2.2 Test objectives

- Verify every in-scope requirement with at least one reproducible test, or record why it cannot be tested yet.
- Verify security controls: authentication, RBAC, ownership, session handling, answer-key protection, server-side timing and grading, input sanitisation, transport security.
- Verify measurable NFR thresholds with evidence; never report a threshold as met without a recorded run.
- Verify error codes and messages as specified in SAD s4.3-4.5.
- Provide traceability from requirement to test and from test to evidence, and keep it consistent as the SRS/SAD change.
- Expose conflicts and unresolved decisions early (Annex F) instead of resolving them silently.

3. Features to be Tested

Features are mapped to requirement IDs. Descriptions retain the v1.0 feature statements, completed from SAD s3.8/s4.3; the per-requirement tests are in Annex C.

3.1 Account Management

ID	Feature to be tested	Owner (SRS App. D)
REQ-ACC-01	Register with full name, e-mail, password and permitted role (Student or Faculty).	Jagadeesh
REQ-ACC-02	Registration validation: empty fields, invalid e-mail, duplicate e-mail (case-insensitive).	Jagadeesh
REQ-ACC-03	Password rule: minimum length (8) with uppercase, lowercase and digit.	Jagadeesh
REQ-ACC-04	Authenticate with valid credentials; access granted within 2 seconds.	Jagadeesh
REQ-ACC-05	Invalid credentials rejected with the single specified message "Invalid email or password".	Jagadeesh
REQ-ACC-06	Account locked for 15 minutes after five consecutive failed log-ins.	Jagadeesh
REQ-ACC-07	Session ends on logout or after 30 minutes of inactivity; token no longer accepted.	Jagadeesh
REQ-ACC-08	View and edit permitted profile information; e-mail and role protected.	Jagadeesh
REQ-ACC-09	Profile validation: name not empty; profile picture not larger than 2 MB.	Jagadeesh
REQ-ACC-10	Authentication and role-based access control for protected functionality.	Jagadeesh

3.2 Material Sharing

ID	Feature to be tested	Owner (SRS App. D)
REQ-MAT-01	Upload material with title, subject, description and file.	Vedh
REQ-MAT-02	File type (PDF, DOCX, PPTX, PNG) and 25 MB size validation.	Vedh
REQ-MAT-03	Owner recorded; material visible in the owner's library within 3 seconds.	Vedh
REQ-MAT-04	Material visible only to its owner or authorized users.	Vedh
REQ-MAT-05	Owner shares material with another registered user.	Vedh
REQ-MAT-06	Authorized user downloads material; download starts within 3 seconds.	Vedh
REQ-MAT-07	Owner updates metadata or replaces the file.	Vedh
REQ-MAT-08	Owner removes material after confirmation.	Vedh
REQ-MAT-09	Owner revokes a user's access; next request by that user is denied.	Vedh
REQ-MAT-10	Unauthorized material operations rejected with "Access denied".	Vedh
REQ-MAT-11	Removed or unretrievable material handled without exposing its details ("Material unavailable").	Vedh

3.3 Question Bank

ID	Feature to be tested	Owner (SRS App. D)
REQ-QB-01	Faculty adds a question (text, four options, correct option, topic, difficulty, explanation).	Alifiya
REQ-QB-02	Question validation (required fields, limits 500/200/50/300, distinct options, correct option among options).	Alifiya
REQ-QB-03	Owner recorded; new question visible in the bank within 3 seconds.	Alifiya
REQ-QB-04	Search and filter the bank by keyword, topic and difficulty (results within 2 s for 1,000 questions).	Alifiya
REQ-QB-05	Faculty edits own question.	Alifiya
REQ-QB-06	Faculty removes own question; published quizzes are unaffected.	Alifiya
REQ-QB-07	Students cannot access the bank; only the owner can edit or remove a question.	Alifiya

3.4 Quizzes

ID	Feature to be tested	Owner (SRS App. D)
REQ-QZ-01	Faculty creates a quiz (title, description, time limit, availability window, 1-50 questions).	Alifiya
REQ-QZ-02	Quiz validation: title, 1-50 questions, time limit 5-180 minutes, end later than start.	Alifiya
REQ-QZ-03	Quiz is saved as Draft; Faculty publishes it (Draft/Published life cycle).	Alifiya
REQ-QZ-04	Selected questions are copied into the quiz at publication.	Alifiya
REQ-QZ-05	Edit and delete rules: Draft editable/deletable; rejected once an attempt has started.	Alifiya
REQ-QZ-06	Student lists available quizzes (without questions or answers).	Vedh
REQ-QZ-07	Start an attempt only inside the availability window and only once per Student.	Vedh
REQ-QZ-08	Questions displayed with a server-based countdown timer.	Vedh
REQ-QZ-09	Student selects one answer per question and submits.	Vedh
REQ-QZ-10	Attempt auto-submitted when the time limit is reached.	Vedh
REQ-QZ-11	Automatic grading of the submitted attempt.	Vedh
REQ-QZ-12	Result available within 2 seconds of submission.	Vedh
REQ-QZ-13	Result details: score, percentage, per-question answers, correct answers, explanations.	Vedh
REQ-QZ-14	Submitted attempt is immutable.	Vedh
REQ-QZ-15	Student views own attempts (title, score, percentage, date).	Vedh
REQ-QZ-16	Faculty views submissions for own quizzes.	Alifiya
REQ-QZ-17	Role and ownership access control for quiz functions.	Alifiya
REQ-QZ-18	Unavailable (removed or non-existent) quiz, question or attempt handled without exposing details.	Alifiya

3.5 Performance Analytics

ID	Feature to be tested	Owner (SRS App. D)
REQ-PA-01	Student views completed quiz attempts with quiz title, score, percentage and submission date.	Pushkar
REQ-PA-02	Student performance calculated and displayed from completed attempts.	Pushkar
REQ-PA-03	Student views quiz performance grouped by topic.	Pushkar
REQ-PA-04	Faculty views performance of Students who submitted attempts for Faculty-owned quizzes.	Pushkar
REQ-PA-05	Class performance shows Student name, score, percentage and submission information.	Pushkar
REQ-PA-06	Faculty access to performance of quizzes they do not own is denied.	Pushkar

3.6 Collaboration

ID	Feature to be tested	Owner (SRS App. D)
REQ-COL-01	Student creates a study group with required group information.	Pushkar
REQ-COL-02	Student joins an existing study group.	Pushkar
REQ-COL-03	Duplicate study-group membership is prevented.	Pushkar

ID	Feature to be tested	Owner (SRS App. D)
REQ-COL-04	Student views the study groups to which they belong.	Pushkar
REQ-COL-05	Admin moderates collaborative content.	Pushkar
REQ-COL-06	Non-Admin users are denied collaborative-content moderation.	Pushkar
REQ-COL-07	Removed or unavailable groups/content handled without exposing their details.	Pushkar

3.7 Security requirements (SEC-*, SO-*)

ID	Feature to be tested	Owner (SRS App. D)
SEC-ACC-01	Authentication enforced for protected functionality.	Jagadeesh
SEC-ACC-02	Generic credential error (no disclosure of whether the e-mail exists).	Jagadeesh
SEC-ACC-03	Passwords stored only as salted bcrypt hashes.	Jagadeesh
SEC-ACC-04	Session termination on logout and inactivity (server-side session record).	Jagadeesh
SEC-ACC-05	A user can change only their own profile; e-mail and role not changeable.	Jagadeesh
SEC-ACC-06	Account lock-out after repeated failed log-ins.	Jagadeesh
SEC-MAT-01	Material access control: only owner and authorized users can view/download.	Vedh
SEC-MAT-02	Only the owner can share material and revoke access.	Vedh
SEC-MAT-03	Only the owner can update or remove material.	Vedh
SEC-MAT-04	Access denied after revocation or removal.	Vedh
SEC-COM-01	TLS 1.2 or higher for all client-server communication; unencrypted connections rejected.	Jagadeesh
SEC-QZ-01	Question-bank and quiz management restricted to Faculty (owner).	Alifiya
SEC-QZ-02	No answer keys or explanations sent before submission.	Vedh
SEC-QZ-03	Results visible only to the attempt owner; submissions only to the quiz owner.	Alifiya
SEC-QZ-04	Quiz timer and availability enforced on the server clock.	Vedh
SEC-QZ-05	Scores computed on the server; submitted attempts immutable.	Vedh
SO-01	Security objective Account confidentiality: Protect user credentials and account data from unauthorized disclosure.	Jagadeesh
SO-02	Security objective Material access control: Ensure study materials are accessible only to their owner and users the owner has authorized.	Jagadeesh
SO-03	Security objective Integrity: Prevent unauthorized modification of accounts, materials and material permissions.	Jagadeesh
SO-04	Security objective Secure authentication and sessions: Ensure protected functionality is reachable only through a valid session and that ended sessions cannot be reused.	Jagadeesh
SO-05	Security objective Secure communication: Protect data in transit between client and server.	Jagadeesh
SO-06	Security objective Assessment integrity: Ensure that quiz answers and results are disclosed only to authorized users and that attempts and scores cannot be tampered with.	Jagadeesh

3.8 Non-functional requirements

ID	Feature to be tested	Owner (SRS App. D)
NFR-PERF-01	At least 95% of page loads within 3 seconds with 200 concurrent users.	Jagadeesh
NFR-PERF-02	200 concurrent users supported with fewer than 1% failed requests.	Jagadeesh
NFR-PERF-03	Sharing and access revocation take effect within 2 seconds.	Vedh
NFR-PERF-04	Grading within 2 seconds with 100 concurrent submissions.	Vedh
NFR-PERF-05	Question-bank search within 2 seconds.	Alifiya
NFR-PERF-06	Quiz questions available within 3 seconds of starting.	Vedh
NFR-SAFE-01	Daily backups; recovery from a backup no more than 24 hours old.	Jagadeesh
NFR-SAFE-02	Interrupted/failed material operation leaves previous state unchanged and is not reported as success.	Vedh

ID	Feature to be tested	Owner (SRS App. D)
NFR-SAFE-03	Downloaded file identical to uploaded file (SHA-256).	Vedh
NFR-QUAL-01	First-time user completes registration, login and upload in under five minutes without instructions.	Jagadeesh
NFR-QUAL-02	At least 99% monthly availability, excluding announced maintenance.	Pushkar
NFR-QUAL-03	Invalid input and unsupported files rejected without a crash or effect on other users.	Jagadeesh
NFR-QUAL-04	Current Chrome, Firefox, Edge, Safari on desktop and mobile.	Jagadeesh
NFR-QUAL-05	Modular, independently testable feature areas.	Pushkar
NFR-QUAL-06	Requirements traceable to test cases.	Pushkar

3.9 Business rules, design constraints, domain and interface requirements

ID	Feature to be tested	Owner (SRS App. D)
BR-01	Business rule BR-01: Protected functions are available only to authenticated users.	Jagadeesh
BR-02	Business rule BR-02: Each study material has exactly one owner: the user who uploaded it.	Vedh
BR-03	Business rule BR-03: Only the owner may share, update, remove or revoke access to a material.	Vedh
BR-04	Business rule BR-04: A user may view or download a material only if they are its owner or have been granted access.	Vedh
BR-05	Business rule BR-05: Materials may be shared only with registered users.	Vedh
BR-06	Business rule BR-06: Users may change only their own profile; email address and role cannot be changed.	Jagadeesh
BR-07	Business rule BR-07: Users may upload only material they have the right to share. The uploader is responsible for copyright compliance. (policy; no mechanism, OD-11)	Vedh
BR-08	Business rule BR-08: Admin accounts are created by platform administrators, not through public registration. (SRS does not say how Admin accounts are created, OD-11)	Jagadeesh
DC-01	MERN stack (MongoDB, Express.js, React, Node.js).	Jagadeesh
DC-02	Defined integrated API.	Alifiya
DC-03	TLS 1.2 or higher.	Pushkar
DC-04	bcrypt password hashing.	Jagadeesh
DC-05	Supported browsers.	Jagadeesh
DC-06	File types, size and field limits.	Vedh
DC-07	Server-side validation and authorization checks.	Jagadeesh
DC-08	Server-side session expiry and lock-out.	Jagadeesh
DC-09	Scope constraint (features outside the SRS are excluded).	Pushkar
DC-10	Requirement IDs unchanged across documents.	Alifiya
DC-11	The system shall be developed by the four-member project team within the course schedule (SRS DC-11). Project constraint; not directly testable.	Vedh
DOM-NOID	Domain requirement without SRS ID: user-facing messages are in English (SAD s3.8). Label DOM-NOID is a test-plan placeholder, not an SRS ID.	Pushkar
UI-01..18	User-interface requirements UI-01 to UI-18 (SRS 3.1.1): registration, login, profile, material upload/library/management, logout, question bank/form, quiz builder/list/attempt/result/submissions, performance, class performance, study groups, content moderation. UI items have no owner in SRS Appendix D (TOD-12).	Pushkar (provisional; not allocated in SRS App. D)
COM-01	Communication-interface requirement COM-01 (SRS 3.1.3): all client-server communication over HTTPS with TLS 1.2 or higher; unencrypted connections rejected (SEC-COM-01). COM-02 to COM-10 are not traced separately (TOD-12).	Pushkar (provisional; not allocated in SRS App. D)

4. Features Not to be Tested

- Question types other than single-correct multiple choice (outside the SRS).
- Restriction of quizzes to study groups (decision: quizzes are open to every Student; SAD s4.6).
- Features not specified in SRS Revision 1.2 (draft), including quiz e-mail/push notifications and notification records (OD-09) and analytics dashboards beyond REQ-PA-01 to 06.
- Admin "manage users" (SAD s3.1): no requirement exists (TOD-08).
- Moderation actions other than deletion, and the listing of collaborative content for moderation, until specified (OD-08, OD-13).
- Re-sharing an already shared material (TOD-09) and rate-limit thresholds (TOD-02) until specified.
- Internals of MongoDB, Node.js, Express, React; third-party hosting, SMTP; physical security of server or device.
- Formats other than PDF, DOCX, PPTX and PNG are tested only for rejection.
- Features of another member are tested once, under their owner; cross-module integration is tested in Annex A.5 and Annex H.
- Real users' data or passwords: never used.

5. Test Approach / Strategy

Applicability: the levels, types and criteria below apply to all feature areas and to the team-level requirements. The strategy is unchanged in kind from v1.0 (Jest, Supertest/Postman, Cypress or Selenium, JMeter) and is made explicit per level.

Level	Purpose / role	Tools	Primary owner	Notes
Unit	Validation, calculation and authorization functions in isolation (28 candidates: UT-ACC/MAT/QZ/PA/COL).	Jest	Each member for own module	Function names are provisional (OD-10). Run on every pull request through GitHub Actions (Part-2 2b).
Integration	REST API + services + MongoDB test database + file storage; status codes, error bodies, permission checks.	Jest + Supertest; Postman	Each member	Tests with Level "Integration (API)" or "Security (API)".
System	End-to-end use cases through the React client.	Cypress or Selenium; Postman	Each member	Tests with Level "System (UI + API)".
Security	Authentication, RBAC, ownership, sessions, injection, answer-key, timing, TLS, logging (s5.1).	Postman, Supertest, openssl, browser tools	Member 1 coordinates	Member 2 for quiz authoring controls; Member 4 for material, attempt and grading controls; Member 3 for analytics/moderation.
Performance / load	NFR-PERF-01 to 06 with measured evidence (s5.2).	JMeter on a separate host	Member 1 (NFR-PERF-01, 02; mixed load); Member 4 (NFR-PERF-03, 04, 06); Member 2 (NFR-PERF-05)	R2 tests; result reported as measured.
Usability and compatibility	NFR-QUAL-01 (3 first-time users), NFR-QUAL-04 (browsers/devices).	Manual, real devices or cloud device service	Member 1 (NFR-QUAL-01, 04)	
Safety and robustness	Backup/restore, interrupted operations, restart, malformed input (NFR-SAFE, NFR-QUAL-03).	mongodump/ mongorestore, Docker, Postman	Member 1 (NFR-SAFE-01, TC-AN-16); Member 4 (NFR-SAFE-02, 03: TC-AN-02, 03, 17)	
Acceptance	Walkthrough of the use cases with the project guide (Annex H).	Manual	All members	Scenarios compose existing tests; they are not counted as extra test cases.
Regression	Smoke set on every build; full set before release candidate and after each defect fix (Annex H).	GitHub Actions + manual list	Each member for own suite	

Types applied: functional (positive, negative, boundary), security, concurrency, failure-recovery, performance, usability, compatibility, inspection and regression.

5.1 Security Validation

v1.0 checks retained; Rev 1.1 added tests and the exact status codes. Security-test results use only test accounts and safe, authorised payloads.

Control / objective	Tests	Requirements
Authentication enforcement, 401 on missing/invalid token	TC-ACC-11, TC-AS-03, TC-QB-12, TC-QZ-17, TC-PA-07, TC-PA-11, TC-COL-10	SEC-ACC-01, REQ-ACC-10, BR-01
Generic credential error; no account enumeration	TC-ACC-06, TC-AS-07	SEC-ACC-02
Password storage (bcrypt, salted)	TC-AS-04, UT-ACC-02	SEC-ACC-03, DC-04
Lock-out (5 failures, 15 min), isolated per account	TC-ACC-07, TC-AS-01, TC-ACC-20, UT-ACC-03	SEC-ACC-06, DC-08
Session end: logout, 30-min inactivity, sliding timer	TC-ACC-08, TC-AS-02, TC-ACC-17, TC-QS-04, UT-ACC-04	SEC-ACC-04, DC-08
Profile only by owner; e-mail/role protected	TC-AS-05, TC-ACC-09	SEC-ACC-05, BR-06
Role checks (Student/Faculty/Admin)	TC-QB-08, TC-QZ-16, TC-QZ-17, TC-ACC-19, TC-COL-06, TC-COL-10, TC-QZ-28	REQ-QB-07, REQ-QZ-17, REQ-COL-06, SEC-QZ-01
Ownership and cross-resource access	TC-QB-09, TC-MAT-06, TC-MAT-12, TC-MS-01 to 05, TC-QZ-24, TC-PA-06, TC-PA-11	SEC-MAT-01 to 03, SEC-QZ-03, REQ-PA-06, REQ-MAT-10
Access revocation effective on the next request (<= 2 s)	TC-MAT-11, TC-MS-02, TC-MP-03, TC-MAT-20	SEC-MAT-04, REQ-MAT-09, NFR-PERF-03
Answer-key confidentiality; immutability; server-side score and clock	TC-QZ-06, TC-QZ-08, TC-QZ-27, TC-QS-06, TC-QZ-13, TC-QZ-31, TC-QS-01, TC-QS-02, UT-QZ-06	SEC-QZ-02, SEC-QZ-04, SEC-QZ-05
NoSQL injection, XSS payloads, malformed ids	TC-QS-03, TC-QS-05, TC-AS-07, TC-AN-05, TC-AN-13, UT-ACC-06	NFR-QUAL-03, DC-07, SEC-QZ-01
Rate limiting (HTTP 429)	TC-AS-09 (blocked: TOD-02)	NFR-PERF-02
Upload validation (type, size, empty, disguised)	TC-MAT-03, TC-MAT-04, TC-MAT-16, TC-AN-05	REQ-MAT-02, DC-06
Transport security (TLS 1.2+)	TC-AS-06	SEC-COM-01, DC-03, COM-01
Security-event logging without secrets	TC-AS-08, TC-COL-14	STRIDE repudiation control (SAD s3.9); OD-06
Server-side enforcement independent of client	TC-AN-12	DC-07, DC-08

Security objectives SO-01 to SO-06 (SRS 3.3.3.1): each objective is realised by the SEC requirements listed in SRS 3.3.3.4, and the tests of those SEC requirements are its evidence set (Annex C, rows SO-01 to SO-06). SO rows are "Partially covered" (design coverage through linked tests), not "Covered", until the linked tests have been reviewed against the objective wording and executed. The SAD s3.9 SO-to-control table was reported missing in the SAD audit (F-24) and is not needed for this mapping.

5.2 Non-functional test strategy

Requirement / threshold	Tests	Environment and workload	Measurement	Acceptance	Evidence
NFR-PERF-01: >= 95% of page loads <= 3 s at 200 concurrent users	TC-AP-01, TC-AP-03	Test server; JMeter on a separate host; 200 virtual users after a documented ramp-up; login, library, profile, material view, quiz list, quiz attempt	Time from request to last byte of the main document or primary data call; browser-level spot check of 10 pages with developer tools	95th percentile <= 3 s (>= 95% of samples within 3 s)	JMeter .jtl and summary; environment specification; run id and build SHA
NFR-PERF-02: 200 users, < 1% failed	TC-AP-02, TC-AP-03	As above	Failed = timeouts, connection errors and HTTP 5xx; expected 4xx business rejections excluded	< 1% failed requests	As above
NFR-PERF-03: share/revoke effective <= 2 s	TC-MP-03	Two sessions (owner, recipient); 20 repetitions	Time from owner confirmation to first successful/denied recipient request	Every repetition <= 2 s	Script log with timestamps
NFR-PERF-04: grading <= 2 s, 100 concurrent submissions	TC-QP-01	100 generated Students with started 50-question attempts; submissions within a 5-second window	Time from submit to result per user	Every result <= 2 s; none lost; scores correct	JMeter results; DB check of 100 stored scores
NFR-PERF-05: bank search <= 2 s	TC-QP-02	Bank of 1,000 questions; 10 keyword and 10 filter runs	Response time per run	Every run <= 2 s	Timing log
NFR-PERF-06: quiz questions <= 3 s	TC-QP-03	50-question quiz; 10 runs	Start request to questions displayed	Every run <= 3 s	Timing log / screen recording
NFR-SAFE-01: daily backup; restore from <= 24 h backup	TC-AN-01	Test DB + file storage; mongodump/mongorestore and file copy	Backup timestamp age; restored data compared	Data restored; backup <= 24 h old	Backup log; before/after record counts
NFR-SAFE-02: interrupted operations keep previous state	TC-AN-02, TC-AN-17	Network interruption; concurrent update/remove	DB and storage inspection	Previous state intact; no success reported	DB/file listing before and after
NFR-SAFE-03: downloaded file identical (SHA-256)	TC-AN-03, TC-MAT-19, UT-MAT-05	PDF, DOCX, PPTX, PNG fixtures	sha256sum / certutil	Checksums identical	Checksum list

Requirement / threshold	Tests	Environment and workload	Measurement	Acceptance	Evidence
NFR-QUAL-01: register, log in, upload < 5 min unaided	TC-AN-04	3 first-time users, no instructions	Stopwatch per user	Each < 5 min (sample of 3 is small; stated in report)	Timing sheet, notes
NFR-QUAL-02: >= 99% monthly availability	TC-AN-07	Monitoring of the test server over the observed period	Uptime / (period - announced maintenance)	Reported as measured; monthly target not demonstrable (TOD-07)	Monitor export
NFR-QUAL-03: invalid input rejected, no crash	TC-AN-05, TC-AN-13, TC-QS-05, TC-MAT-16	Postman malformed/oversized/disguised input; second user active	Server state and other session	No crash; other user unaffected	Postman run, server log
NFR-QUAL-04: current Chrome, Firefox, Edge, Safari, desktop and mobile	TC-AN-06	Real devices or cloud device service (TOD-10)	Pass/fail per browser/device for register, log in, upload, download	All combinations work	Matrix with versions, screenshots
NFR-QUAL-05: modular, independently testable	TC-AN-08	Service/API boundaries	Per-module test run without other modules' internals	Modules separately executable	Test run logs
NFR-QUAL-06: traceable to tests	TC-AN-09	Annex C vs SRS Appendix C	Every requirement has a mapped test	No unmapped ID	Review record

No performance target is reported as met in this document. If the load or device environment is unavailable, the test is recorded as Not executed with the reason (TOD-10).

5.3 Entry and exit criteria

Entry criteria (revised: v1.0 required an "approved SRS and SAD", but both are drafts):

- SRS Rev 1.2 and SAD Rev 1.1 (or its successor) baselined by the team, or the affected tests marked Blocked with the decision named.
- A build identified by repository tag and commit SHA is deployed to the test server; smoke test passes.
- Test accounts and fixtures (s6.3, s6.4) loaded; test clock available for timer tests (s6.5).
- Test cases of the sprint reviewed by a second member.

Exit criteria (revised so that blocked tests do not make the criteria impossible):

- All Ready (R1/R2/R4) tests of the final build executed; each Blocked (R3) test either executed after its decision or listed with its open decision in the Test Summary Report.
- No Critical or High defect open.
- Every non-blocked requirement has at least one Passed test with evidence; Blocked and Partially covered requirements are listed explicitly.
- NFR-PERF-01 to 06 each reported as met, not met or not executed against the threshold with attached evidence.
- Deviations from the submitted SRS, SAD and STP documented (Part-2 2i).

5.4 Test design, priority and status conventions

- ID format: TC-<area>-<nn>; unit tests UT-<area>-<nn>. IDs are never reused or renumbered (s13.1).
- Every test carries: requirement(s), use case, SAD reference, level, type, priority with rationale, role, preconditions, data, numbered steps, expected result (with status code/message where specified), pass/fail criteria, dependencies, owner, status and evidence.
- Priority: High = security control, core use-case flow, data integrity or specified NFR threshold; Medium = other functional; Low = review/inspection of low risk.
- Boundary values use the specified limits only: password 8 characters; 5 failures / 15 min; 30 min inactivity; picture 2 MB; material 25 MB (PDF, DOCX, PPTX, PNG); title 100, description 500 (material); 1-50 questions; time limit 5-180 min; question text 500, option 200, topic 50, explanation 300; NFR thresholds as in s5.2. "MB" = 1,048,576 bytes with adjacent values also tested (TOD-06).

5.5 Defect handling

Severity	Definition	Target action
Critical	Security breach, data loss/corruption, build unusable, core flow blocked	Stop and fix; suspension rules s12 apply
High	Requirement not met; wrong authorization or grading; threshold clearly missed	Fix before the release candidate
Medium	Incorrect message/status; workaround exists	Fix in the sprint or record as deviation
Low	Cosmetic or documentation	Backlog

Defects are logged as GitHub Issues (Part-2 requires the GitHub project; v1.0 named Jira or GitHub Issues) with: test case ID, requirement, build SHA, steps, expected/actual, severity, evidence link. Lifecycle: New, Confirmed, Fixed (pull request linked), Retested, Closed or Reopened. Each member fixes and retests defects in their own module; Member 3 tracks ageing.

5.6 Regression and acceptance

The smoke set (High priority, Ready tests) runs on every build; the full regression set (all Ready and Environment-dependent automated tests) runs on the release candidate and after each defect fix. Acceptance scenarios AT-01 to AT-08 (Annex H) are walked through with the project guide on the final build.

6. Test Environment

6.1 Hardware and software

Hardware: a test server or cloud VM hosting the build, MongoDB test database and file storage; a separate load-generator machine for JMeter; desktop/laptop clients; Android and iOS devices or a cloud device service. Software: platform build (React client, Node.js 20 / Express 4 API, MongoDB 7, Nginx with TLS 1.2+, Docker per SAD s3.6); browsers: current Chrome, Firefox, Edge and Safari on desktop and mobile (v1.0 s6 listed only Chrome and Firefox in one place, which contradicted NFR-QUAL-04; corrected). No environment is verified as operational; availability is an entry criterion.

6.2 Tools

Jest, Supertest, Postman, Cypress or Selenium, JMeter, browser developer tools, GitHub Issues/Projects and GitHub Actions (Part-2), sha256sum / certutil, mongodump / mongorestore, openssl s_client. None is claimed to be installed.

6.3 Test accounts

Account	Role	Purpose
F1, F2	Faculty	Owners of questions, quizzes, materials; cross-owner denial
S1, S2, S3	Student	Attempts, sharing recipients, groups; S3 has no attempts
A1	Admin	Moderation and role-restriction tests; seeded (creation mechanism: OD-11)
100/200 generated Students	Student	Load tests (NFR-PERF-01 to 04)
3 first-time users	Student/Faculty	Usability test (NFR-QUAL-01)

Passwords are synthetic, created per environment, held outside the repository, and never taken from real users. Values such as "Study2026" in test steps are test-only.

6.4 Test data and fixtures

Fixture	Content
FX-FILES	Valid PDF, DOCX, PPTX, PNG; TXT; 0-byte file; .exe renamed .pdf; files of 26,214,400 and 26,214,401 bytes (25 MB boundary); pictures of 2,097,152 and 2,097,153 bytes (2 MB boundary); PNG/JPEG/GIF/WebP pictures (OD-13)
FX-META	Title of 100/101 characters, description of 500/501; full name 100/101; e-mail 254/255; passwords satisfying and violating the rule
FX-QB20 / FX-QB1000	Question bank of 20 questions in three topics, and a generated bank of 1,000 questions
FX-QUIZ	Quizzes in open, upcoming, closed windows; 1-, 5-, 10- and 50-question quizzes; 5-minute quiz; known answer keys
FX-MAT	Materials owned by F1 shared with S1; owned by F2 not accessible to F1/S1; removed and revoked materials
FX-GRP	Groups G1 (created by S1), G2 (several Students), G3 removed; content C1 available, C2 removed
FX-LOAD	200 generated users; 100 started attempts for the grading test
FX-BACKUP	Backup taken before M1 is deleted

Named objects (M1, Z1, G1, C1, ...) are defined by the precondition of each test; the seed script is parameterised per test, so the same name may denote different content in different tests. Reset: before each test run the seed script restores the database and file storage to the fixture state; after the run, logs and evidence are exported and the data is discarded.

6.5 Test clock

Timer tests (15-minute lock-out, 30-minute inactivity, quiz windows and deadlines) need a server clock that can be advanced or mocked. The SAD does not define such a facility; the team must provide one in the test build (for example, an environment-controlled time offset). Until then, those tests are R1 but cannot be completed in real time within the sprint.

6.6 Build identification and evidence

Every execution record states build tag and commit SHA, environment specification, date, tester, result, and an evidence reference. Evidence folder convention: evidence/<TC-ID>/<yyyymmdd>-<sha>-<tester>/ with logs, Postman/JMeter exports, screenshots or short screen recordings; evidence must not contain real credentials or tokens.

6.7 Browsers and devices (NFR-QUAL-04)

Desktop: Chrome, Firefox, Edge, Safari (macOS). Mobile: Chrome, Firefox, Edge, Safari on Android and iOS. Versions recorded at execution.

7. Test Schedule

v1.0 planned 03-10-2026 to 07-10-2026. That window has passed and is replaced by the Deliverables Part-2 plan. All dates below are planned milestones, not completed work; backlog creation (ETA 12-Oct-2026) cannot be confirmed from the supplied material.

Milestone / activity	Date (2026)	Responsible	Status
Backlog created in GitHub from FR/NFR, story points, stories assigned (Part-2 s1); test stories linked to TC IDs	12-Oct (ETA); changes allowed until 19-Oct	Each member	Backlog CSV prepared (Team17_Part2_Backlog.csv); GitHub Project creation and the 12-Oct ETA not verified
Sprint dry run (optional, Part-2 s1f)	12-16 Oct	All	Unverified
STP Rev 1.2 draft prepared; team review; resolve OD-03, OD-07, OD-08 as far as possible and supply the Member 4 SRN	18-Oct (Sun) to 19-Oct	All; Member 3 merges	Planned - draft prepared
Preparation: test environment, fixtures, seed scripts, test clock, GitHub Actions workflow for Jest/Supertest, smoke set	19-20 Oct	Each member for own part; Member 1 pipeline	Planned
Sprint 1 execution: tests for Sprint 1 stories as features are merged; each pull request links its TC IDs and evidence	19-23 Oct	Each member	Planned
Sprint 1 video (1 minute, working product) uploaded to the repository (Part-2 s2g-h)	by 23-Oct	All	Planned
Sprint 1 test status report; defects triaged; deviations list started	23-Oct	Member 3	Planned
Sprint 2 execution: remaining functional tests; security tests	26-28 Oct	Each member	Planned
Performance, safety, usability, compatibility and availability tests (NFR)	27-29 Oct	Members 1, 2, 4 as in s5.2	Planned
Regression on the release candidate (smoke + full)	29-Oct	All	Planned
Acceptance walkthrough with the project guide (AT-01 to AT-08)	29-30 Oct (date to be agreed)	All; project guide	Planned
Sprint 2 video (2 minutes); documentation complete (Test Summary Report, deviation log, final STP); development freeze (Part-2 s3)	30-Oct	All; Member 3 report	Planned
Final demo	Date not stated in Part-2	All	To be confirmed

Proposed split of tests by sprint (dependency-driven; aligned with Team17_Part2_Backlog.csv, which is a prepared import file, not evidence that the GitHub Project exists): Sprint 1 - Account, Material, Question Bank and quiz authoring, and study groups (REQ-COL-01 to 04, which depend on Account only); tests flagged "Sprint 1" in Annex B. Sprint 2 - quiz execution and grading, analytics, moderation and content availability (REQ-COL-05 to 07), NFR and cross-module tests. Order of dependency: Account -> Material and Question Bank -> Quiz -> Performance Analytics; Collaboration depends on Account only. If a story slips, its tests move with it and the change is recorded in the status report.

8. Test Deliverables

Deliverable	Description	Status
Software Test Plan	This document (Revision 1.2 draft)	Draft - review pending
Test cases	160 test cases (97 retained from v1.0, 63 new) in Annex A; 117 are Ready, 24 Environment-dependent, 13 Blocked, 6 Manual	Designed; not executed
Unit-test candidates	28 Jest candidates (20 retained, 8 new)	Specified; scripts not written
Traceability and inventory	Annexes B-E and the Excel workbook	Mapped; execution column empty
Final report and deviation log	Team17_Submission_Set_Final_Report.docx (changes, unresolved decisions, consistency check); Team17_Deviation_Log.docx	Draft

Deliverable	Description	Status
Open-decision and blocked-test register	Annex F	Open
Test data / seed scripts / test clock	s6	To be prepared
Execution results and evidence	Per test (s6.6)	Not available - nothing executed
Defect reports	GitHub Issues	None logged
Sprint videos	1-minute (Sprint 1), 2-minute (Sprint 2), in the repository	Planned
Test Summary Report and deviation log	Member 3 (document deviations: Team17_Deviation_Log.docx)	Planned for 30-Oct

9. Roles and Responsibilities

Ownership follows SRS Revision 1.2 Appendix D (four members; baseline workload decision confirmed by the team on 18-10-2026). The owner of a test case or unit-test candidate is the Appendix D primary owner of its first-listed requirement; cross-module tests list the other members' requirements as supporting (SRS D.2, D.4). UI-01 to UI-18 and COM-01 are not allocated in Appendix D and stay provisionally with Member 3 (TOD-12). Member 4's SRN is still to be supplied. Annex B and C carry the owner of every test.

Role	Name	Responsibility
Member 1 - Account Management, authentication, sessions, access-control middleware; security coordination	Jagadeesh (PES1UG24CS347)	Designs, executes and logs defects for REQ-ACC-01 to 10, SEC-ACC-01 to 06, SEC-COM-01, NFR-PERF-01, 02, NFR-SAFE-01, NFR-QUAL-01, 03, 04, BR-01, 06, 08, DC-01, 04, 05, 07, 08; coordinates SO-01 to SO-06 (SRS D.4). Tests: TC-ACC-01-20; TC-AN-01, 04-06, 10, 12-13, 16; TC-AP-01-03; TC-AS-01-09; TC-QS-05; UT-ACC-01, UT-ACC-02, UT-ACC-03, UT-ACC-04, UT-ACC-05, UT-ACC-06, UT-MAT-04. Qualitative property (from SRS NFRs; team to confirm): confidentiality of account credentials and sessions (SEC-ACC-03, SEC-ACC-04; SO-01, SO-04). Maintains shared middleware and the GitHub Actions test pipeline.
Member 2 - Question Bank and Quiz Authoring	Alifiya (PES1UG25CD841)	REQ-QB-01 to 07, REQ-QZ-01 to 05, 16, 17, 18, SEC-QZ-01, 03, NFR-PERF-05, DC-02, DC-10. Tests: TC-AN-11; TC-QB-01-12; TC-QP-02, 04; TC-QS-03-04; TC-QZ-01-05, 15-22, 24, 28; UT-QZ-02, UT-QZ-03, UT-QZ-07, UT-QZ-08. Qualitative property: question-bank search within 2 seconds for up to 1,000 questions (NFR-PERF-05, REQ-QB-04). Checks that tests follow SAD components (Question Bank Service, Quiz Engine).
Member 3 - Performance Analytics, Collaboration, study groups and moderation; test merge and V&V	Pushkar (PES1UG24CS350)	REQ-PA-01 to 06, REQ-COL-01 to 07, NFR-QUAL-02, 05, 06, DC-03, DC-09, domain requirement. Tests: TC-AN-07-09, 14-15; TC-COL-01-14; TC-PA-01-13; UT-COL-01, UT-COL-02, UT-COL-03, UT-COL-04, UT-PA-01, UT-PA-02, UT-PA-03. Qualitative property: features implemented as separate, independently testable modules (NFR-QUAL-05). Merges the final STP; maintains traceability, metrics and the Test Summary Report; decides suspension/resumption after consulting the owner.

Role	Name	Responsibility
Member 4 - Material Management and Sharing, File Storage, Quiz Execution, attempts, timing, submission and grading	Vedh ([SRN not yet supplied])	REQ-MAT-01 to 11, REQ-QZ-06 to 15, SEC-MAT-01 to 04, SEC-QZ-02, 04, 05, NFR-PERF-03, 04, 06, NFR-SAFE-02, 03, BR-02 to 05, 07, DC-06, DC-11. Tests: TC-AN-02-03, 17; TC-MAT-01-22; TC-MP-01-03; TC-MS-01-05; TC-QP-01, 03; TC-QS-01-02, 06; TC-QZ-06-14, 23, 25-27, 29-32; UT-MAT-01, UT-MAT-02, UT-MAT-03, UT-MAT-05, UT-MAT-06, UT-QZ-01, UT-QZ-04, UT-QZ-05, UT-QZ-06, UT-QZ-09. Qualitative property: downloaded file identical to the uploaded file, checked by checksum (NFR-SAFE-03). Supplies the attempt-started state to quiz authoring and the shared attempt-query interface to analytics (SRS D.4).
Project guide	Ashok Kumar Patil	Reviews the plan, takes part in acceptance, signs off.

Platform roles (Student, Faculty, Admin) are test personas, not team members. Every member takes part in review, integration and regression; the table shows primary ownership. Where the SAD assigns an item to "team agreement" (OD-05, OD-06) the owner of the test that needs it prepares it.

10. Risks and Mitigation

Risk	Mitigation
Two-week sprint window (19-30 Oct) leaves little time between feature merge and freeze	Run smoke tests on every pull request; run High-priority and security tests first; test stories linked to backlog items
Test environment instability or downtime	Seed scripts and Docker configuration kept so the environment can be rebuilt; back up before each cycle; local fallback
Insufficient or inconsistent test data	Seed scripts per fixture (s6.4); shared accounts across members; verify data before execution
Integration defects between modules (quiz attempts feeding analytics; AccessGrant checks; moderation)	Early API integration tests; TC-PA-10, TC-PA-13, TC-AN-11; regression after fixes
Load generation limited by the test server, load generator or short test period (TOD-10)	JMeter on a separate host, gradual ramp-up, record specifications; report as measured; mark Not executed if impossible
Browser/device coverage (Safari, iOS)	Cloud device service; record versions; smoke checks first
Third-party infrastructure outage	Check before each cycle; classify as environment issue
Late change to SRS/SAD IDs or limits breaking traceability	Freeze IDs for the baseline (DC-10); TC-AN-09 and Annex C re-checked before sign-off
SAD file not supplied to this revision (TOD-01) - status codes, error-code strings and messages taken from STP Rev 1.1 may differ from the SAD	Every SAD-derived expectation is tagged with its source; a verification pass against the SAD before execution
Member 4 SRN not yet supplied; workload imbalance (Member 4 owns the most requirements under the baseline allocation)	Ownership is data-driven (Annex B/C); SRN to be supplied before submission; Option B in the ownership workbook is available if the team revisits the split
Blocked tests (13) hide untested behaviour (OD-03, OD-07, OD-08, OD-13, TOD-02/03/04)	Register in Annex F with decision owner; re-baseline tests when decided
Time-dependent tests (15 min, 30 min) cannot run in real time	Injectable test clock (s6.5); agree with developers in Sprint 1
Flaky timing and concurrency tests	Repeat runs, report distribution, fix environment noise before logging defects

11. Assumptions and Dependencies

Assumptions:

- Requirement, use-case, security and NFR IDs of SRS Rev 1.2 remain stable during testing (DC-10).
- A stable build (React, Node.js/Express, MongoDB) is deployed on the test server before execution; its existence is not verified here.
- Test accounts F1, F2, S1, S2, S3, A1 and the fixtures of s6.4 are provided before execution.
- Testing is performed only on the test environment.
- Dates in s7 follow Deliverables Part-2 and are planned, not achieved.

Dependencies:

- MongoDB test database and backup/restore tooling; file storage.
- Browsers/devices; Jest, Supertest, Postman, Cypress or Selenium, JMeter, openssl, sha256sum.
- A test clock and fault-injection hooks (grading failure, MongoDB stop) in the test build.
- Decisions OD-03, OD-07, OD-08, OD-13, TOD-02, TOD-03, TOD-04 for the blocked tests (Annex F).
- Project guide availability for acceptance.

12. Suspension and Resumption Criteria

Suspend testing if: the test environment (server, MongoDB test database or file storage) is unavailable for more than 2 hours; the build blocks more than 30% of the planned tests of the current cycle; a critical defect blocks meaningful execution (cannot log in, upload, start or submit a quiz); required data or services cannot be restored within 2 hours. (The 2-hour period shortens the template's 4 hours; retained from v1.0.)

Resume when: blocking defects are fixed or a new stable build passes the smoke test; the environment is verified; data and services are restored; blocked and affected tests are re-run with regression tests of related modules. Member 3 (V&V lead) decides after consulting the owner of the affected part; the decision, time lost and any change to s7 dates are recorded in the status report. Tests blocked by an open decision (R3) are not a suspension trigger; they are tracked in Annex F.

13. Test Case Management and Traceability

13.1 ID policy

All 97 test IDs and 20 unit-test IDs of v1.0 are retained unchanged; none was split, replaced or renumbered (Annex G). New tests continue each series (for example TC-ACC-12 onward) and are marked "New in Rev 1.1". IDs are unique across the document (checked by script; Annex G records the check). One test ID is never reused for different content. Unit test UT-MAT-04 validates profile updates (REQ-ACC-08/09); its prefix is retained to avoid renumbering and is noted as an inconsistency, not corrected.

13.2 Traceability

Annex C has one row for each of the 118 requirement rows (115 SRS IDs plus the domain requirement, the UI group and COM-01). Annex D is the reverse index. Annex E summarises coverage. A requirement is mapped when a test lists it; each test lists its primary requirement first, and the test owner is the SRS Appendix D primary owner of that first-listed requirement. SO-01 to SO-06 are mapped through the SEC requirements named in SRS 3.3.3.4. The v1.0 RTM contained three inconsistencies that are corrected here: REQ-PA-04 and SEC-ACC-02 were mapped to tests that did not list them (TC-PA-06, TC-ACC-06), and TC-QS-04 listed SEC-ACC-04 that the RTM omitted.

13.3 Maintenance

Pull requests reference the TC IDs they affect; Member 3 re-checks Annex C against the SRS RTM (Appendix C) whenever an ID or limit changes; the spreadsheet companion is the working copy for execution status.

14. Test Metrics and Reporting

No test has been executed as at 18-10-2026; all execution metrics are zero/not available. Planned baselines: 160 test cases (28 unit candidates); 118 requirement rows: 91 Covered, 23 Partially covered, 1 Blocked pending clarification, 3 Not directly testable, 0 Not covered.

Metric	Definition / target
Tests planned / executed	Number and % of the 160 test cases and 28 unit candidates executed; blocked tests reported separately
Pass / fail / blocked / not executed %	Per member and team; no pass without evidence
Requirement coverage	% of non-blocked requirements with at least one Passed test; list of Blocked/Partially covered
Defects by severity and module; ageing	Critical/High/Medium/Low; days from logging to closure
NFR compliance	Each of NFR-PERF-01 to 06 met / not met / not executed
Security coverage	% of the 16 SEC requirements with an executed security test, plus s5.1 checklist
Regression status	% of the regression suite passing on the final build; reopened defects
Open decisions	Count open per Annex F; blocked tests unblocked

Reports: sprint test status report (23-Oct), Test Summary Report (30-Oct), defect reports (GitHub Issues), coverage/traceability report (spreadsheet). Deviations from the SRS, SAD and STP are recorded in the log of Annex H.

15. Approvals

No approval has been given. Signature cells are intentionally blank.

Role	Name	Signature / Date
Member 1	Jagadeesh	
Member 2	Alifiya	
Member 3	Pushkar	
Member 4	Vedh	
Project guide / product owner	Ashok Kumar Patil	

Annex A. Test Cases

160 test cases. Each card is a complete, independent specification. Retained cases keep their v1.0 wording; "Rev 1.1 clarification" adds specified status codes and messages from SAD s4.3-4.5. Execution status is "Not executed" for all.

A.1 Account Management

TC-ACC-01 - Register a new Student account			
Requirements	REQ-ACC-01		
Use case	UC-ACC-01	**SAD reference**	Components: Auth & Session Service; API Layer (validation) | API: POST /auth/register
Test level	System (UI + API)	**Test type**	Functional – Positive
Priority	High - Core use-case flow or integrity control; failure blocks the use case.	**Assigned to**	Jagadeesh (M1)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	New user is not registered; Register Account page is open.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Open Register Account. 2. Enter a valid full name (e.g. "Asha Rao"). 3. Enter a unique valid email (e.g. asha.rao@example.com). 4. Enter a password satisfying the password rules (e.g. "Study2026"). 5. Select Student. 6. Submit.		
Expected result	Account is created successfully with the selected role (Student). The user receives a confirmation. Rev 1.1 clarification: Response contract (SAD s4.3): HTTP 201 {id, email, role}; password never returned.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-ACC-02 - Registration: valid, duplicate (case-insensitive), empty and invalid-email inputs			
Requirements	REQ-ACC-01, REQ-ACC-02		
Use case	UC-ACC-01	**SAD reference**	Components: Auth & Session Service; API Layer (validation) | API: POST /auth/register
Test level	System (UI + API)	**Test type**	Functional – Positive/Negative
Priority	High - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Jagadeesh (M1)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Register Account page is open.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Register using a valid Student account. 2. Attempt registration again using the same email with different letter casing. 3. Attempt registration with an empty required field. 4. Attempt registration with an invalid email.		
Expected result	Valid registration succeeds. Duplicate, empty and invalid inputs are rejected and the failing field/rule is identified. Rev 1.1 clarification: Duplicate e-mail: HTTP 400 VALIDATION_ERROR naming the e-mail field (SAD s4.3).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-ACC-03 - Registration validation: each required field empty; invalid e-mail formats; duplicate e-mail with different capitalisation			
Requirements	REQ-ACC-02		
Use case	UC-ACC-01	**SAD reference**	Components: Auth & Session Service; API Layer (validation) | API: POST /auth/register

Test level	System (UI + API)	**Test type**	Functional – Boundary
Priority	Medium - Boundary values defined in the specification are a common defect location.	**Assigned to**	Jagadeesh (M1)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Register Account page is open; one account already exists for s1@example.com.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Submit registration with each required field (full name, email, password, role) individually empty. 2. Submit an invalid email format (e.g. "abc", "abc@"). 3. Submit a duplicate email using different capitalization (e.g. S1@EXAMPLE.COM).		
Expected result	Each invalid registration is rejected and the failing field is named. No duplicate account is created. Rev 1.1 clarification: Each failure: HTTP 400 VALIDATION_ERROR naming the failing field and rule.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-ACC-04 - Password rule boundary (8 vs 7 characters; missing character classes)			
Requirements	REQ-ACC-03		
Use case	UC-ACC-01	**SAD reference**	Components: Auth & Session Service | API: POST /auth/register
Test level	System (UI + API)	**Test type**	Functional – Boundary
Priority	High - Boundary values defined in the specification are a common defect location.	**Assigned to**	Jagadeesh (M1)
Required role	Public / any	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Register Account page is open.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Test a password of exactly 8 characters satisfying uppercase, lowercase and digit (e.g. "Abcdef1g"). 2. Test a 7-character password (e.g. "Abcde1g"). 3. Test passwords missing an uppercase letter, a lowercase letter or a digit.		
Expected result	The valid boundary password is accepted. All invalid passwords are rejected and the password rule is identified. Rev 1.1 clarification: Password failures: HTTP 400 VALIDATION_ERROR naming the password rule.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-ACC-05 - Valid log-in grants access within 2 seconds			
Requirements	REQ-ACC-04		
Use case	UC-ACC-02	**SAD reference**	Components: Auth & Session Service | API: POST /auth/login
Test level	Performance (JMeter / timing)	**Test type**	Functional / Performance
Priority	High - Verifies a measurable NFR threshold; failure is an NFR breach.	**Assigned to**	Jagadeesh (M1)
Required role	Student	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	Student S1 is registered.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Log in using valid credentials (10 runs). 2. Measure the time from the login request to successful access.		
Expected result	Authentication succeeds and access is granted within 2 seconds in every run. Rev 1.1 clarification: Success response HTTP 200 {token, expiresIn: 1800}. Measure with a scripted client (10 runs); the 2-second limit is applied to every run.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		
Dependencies	Tooling/environment per s6 (JMeter, load generator, devices, backup or TLS host as applicable). All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-ACC-06 - Invalid credentials return the single generic message

Requirements	REQ-ACC-05, SEC-ACC-02		
Use case	UC-ACC-02	**SAD reference**	Components: Auth & Session Service | API: POST /auth/login
Test level	System (UI + API)	**Test type**	Functional – Negative
Priority	High - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Jagadeesh (M1)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 is registered; the email nobody@example.com is not registered.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Enter an incorrect password for the registered account S1. 2. Repeat using the unregistered email. 3. Inspect the response.		
Expected result	Both cases return exactly: "Invalid email or password". No account-existence information is disclosed (same message and status in both cases). Rev 1.1 clarification: HTTP 401 INVALID_CREDENTIALS with message "Invalid email or password"; identical status, code, message and body shape for wrong password and unknown e-mail (SEC-ACC-02).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-ACC-07 - Account lock-out after five consecutive failed log-ins (15 minutes)			
Requirements	REQ-ACC-06, SEC-ACC-06, DC-08		
Use case	UC-ACC-02	**SAD reference**	Components: Auth & Session Service | API: POST /auth/login
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Jagadeesh (M1)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S2 is registered and not locked.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Enter an incorrect password five consecutive times. 2. Attempt login using the correct password immediately afterward. 3. Attempt login again after the 15-minute lockout period (server test clock advanced).		
Expected result	The account is locked after five consecutive failures ("Account temporarily locked. Try again in 15 minutes."). Authentication remains unavailable during the lockout and becomes available after the defined lockout period. Rev 1.1 clarification: Locked response: HTTP 423 ACCOUNT_LOCKED, message "Account temporarily locked. Try again in 15 minutes." The correct password is also refused during lock-out. Boundary: sixth attempt after exactly 15:00 is accepted (clock semantics: TOD-04).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Server test clock that can be advanced (s6.5). All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: TOD-04.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-ACC-08 - Session ends on logout and after 30 minutes of inactivity			
Requirements	REQ-ACC-07		
Use case	UC-ACC-03 to 05 (UC ID not individually verifiable without SRS)	**SAD reference**	Components: Auth & Session Service; API Layer (authentication middleware) | API: POST /auth/logout; all protected endpoints
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Jagadeesh (M1)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 is registered.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Log in successfully and note the token. 2. Log out.		

	3. Reuse the previous authentication token for a protected API request. 4. Separately, leave a session inactive for 30 minutes (server test clock advanced) and repeat a protected request.		
Expected result	Both requests are rejected (HTTP 401) because the session/token is no longer valid. The idle case shows "Session expired. Please log in again." Rev 1.1 clarification: After logout and after idle expiry: HTTP 401 NOT_AUTHENTICATED; idle message "Session expired. Please log in again." Logout returns HTTP 204.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Server test clock that can be advanced (s6.5). All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-ACC-09 - View and update permitted profile fields; e-mail and role protected			
Requirements	REQ-ACC-08, BR-06, DC-06		
Use case	UC-ACC-03 to 05 (UC ID not individually verifiable without SRS)	**SAD reference**	Components: Account Service; File Storage; API Layer | API: GET, PUT /account/profile
Test level	System (UI + API)	**Test type**	Functional – Positive
Priority	Medium - Verifies the stated functional requirement under normal conditions.	**Assigned to**	Jagadeesh (M1)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 is registered.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Log in as S1. 2. Open the profile. 3. Change name, institution and profile picture (PNG under 2 MB). 4. Save. 5. Inspect the email and role fields.		
Expected result	Permitted profile fields are updated and shown on reload. Email and role remain non-editable. Rev 1.1 clarification: PUT /account/profile accepts fullName, institution and profilePicture only (HTTP 200); a request containing e-mail or role is rejected with HTTP 400 (SAD s4.3). Picture type assumption: PNG (accepted types are OD-13).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: OD-13.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-ACC-10 - Profile validation: empty name; picture at and above 2 MB			
Requirements	REQ-ACC-09, DC-06		
Use case	UC-ACC-03 to 05 (UC ID not individually verifiable without SRS)	**SAD reference**	Components: Account Service; File Storage; API Layer | API: PUT /account/profile
Test level	System (UI + API)	**Test type**	Functional – Negative/Boundary
Priority	Medium - Boundary values defined in the specification are a common defect location.	**Assigned to**	Jagadeesh (M1)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 is logged in with a valid saved profile.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Attempt a profile update with an empty name. 2. Attempt upload of a profile picture slightly greater than 2 MB. 3. Upload a profile picture of exactly 2 MB.		
Expected result	Steps 1 and 2 are rejected and the failing field/rule is identified; previous valid profile information remains unchanged. Step 3 is accepted (limit is "larger than 2 MB"). Rev 1.1 clarification: Failures: HTTP 400 VALIDATION_ERROR naming field and rule. 1 MB is taken as 1,048,576 bytes (2 MB = 2,097,152 bytes); also test 2,097,153 bytes (TOD-06).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: TOD-06.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-ACC-11 - Protected endpoints reject missing authentication and unauthorized roles

Requirements	REQ-ACC-10, BR-01		
Use case	UC-ACC-02	**SAD reference**	Components: API Layer; Auth & Session Service; API Layer (authentication middleware) | API: All protected endpoints
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Jagadeesh (M1)
Required role	Faculty, Student, Unauthenticated	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	No valid session; Student S1 session available for the role check.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Call a protected account endpoint without authentication. 2. Call a protected material endpoint without authentication. 3. Attempt a role-restricted operation using an unauthorized role (e.g. S1 calls a Faculty-only endpoint).		
Expected result	Requests are rejected (401 / 403) and protected information or operations are not exposed. Rev 1.1 clarification: Resolves the "401 / 403" ambiguity of STP v1.0: missing/invalid token -> HTTP 401 NOT_AUTHENTICATED; valid token with the wrong role -> HTTP 403 ACCESS_DENIED "Access denied".		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-ACC-12 - Registration rejects the Admin role (BR-08)			
Requirements	REQ-ACC-01, BR-08		
Use case	UC-ACC-01	**SAD reference**	Components: Auth & Session Service; API Layer (validation) | API: POST /auth/register
Test level	Integration (API)	**Test type**	Security - Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Jagadeesh (M1)
Required role	Public (unauthenticated)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Registration endpoint reachable; no account exists for admin.test@example.com.		
Test data	admin.test@example.com; password Study2026		
Steps	1. POST /api/auth/register with {fullName:"Test Admin", email:"admin.test@example.com", password:"Study2026", role:"Admin"}. 2. Repeat with role "Teacher", role "" (empty) and with the role field omitted. 3. Query the users collection for admin.test@example.com.		
Expected result	Steps 1-2: HTTP 400 VALIDATION_ERROR naming the role field and the rule (Student or Faculty only); no token or id returned. Step 3: no account exists.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Registration endpoint; MongoDB test database.. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: OD-11.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-ACC-13 - Registration length boundaries (full name, e-mail)			
Requirements	REQ-ACC-02		
Use case	UC-ACC-01	**SAD reference**	Components: Auth & Session Service; API Layer (validation) | API: POST /auth/register
Test level	Integration (API)	**Test type**	Functional - Boundary
Priority	Medium - Boundary values defined in the specification are a common defect location.	**Assigned to**	Jagadeesh (M1)
Required role	Public	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Registration endpoint reachable; e-mail addresses below are unused.		
Test data	Generated strings of 1/100/101 characters; e-mails of 254/255 characters		
Steps	1. Register with a full name of exactly 1 character and a valid unique e-mail. 2. Register with a full name of exactly 100 characters. 3. Register with a full name of 101 characters. 4. Register with a valid e-mail address of exactly 254 characters in total (unique). 5. Register with a valid-format e-mail address of 255 characters.		
Expected result	Steps 1, 2 and 4: HTTP 201 {id, email, role}. Steps 3 and 5: HTTP 400 VALIDATION_ERROR naming the failing field and limit; no account created. Limits are those of SAD s4.3 (full name 1-100, e-mail at most 254); (confirmed against SRS Appendix B.1).		

Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-ACC-14 - View own profile			
Requirements	REQ-ACC-08, REQ-ACC-10, SEC-ACC-03		
Use case	UC-ACC-03 to 05 (not individually verifiable)	**SAD reference**	Components: Account Service; File Storage; API Layer | API: GET, PUT /account/profile; All protected endpoints; POST /auth/register
Test level	System (UI + API)	**Test type**	Functional - Positive
Priority	Medium - Verifies the stated functional requirement under normal conditions.	**Assigned to**	Jagadeesh (M1)
Required role	Student (S1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 is registered and logged in.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Open the Profile screen. 2. Call GET /api/account/profile with S1's token. 3. Inspect the JSON response.		
Expected result	HTTP 200 with fullName, institution, profilePicture, email and role only. No password, hash, session, token or lock-out counter appears in the response. The screen shows e-mail and role as read-only.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-ACC-15 - Profile text-field boundaries			
Requirements	REQ-ACC-09		
Use case	UC-ACC-03 to 05 (not individually verifiable)	**SAD reference**	Components: Account Service; File Storage | API: PUT /account/profile
Test level	Integration (API)	**Test type**	Functional - Boundary
Priority	Medium - Boundary values defined in the specification are a common defect location.	**Assigned to**	Jagadeesh (M1)
Required role	Student (S1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 is logged in with a valid profile.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. PUT /api/account/profile with fullName of exactly 100 characters. 2. PUT with fullName of 101 characters. 3. PUT with institution of exactly 100 characters. 4. PUT with institution of 101 characters. 5. GET the profile.		
Expected result	Steps 1 and 3: HTTP 200 and saved. Steps 2 and 4: HTTP 400 VALIDATION_ERROR naming field and limit; stored values unchanged. Step 5 shows the last valid values. Limits per SAD s4.3 (name 1-100, institution at most 100); confirmed against SRS Appendix B.3.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-ACC-16 - Profile picture: accepted file types			
Requirements	REQ-ACC-09, REQ-ACC-08		
Use case	UC-ACC-03 to 05 (not individually verifiable)	**SAD reference**	Components: Account Service; File Storage | API: PUT /account/profile; GET, PUT /account/profile
Test level	Integration (API)	**Test type**	Functional - Boundary
Priority	Medium - Boundary values defined in the specification are a common defect location.	**Assigned to**	Jagadeesh (M1)
Required role	Student (S1)	**Readiness**	Blocked - expected result not agreed
Preconditions	S1 is logged in. Picture fixtures below are each under 2 MB.		
Test data	PNG, JPEG, GIF, WebP and fake-PNG fixtures (each < 2 MB)		

Steps	1. Upload a PNG picture. 2. Upload a JPEG picture. 3. Upload a GIF picture. 4. Upload a WebP picture. 5. Upload a text file renamed to .png.		
Expected result	BLOCKED (OD-13): the accepted picture types are not specified. Record the actual outcome of each step. Agreed expectation to be approved before execution is closed; the only expectation fixed by the SAD is that an unsupported picture type returns HTTP 400 and that a non-image file must not be stored.		
Pass / fail criteria	Cannot pass or fail until the linked decision is recorded (Blocked). When the decision is approved, update the expected result and re-baseline.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: OD-13.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-ACC-17 - Inactivity timer restarts on activity (sliding 30-minute expiry)			
Requirements	REQ-ACC-07, SEC-ACC-04, DC-08		
Use case	UC-ACC-03 to 05 (not individually verifiable)	**SAD reference**	Components: Auth & Session Service; API Layer (authentication middleware); API Layer | API: POST /auth/logout; all protected endpoints
Test level	Security (API)	**Test type**	Security - Boundary
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Jagadeesh (M1)
Required role	Student (S1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 is registered; the server test clock can be advanced.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Log in as S1 (time T0). 2. Advance the clock by 20 minutes; call GET /api/account/profile. 3. Advance by a further 20 minutes (40 minutes after login, 20 minutes after the last request); call the endpoint again. 4. Advance by 31 minutes with no request; call the endpoint again.		
Expected result	Steps 2 and 3: HTTP 200 (inactivity, not total age, ends the session). Step 4: HTTP 401 NOT_AUTHENTICATED with message "Session expired. Please log in again." Sliding behaviour is implied by "30 minutes of inactivity" (SAD s3.4, s4.3); consistent with SRS REQ-ACC-07 (30 minutes of inactivity).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Injectable server clock or clock-mocking in the test build (see s6.5).. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-ACC-18 - Log-in request validation and success contract			
Requirements	REQ-ACC-04, REQ-ACC-05		
Use case	UC-ACC-02	**SAD reference**	Components: Auth & Session Service | API: POST /auth/login
Test level	Integration (API)	**Test type**	Functional - Positive / Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Jagadeesh (M1)
Required role	Public	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 is registered.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. POST /api/auth/login with the e-mail missing. 2. POST with the password missing. 3. POST with valid credentials. 4. Use the returned token on GET /api/account/profile.		
Expected result	Steps 1-2: HTTP 400 VALIDATION_ERROR naming the missing field. Step 3: HTTP 200 {token, expiresIn: 1800}. Step 4: HTTP 200.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-ACC-19 - Seeded Admin can authenticate and is role-restricted

Requirements	REQ-ACC-10, BR-08		
Use case	UC-ACC-02	**SAD reference**	Components: API Layer; Auth & Session Service | API: All protected endpoints
Test level	Security (API)	**Test type**	Security - Negative
Priority	Medium - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Jagadeesh (M1)
Required role	Admin (A1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Admin account A1 exists (provisioned by seed script; the creation mechanism is unspecified, OD-11).		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Log in as A1. 2. Call GET /api/quizzes (Student-only) with A1's token. 3. Call POST /api/questions (Faculty-only) with A1's token. 4. Call GET /api/performance/personal (Student-only) with A1's token.		
Expected result	Step 1: HTTP 200 with a token. Steps 2-4: HTTP 403 ACCESS_DENIED "Access denied"; no data returned.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: OD-11.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-ACC-20 - Lock-out is isolated to the failing account			
Requirements	REQ-ACC-06, SEC-ACC-06		
Use case	UC-ACC-02	**SAD reference**	Components: Auth & Session Service | API: POST /auth/login
Test level	Security (API)	**Test type**	Security - Negative
Priority	Medium - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Jagadeesh (M1)
Required role	Student (S1, S2)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 and S2 are registered and not locked.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Submit five consecutive invalid log-ins for S2. 2. Log in as S1 with the correct password. 3. Log in as S2 with the correct password.		
Expected result	Step 2: HTTP 200 (S1 unaffected). Step 3: HTTP 423 ACCOUNT_LOCKED "Account temporarily locked. Try again in 15 minutes."		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

A.2 Material Sharing

TC-MAT-01 - Upload a valid material			
Requirements	REQ-MAT-01, BR-02		
Use case	UC-MAT-01	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: POST /materials
Test level	System (UI + API)	**Test type**	Functional – Positive
Priority	High - Core use-case flow or integrity control; failure blocks the use case.	**Assigned to**	Vedh (M4)
Required role	Faculty	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Faculty F1 is logged in.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Log in as F1. 2. Open Upload Material. 3. Enter a valid title, subject and description. 4. Upload a valid PDF. 5. Submit.		
Expected result	Material is uploaded and recorded under F1 (owner). Rev 1.1 clarification: Response HTTP 201 {id, owner, title, subject}. SAD s4.3 restricts upload to Student and Faculty (OD-07 for Admin: TC-MAT-14).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any		

	difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: OD-07.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-MAT-02 - Material metadata length boundaries (title 100, description 500)			
Requirements	REQ-MAT-01		
Use case	UC-MAT-01	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: POST /materials
Test level	System (UI + API)	**Test type**	Functional – Boundary
Priority	Medium - Boundary values defined in the specification are a common defect location.	**Assigned to**	Vedh (M4)
Required role	Faculty	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 is logged in.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Upload material with a title of exactly 100 characters. 2. Upload material with a description of exactly 500 characters. 3. Repeat with a 101-character title and a 501-character description.		
Expected result	Values at the limits are accepted. Values exceeding the limits are rejected with the applicable rule identified. Rev 1.1 clarification: Failures: HTTP 400 VALIDATION_ERROR naming the field and limit (title 1-100, description 1-500 per SAD s4.3).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-MAT-03 - Supported and unsupported file types; oversized file			
Requirements	REQ-MAT-02, DC-06		
Use case	UC-MAT-01	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer; Material Service | API: POST /materials; PUT /materials/:id
Test level	System (UI + API)	**Test type**	Functional – Negative
Priority	High - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Vedh (M4)
Required role	Faculty	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 is logged in.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Upload a valid PDF. 2. Repeat with DOCX, PPTX and PNG. 3. Attempt an unsupported file type (e.g. TXT). 4. Attempt a file larger than 25 MB.		
Expected result	Supported types within the size limit are accepted. Unsupported or oversized files are rejected, the failing rule is identified and nothing is stored. Rev 1.1 clarification: Rejections: HTTP 400 VALIDATION_ERROR (unsupported type or size); the file is rejected before storage and no Material record exists.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-MAT-04 - 25 MB size boundary			
Requirements	REQ-MAT-02, DC-06		
Use case	UC-MAT-01	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer; Material Service | API: POST /materials; PUT /materials/:id
Test level	System (UI + API)	**Test type**	Functional – Boundary
Priority	High - Boundary values defined in the specification are a common defect location.	**Assigned to**	Vedh (M4)
Required role	Faculty	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 is logged in.		

Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Upload a file of exactly 25 MB. 2. Upload a file slightly above 25 MB.		
Expected result	The 25 MB file is accepted. The larger file is rejected ("File exceeds the 25 MB limit."). Rev 1.1 clarification: 25 MB = 26,214,400 bytes accepted; 26,214,401 bytes rejected (TOD-06 for the MB definition).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: TOD-06.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-MAT-05 - Uploaded material visible in owner library within 3 seconds			
Requirements	REQ-MAT-03		
Use case	UC-MAT-01	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: POST /materials; GET /materials
Test level	Performance (JMeter / timing)	**Test type**	Functional / Performance
Priority	Medium - Verifies a measurable NFR threshold; failure is an NFR breach.	**Assigned to**	Vedh (M4)
Required role	Faculty	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	F1 is logged in.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Upload a valid material of 25 MB or smaller. 2. Measure the time until it appears in the owner's library.		
Expected result	The material appears in the owner's library within 3 seconds.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		
Dependencies	Tooling/environment per s6 (JMeter, load generator, devices, backup or TLS host as applicable). All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-MAT-06 - User without access cannot view material			
Requirements	REQ-MAT-04		
Use case	UC-MAT-02	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: GET /materials; GET /materials/:id
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Vedh (M4)
Required role	Faculty, Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 owns M1; S1 has no access to M1.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Log in as S1. 2. Request M1 through the UI. 3. Directly request the material API for M1.		
Expected result	Access is denied and no material details or file are returned. Rev 1.1 clarification: HTTP 403 ACCESS_DENIED "Access denied" for both the UI route and GET /api/materials/:id.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-MAT-07 - Owner shares material with a registered user			
Requirements	REQ-MAT-05, BR-05, BR-03		
Use case	UC-MAT-03	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: POST /materials/:id/share
Test level	System (UI + API)	**Test type**	Functional – Positive
Priority	High - Core use-case flow or integrity control;	**Assigned to**	Vedh (M4)

	failure blocks the use case.		
Required role	Faculty, Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 owns M1; S1 is a registered user without access.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Log in as F1. 2. Select owned material M1. 3. Share M1 with registered user S1. 4. Log in as S1.		
Expected result	S1 receives access to M1 immediately after the successful share operation.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-MAT-08 - Authorized user downloads material; download starts within 3 seconds			
Requirements	REQ-MAT-06		
Use case	UC-MAT-04	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: GET /materials/:id/download
Test level	Performance (JMeter / timing)	**Test type**	Functional / Performance
Priority	High - Verifies a measurable NFR threshold; failure is an NFR breach.	**Assigned to**	Vedh (M4)
Required role	Student	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	S1 has access to M1 (file up to 25 MB).		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Log in as S1. 2. Download M1. 3. Measure the time until the download begins.		
Expected result	Download begins within 3 seconds for a file up to 25 MB.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		
Dependencies	Tooling/environment per s6 (JMeter, load generator, devices, backup or TLS host as applicable). All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-MAT-09 - Owner updates metadata and replaces the file			
Requirements	REQ-MAT-07		
Use case	UC-MAT-05/06 (update, remove; order not verified)	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: PUT /materials/:id
Test level	System (UI + API)	**Test type**	Functional – Positive/Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Vedh (M4)
Required role	Faculty	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 owns M1.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. As owner F1, edit the M1 title, subject and description. 2. Replace the file with a supported file. 3. Attempt to replace it with an unsupported file.		
Expected result	Valid changes are saved. The unsupported replacement is rejected and the previous valid material remains unchanged.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-MAT-10 - Owner removes material after confirmation			
Requirements	REQ-MAT-08		
Use case	UC-MAT-05/06 (update, remove; order not verified)	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: DELETE /materials/:id
Test level	System (UI + API)	**Test type**	Functional – Positive/Negative
Priority	High - Verifies rejection behaviour and error	**Assigned to**	Vedh (M4)

	messages specified for invalid use.		
Required role	Faculty, Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 owns M1; S1 has access to M1.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Select Remove on M1. 2. Cancel the confirmation. 3. Verify M1 remains available. 4. Select Remove again and confirm. 5. Request M1 as an authorized user (S1).		
Expected result	Cancellation leaves M1 unchanged. Confirmed removal prevents further access. Rev 1.1 clarification: Removal returns HTTP 204; later requests return HTTP 404 "Material unavailable".		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-MAT-11 - Owner revokes access; next request denied			
Requirements	REQ-MAT-09		
Use case	UC-MAT-07	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: DELETE /materials/:id/access/:userId
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Vedh (M4)
Required role	Faculty, Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 owns M1.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. F1 shares M1 with S1. 2. F1 revokes S1's access. 3. S1 requests M1 again.		
Expected result	The next request from S1 is denied. Rev 1.1 clarification: Revocation returns HTTP 204; the next request by S1 returns HTTP 403 "Access denied" (REQ-MAT-10) or 404 if the material no longer exists.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-MAT-12 - Non-owner share/update/remove/revoke rejected with "Access denied"			
Requirements	REQ-MAT-10, BR-03, BR-04, SEC-MAT-02, SEC-MAT-03		
Use case	UC-MAT-02 to 07	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: All /materials endpoints; POST /materials/:id/share; DELETE /materials/:id/access/:userId
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Vedh (M4)
Required role	Faculty, Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 owns M1; non-owners F2 and S1 (S1 has been granted access).		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. As a non-owner, attempt to share another user's material. 2. Attempt to update it. 3. Attempt to remove it. 4. Attempt to revoke another user's access.		
Expected result	Every operation is rejected with: "Access denied". The material remains unchanged. Rev 1.1 clarification: Every operation returns HTTP 403 ACCESS_DENIED "Access denied".		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		

Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-MAT-13 - Removed or unretrievable material reported as unavailable			
Requirements	REQ-MAT-11		
Use case	UC-MAT-02, 04, 07	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: GET /materials/:id; GET /materials/:id/download
Test level	System (UI + API)	**Test type**	Functional – Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Vedh (M4)
Required role	Public / any	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	M2 has been removed; M3 exists but its stored file is unavailable (deleted from test storage).		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Request the removed material M2. 2. Request material M3 whose stored file is unavailable.		
Expected result	The system reports "Material unavailable" and does not expose its metadata or file details. Rev 1.1 clarification: HTTP 404 "Material unavailable" with no metadata, owner or file name in the body.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-MAT-14 - Admin upload (role scope)			
Requirements	REQ-MAT-01		
Use case	UC-MAT-01	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: POST /materials
Test level	Integration (API)	**Test type**	Functional - Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Vedh (M4)
Required role	Admin (A1)	**Readiness**	Blocked - expected result not agreed
Preconditions	Admin A1 is logged in; a valid PDF fixture is available.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. POST /api/materials as A1 with valid title, subject, description and PDF. 2. Record status, body and whether a Material record was created.		
Expected result	BLOCKED (OD-07): SRS REQ-MAT-01 and UC-MAT-01 name any authenticated user (Admin included); SAD s4.3 returns HTTP 403 for roles other than Student and Faculty. The expected result must be agreed before this test can pass or fail.		
Pass / fail criteria	Cannot pass or fail until the linked decision is recorded (Blocked). When the decision is approved, update the expected result and re-baseline.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: OD-07.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-MAT-15 - Library lists only owned and shared materials; pagination			
Requirements	REQ-MAT-03, REQ-MAT-04		
Use case	UC-MAT-02	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: POST /materials; GET /materials; GET /materials/:id
Test level	Integration (API)	**Test type**	Functional - Positive
Priority	Medium - Verifies the stated functional requirement under normal conditions.	**Assigned to**	Vedh (M4)
Required role	Student (S1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 owns M5, has an access grant to M1 (owned by F1) and has no access to M6 (owned by F2).		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. GET /api/materials as S1. 2. GET /api/materials?page=1&limit=1 and then ?page=2&limit=1. 3. Compare the union of both pages with step 1.		
Expected result	Step 1: HTTP 200; items are M5 and M1 only; total = 2; M6 absent. Steps 2-3: each page holds one item, no duplicates, union equals step 1.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		

Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-MAT-16 - Upload with missing metadata or file			
Requirements	REQ-MAT-01, REQ-MAT-02, NFR-QUAL-03		
Use case	UC-MAT-01	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: POST /materials; PUT /materials/:id
Test level	Integration (API)	**Test type**	Functional - Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Vedh (M4)
Required role	Faculty (F1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 is logged in.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. POST /api/materials without a title. 2. Without a subject. 3. Without a description. 4. Without a file.		
Expected result	Each request: HTTP 400 VALIDATION_ERROR naming the missing field; no Material record and no file stored; library unchanged.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-MAT-17 - Share target validation			
Requirements	REQ-MAT-05, BR-05		
Use case	UC-MAT-03	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: POST /materials/:id/share
Test level	Integration (API)	**Test type**	Functional - Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Vedh (M4)
Required role	Faculty (F1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 owns M1.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. POST /api/materials/M1/share with no userId. 2. With userId "not-an-id". 3. With a well-formed id that belongs to no user. 4. Check the AccessGrant collection.		
Expected result	Steps 1-2: HTTP 400 VALIDATION_ERROR. Step 3: HTTP 404 (recipient is not a registered user). Step 4: no AccessGrant was created. Re-sharing an already shared material has no specified result and is not tested (TOD-09).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-MAT-18 - Owner lists users who hold access			
Requirements	REQ-MAT-09		
Use case	UC-MAT-07	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: DELETE /materials/:id/access/:userId
Test level	System (UI + API)	**Test type**	Functional - Positive
Priority	Medium - Verifies the stated functional requirement under normal conditions.	**Assigned to**	Vedh (M4)
Required role	Faculty (F1)	**Readiness**	Blocked - expected result not agreed
Preconditions	F1 owns M1 shared with S1 and S2.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Open the access-management view for M1.		

	2. Identify the users who hold access.		
Expected result	BLOCKED (OD-08): no API contract exists for listing the users who hold access to a material (UC-MAT-07). The expected list, fields and error behaviour must be specified first.		
Pass / fail criteria	Cannot pass or fail until the linked decision is recorded (Blocked). When the decision is approved, update the expected result and re-baseline.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: OD-08.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-MAT-19 - File replacement serves only the new file			
Requirements	REQ-MAT-07, NFR-SAFE-03		
Use case	UC-MAT-05/06	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer; Material Service | API: PUT /materials/:id
Test level	Integration (API)	**Test type**	Functional - Positive
Priority	High - Core use-case flow or integrity control; failure blocks the use case.	**Assigned to**	Vedh (M4)
Required role	Faculty (F1), Student (S1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 owns M7 (file A); M7 is shared with S1.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Record the SHA-256 of file A, then PUT /api/materials/M7 with file B (different content, PDF); record the SHA-256 of file B. 2. Download M7 as S1; compute the checksum. 3. Download M7 as F1; compute the checksum.		
Expected result	HTTP 200 {id, updatedAt} for step 1. Both downloads equal file B (checksum B); neither equals file A; title/subject/description unchanged unless sent.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-MAT-20 - Revocation edge cases			
Requirements	REQ-MAT-09		
Use case	UC-MAT-07	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: DELETE /materials/:id/access/:userId
Test level	Integration (API)	**Test type**	Functional - Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Vedh (M4)
Required role	Faculty (F1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 owns M1; M1 shared with S1 and S2; S3 has no grant.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. DELETE /api/materials/M1/access/S3. 2. Revoke S1; then request M1 as S2. 3. Remove M1, then DELETE /api/materials/M1/access/S2.		
Expected result	Step 1: HTTP 404 (access grant unavailable). Step 2: HTTP 204; S2 still has access (HTTP 200). Step 3: HTTP 404 (material unavailable).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-MAT-21 - Library shows owner name and access status			
Requirements	REQ-MAT-04		
Use case	UC-MAT-02	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: GET /materials; GET /materials/:id
Test level	System (UI + API)	**Test type**	Functional - Positive
Priority	Low - Verifies the stated functional requirement under normal conditions.	**Assigned to**	Vedh (M4)
Required role	Student (S1)	**Readiness**	Blocked - expected result not agreed
Preconditions	S1 owns M5 and has access to M1 (owned by F1).		

Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Open the material library. 2. Inspect owner name and access status ("Owner" / "Shared with me") for each row.		
Expected result	BLOCKED (OD-13): SRS Appendix B.6 lists owner name and access status, but the GET /materials contract (SAD s4.3) returns only {id, title, subject, owner}. The contract must be updated before expected values can be asserted.		
Pass / fail criteria	Cannot pass or fail until the linked decision is recorded (Blocked). When the decision is approved, update the expected result and re-baseline.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: OD-13.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-MAT-22 - Student can upload and own material			
Requirements	REQ-MAT-01, REQ-MAT-03, BR-02		
Use case	UC-MAT-01	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: POST /materials; GET /materials
Test level	System (UI + API)	**Test type**	Functional - Positive
Priority	Medium - Verifies the stated functional requirement under normal conditions.	**Assigned to**	Vedh (M4)
Required role	Student (S1), Faculty (F1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 and F1 are logged in (separately).		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. As S1, upload a valid DOCX with title, subject and description. 2. Open S1's library. 3. As F1 (no grant), GET /api/materials/<new id>.		
Expected result	Step 1: HTTP 201 with owner = S1. Step 2: the material appears with S1 as sole owner. Step 3: HTTP 403 ACCESS_DENIED.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

A.3 Security: Account and Material

TC-AS-01 - Lock-out counter reset, lock-out and release after 15 minutes			
Requirements	SEC-ACC-06, REQ-ACC-06, DC-08		
Use case	UC-ACC-02	**SAD reference**	Components: Auth & Session Service | API: POST /auth/login
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Jagadeesh (M1)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S2 is registered; test clock can be advanced.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Submit four invalid logins, then a valid one (counter resets). 2. Submit five consecutive invalid logins. 3. Submit a valid login during the lockout. 4. Advance the clock by 15 minutes and log in with valid credentials.		
Expected result	Step 1: login succeeds. Steps 2–3: the account is locked and the valid login is refused. Step 4: login succeeds after the 15-minute lockout period. Rev 1.1 clarification: Locked login returns HTTP 423 ACCOUNT_LOCKED.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Server test clock that can be advanced (s6.5). All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-MS-01 - Direct material access without authorization denied			
Requirements	SEC-MAT-01, REQ-MAT-04		
Use case	UC-MAT-02	**SAD reference**	Components: Material Service, AccessGrant, File

			Storage; API Layer | API: GET /materials/:id; GET /materials/:id/download; GET /materials
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Vedh (M4)
Required role	Faculty, Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 owns M1; S3 has no access grant.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. As S3, view M1 in the UI and through the API. 2. Request the M1 download URL directly without authorization.		
Expected result	Access is denied ("Access denied") and the file is not returned.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-AS-02 - Logged-out and idle tokens cannot be reused			
Requirements	SEC-ACC-04, REQ-ACC-07, DC-08		
Use case	UC-ACC-03 to 05 (UC ID not individually verifiable without SRS)	**SAD reference**	Components: API Layer; Auth & Session Service; API Layer (authentication middleware) | API: POST /auth/logout; all protected endpoints
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Jagadeesh (M1)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 has an active session.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Log out, then reuse the old token on a protected endpoint. 2. Log in again, stay inactive for 30 minutes, then call a protected endpoint with the same token.		
Expected result	Both requests are rejected (401); the session record is invalid and the token cannot be reused. Rev 1.1 clarification: HTTP 401 NOT_AUTHENTICATED in both cases.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Server test clock that can be advanced (s6.5). All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-MS-02 - Revoked user denied on the very next request			
Requirements	SEC-MAT-04, REQ-MAT-09		
Use case	UC-MAT-07	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: GET /materials/:id; DELETE /materials/:id/access/:userId
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Vedh (M4)
Required role	Faculty, Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 owns M1 shared with S1.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. S1 opens M1 successfully. 2. F1 revokes S1's access. 3. S1 requests M1 view and download again.		
Expected result	The revoked user is denied on the very next request.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-AS-03 - Unauthenticated and malformed-token requests rejected			
Requirements	SEC-ACC-01, REQ-ACC-10, BR-01		
Use case	UC-ACC-02	**SAD reference**	Components: API Layer; Auth & Session Service; API Layer (authentication middleware) | API: All protected endpoints

Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Jagadeesh (M1)
Required role	Unauthenticated	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	No authenticated session.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Call protected account endpoints without a token. 2. Call protected material endpoints without a token. 3. Repeat with a malformed token.		
Expected result	Every request is rejected (401) and no account or material data is returned. Rev 1.1 clarification: HTTP 401 NOT_AUTHENTICATED; body {error:{code,message}} with no stack trace.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-MS-03 - Removed material not retrievable by any user or URL			
Requirements	SEC-MAT-04, REQ-MAT-11		
Use case	UC-MAT-02, 04, 07	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: GET /materials/:id; DELETE /materials/:id/access/:userId; GET /materials/:id/download
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Vedh (M4)
Required role	Faculty, Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 owns M1 shared with S1.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. F1 removes M1 after confirmation. 2. S1 and F1 request M1 again. 3. Request the previous download URL.		
Expected result	Previous users cannot retrieve M1; no details or file are exposed.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-AS-04 - Passwords stored only as salted bcrypt hashes			
Requirements	SEC-ACC-03, DC-04		
Use case	-	**SAD reference**	Components: Auth & Session Service | API: POST /auth/register
Test level	Security (API / system)	**Test type**	Security – Inspection
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Jagadeesh (M1)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Accounts S1 and S3 are registered with the same password.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Inspect the users collection in MongoDB. 2. Compare the stored password values of S1 and S3. 3. Inspect API responses and server logs for registration and login.		
Expected result	Passwords are stored only as salted bcrypt hashes (identical passwords give different hashes). No plaintext password or hash appears in any response or log.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-MS-04 - Non-owner cannot share or revoke	
Requirements	SEC-MAT-02, REQ-MAT-10, BR-03, BR-04

Use case	UC-MAT-02 to 07	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: POST /materials/:id/share; DELETE /materials/:id/access/:userId; All /materials endpoints
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Vedh (M4)
Required role	Faculty, Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 owns M1; S1 has access to M1; F2 is a non-owner.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. As S1, attempt to share M1 with S2. 2. As F2, attempt to revoke S1's access to M1.		
Expected result	Both are rejected with "Access denied"; no grant is created or removed.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-AS-05 - Cross-user profile update and e-mail/role change rejected			
Requirements	SEC-ACC-05		
Use case	-	**SAD reference**	Components: Account Service | API: PUT /account/profile
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Jagadeesh (M1)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 and S2 are logged in.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. As S2, call the profile update endpoint with S1's user id. 2. As S2, submit a profile update containing a changed email or role.		
Expected result	Step 1 is rejected (403) and S1's profile is unchanged. Step 2: email and role are not modified. Rev 1.1 clarification: SAD s4.3 defines PUT /account/profile for the caller's own profile only (no user-id parameter). Expected: an injected user id (body, query or path variant) never changes S1's profile; the response is HTTP 403 ACCESS_DENIED, or the id is ignored and only S2's own profile is affected. Record which; either satisfies SEC-ACC-05 provided S1 is unchanged.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-MS-05 - Non-owner cannot update or remove			
Requirements	SEC-MAT-03, REQ-MAT-07, REQ-MAT-08, REQ-MAT-10, BR-03, BR-04		
Use case	UC-MAT-05/06 (update, remove; order not verified)	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: PUT, DELETE /materials/:id; PUT /materials/:id; DELETE /materials/:id
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Vedh (M4)
Required role	Faculty, Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 owns M1; S1 has access to M1; F2 is a non-owner.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. As S1, send an update (metadata and file replacement) for M1. 2. As F2, send a remove request for M1.		
Expected result	Both are rejected with "Access denied"; M1 is unchanged.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-AS-06 - TLS 1.2+ enforced; plain HTTP and TLS 1.0/1.1 refused			
Requirements	SEC-COM-01, DC-03, COM-01		
Use case	-	**SAD reference**	Components: Nginx reverse proxy (TLS); Nginx reverse proxy; Nginx | API: All endpoints
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Jagadeesh (M1)
Required role	Public / any	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	Test server deployed with TLS.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Request the application and API over plain HTTP. 2. Attempt TLS 1.0 and TLS 1.1 handshakes (e.g. openssl s_client). 3. Attempt TLS 1.2 and TLS 1.3 handshakes.		
Expected result	Unencrypted requests are rejected or not served with protected data. TLS 1.0/1.1 are refused; TLS 1.2 or higher is accepted.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		
Dependencies	Tooling/environment per s6 (JMeter, load generator, devices, backup or TLS host as applicable). All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: TOD-10.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-AS-07 - NoSQL-operator injection on authentication endpoints			
Requirements	SEC-ACC-01, SEC-ACC-02, NFR-QUAL-03, DC-07		
Use case	UC-ACC-01, UC-ACC-02	**SAD reference**	Components: API Layer; Auth & Session Service; API Layer middleware | API: All protected endpoints; POST /auth/login
Test level	Security (API)	**Test type**	Security - Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Jagadeesh (M1)
Required role	Public	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 is registered; the API is running.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. POST /api/auth/login with {"email":{"$ne":""},"password":{"$ne":""}}. 2. POST with {"email":"s1@example.com","password":{"$gt":""}}. 3. POST /api/auth/register with email {"$ne":null}. 4. Send the query-string variant email[$ne]=x on the login endpoint.		
Expected result	Every request is rejected with HTTP 400 or 401; no HTTP 200, no token. No account is created; no stack trace or query text in the body; the server stays available.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-AS-08 - Security events logged without sensitive data			
Requirements	SEC-ACC-01, SEC-ACC-06, SEC-MAT-01		
Use case	-	**SAD reference**	Components: API Layer; Auth & Session Service; Material Service, AccessGrant, File Storage | API: All protected endpoints; POST /auth/login; GET /materials/:id
Test level	Security (API)	**Test type**	Security - Inspection
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Jagadeesh (M1)
Required role	All roles	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Server log location is known; logging enabled; test passwords and tokens are recorded privately for the search in step 3.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Perform: successful and failed log-in, five-failure lock-out, logout, idle-session rejection, one 403 denial (S3 views M1), material share/revoke/remove, an unsupported-file upload, quiz publication, attempt start and submission, and an Admin moderation action. 2. Read the server log for the period. 3. Search the log for the test passwords, token strings, and answer-key text (correct options, explanations).		
Expected result	Each event in step 1 has one entry with timestamp, event type, user id, resource id and outcome (SAD s4.4).		

	Step 3: no password, session token or answer key appears in any entry. No SRS requirement ID has been verified for audit logging; owner is unassigned (OD-06). Result is reported against the SAD s3.9 / s4.4 control.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Access to server logs in the test environment.. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: OD-06.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-AS-09 - Rate limiting returns HTTP 429			
Requirements	NFR-PERF-02, NFR-QUAL-03		
Use case	-	**SAD reference**	Components: API Layer (rate limiting); Nginx; MongoDB | API: n/a
Test level	Security (API)	**Test type**	Security - Negative
Priority	Medium - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Jagadeesh (M1)
Required role	Public / Student	**Readiness**	Blocked - expected result not agreed
Preconditions	Rate-limit configuration of the test environment is recorded.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Send requests to POST /api/auth/login from one client until the configured limit is exceeded. 2. Send a normal request from a second client. 3. Wait for the limit window to end and repeat step 1's first request.		
Expected result	BLOCKED (TOD-02): the SAD names rate limiting and the code RATE_LIMITED (HTTP 429) but no threshold or window is specified. Once agreed: step 1 ends with HTTP 429 {error:{code:"RATE_LIMITED"}}; step 2 succeeds; step 3 succeeds.		
Pass / fail criteria	Cannot pass or fail until the linked decision is recorded (Blocked). When the decision is approved, update the expected result and re-baseline.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: TOD-02.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

A.4 Performance: Account and Material

TC-AP-01 - 200 concurrent users: 95% of page loads within 3 s			
Requirements	NFR-PERF-01		
Use case	-	**SAD reference**	Components: Web Client; API Layer; Nginx | API: n/a
Test level	Performance (JMeter / timing)	**Test type**	Performance (NFR)
Priority	High - Verifies a measurable NFR threshold; failure is an NFR breach.	**Assigned to**	Jagadeesh (M1)
Required role	Student	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	200 generated Student accounts; JMeter scenario.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Generate 200 concurrent users. 2. Execute representative page/API requests (login, library, profile, material view). 3. Record response times.		
Expected result	At least 95% of page loads complete within 3 seconds.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		
Dependencies	Tooling/environment per s6 (JMeter, load generator, devices, backup or TLS host as applicable). All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: TOD-10.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-MP-01 - Upload of 25 MB file visible within 3 s (10 runs)			
Requirements	REQ-MAT-03		
Use case	UC-MAT-01	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: POST /materials; GET /materials
Test level	Performance (JMeter / timing)	**Test type**	Performance (NFR)
Priority	Medium - Verifies a measurable NFR threshold; failure is an NFR breach.	**Assigned to**	Vedh (M4)
Required role	Faculty	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	F1 is logged in; 25 MB PDF available.		

Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Upload the 25 MB file (10 runs). 2. Measure the time from upload completion to the material appearing in the library.		
Expected result	The material is visible in the owner's library within 3 seconds in every run.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		
Dependencies	Tooling/environment per s6 (JMeter, load generator, devices, backup or TLS host as applicable). All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-AP-02 - 200 concurrent users: fewer than 1% failed requests			
Requirements	NFR-PERF-02		
Use case	-	**SAD reference**	Components: API Layer (rate limiting); Nginx; MongoDB | API: n/a
Test level	Performance (JMeter / timing)	**Test type**	Performance (NFR)
Priority	High - Verifies a measurable NFR threshold; failure is an NFR breach.	**Assigned to**	Jagadeesh (M1)
Required role	Public / any	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	Same load setup as TC-AP-01.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Execute the defined workload using 200 concurrent users. 2. Record successful and failed requests.		
Expected result	Fewer than 1% of requests fail.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		
Dependencies	Tooling/environment per s6 (JMeter, load generator, devices, backup or TLS host as applicable). All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: TOD-10.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-MP-02 - Download of 25 MB file starts within 3 s (10 runs)			
Requirements	REQ-MAT-06		
Use case	UC-MAT-04	**SAD reference**	Components: Material Service, AccessGrant, File Storage; API Layer | API: GET /materials/:id/download
Test level	Performance (JMeter / timing)	**Test type**	Performance (NFR)
Priority	Medium - Verifies a measurable NFR threshold; failure is an NFR breach.	**Assigned to**	Vedh (M4)
Required role	Student	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	S1 has access to a 25 MB material.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. S1 starts the download (10 runs). 2. Measure the time until the download begins.		
Expected result	The download begins within 3 seconds in every run.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		
Dependencies	Tooling/environment per s6 (JMeter, load generator, devices, backup or TLS host as applicable). All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-MP-03 - Share and revocation effective within 2 s			
Requirements	NFR-PERF-03		
Use case	-	**SAD reference**	Components: Material Service; AccessGrant | API: n/a
Test level	Performance (JMeter / timing)	**Test type**	Performance (NFR)
Priority	High - Verifies a measurable NFR threshold; failure is an NFR breach.	**Assigned to**	Vedh (M4)
Required role	Faculty, Student	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	F1 owns M1.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. F1 shares material M1 with S1. 2. Record the time from the owner's confirmation to successful access for S1.		

	3. Revoke S1's access. 4. Record the time until the revoked request is rejected.		
Expected result	Material share and access revocation take effect within 2 seconds.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		
Dependencies	Tooling/environment per s6 (JMeter, load generator, devices, backup or TLS host as applicable). All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-AP-03 - Mixed workload with 200 users including quiz submission			
Requirements	NFR-PERF-01, NFR-PERF-02, NFR-PERF-04		
Use case	-	**SAD reference**	Components: Web Client; API Layer; Nginx | API: n/a
Test level	Performance (JMeter)	**Test type**	Performance (NFR)
Priority	Medium - Verifies a measurable NFR threshold; failure is an NFR breach.	**Assigned to**	Jagadeesh (M1)
Required role	Generated Students/Faculty	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	JMeter plan with 200 concurrent virtual users; quiz Z1 with 50 questions; load generator on a separate machine (TOD-10).		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Ramp up to 200 users running login, library, profile, quiz list, attempt start, answer save and submit. 2. Record page-load times, failed requests and grading latency for the whole run.		
Expected result	At least 95% of page loads within 3 s; fewer than 1% failed requests; every grading result within 2 s. Report results as measured with the environment specification; do not mark Pass without the JMeter results.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		
Dependencies	JMeter; separate load-generator host; seed scripts.. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: TOD-10.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

A.5 Safety, Quality and Team-level

TC-AN-01 - Daily backup and restore (backup no older than 24 h)			
Requirements	NFR-SAFE-01		
Use case	-	**SAD reference**	Components: MongoDB; File Storage; Docker deployment | API: n/a
Test level	System (environment)	**Test type**	Non-functional – Safety (backup/restore)
Priority	High - Core use-case flow or integrity control; failure blocks the use case.	**Assigned to**	Jagadeesh (M1)
Required role	Faculty, Student	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	F1 and S1 accounts exist; daily backup job configured on the test server; M1 (PDF) uploaded by F1.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Upload material M1 as F1. 2. Run the daily backup and note its timestamp. 3. Delete M1 (and its stored file) from the system. 4. Restore from that backup. 5. Check M1, its file and the F1/S1 account records.		
Expected result	M1, its file and the user accounts are back after the restore. The backup used is no more than 24 hours old.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		
Dependencies	Tooling/environment per s6 (JMeter, load generator, devices, backup or TLS host as applicable). All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: OD-06.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-AN-02 - Interrupted upload/update/removal leaves previous state			
Requirements	NFR-SAFE-02		
Use case	-	**SAD reference**	Components: Material Service; File Storage | API: n/a
Test level	System (environment)	**Test type**	Non-functional – Safety (Negative)
Priority	High - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Vedh (M4)

Required role	Faculty	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	F1 owns M1 (known title and file); network can be interrupted (e.g. browser offline mode, proxy drop).		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Start uploading a new material and interrupt it midway. 2. Start an update of M1 (metadata and file replacement) and interrupt it midway. 3. Start the removal of M1 and interrupt it before completion. 4. Reload the library and inspect the database and file storage.		
Expected result	No success message is shown for any interrupted operation. No partial record or partial file exists. The previous state of M1 is unchanged.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		
Dependencies	Tooling/environment per s6 (JMeter, load generator, devices, backup or TLS host as applicable). All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-AN-03 - Downloaded file identical to uploaded file (SHA-256)			
Requirements	NFR-SAFE-03		
Use case	-	**SAD reference**	Components: Material Service; File Storage | API: n/a
Test level	System (environment)	**Test type**	Non-functional – Data integrity
Priority	High - Core use-case flow or integrity control; failure blocks the use case.	**Assigned to**	Vedh (M4)
Required role	Faculty, Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 and S1 exist; test files of each supported type available.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Compute the SHA-256 checksum of a test file (PDF, then DOCX, PPTX and PNG). 2. Upload it as F1 and share it with S1. 3. Download it as S1. 4. Compute the SHA-256 checksum of the downloaded file and compare.		
Expected result	The checksums are identical for every file type.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-AN-04 - First-time users complete register, log in, upload in under 5 minutes			
Requirements	NFR-QUAL-01		
Use case	-	**SAD reference**	Components: Web Client | API: n/a
Test level	Usability (manual)	**Test type**	Non-functional – Usability
Priority	Medium - Verifies the stated quality or inspection requirement.	**Assigned to**	Jagadeesh (M1)
Required role	Public / any	**Readiness**	Manual / inspection
Preconditions	Three first-time users (no prior use of the platform); no instructions or training given; a test PDF is available.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Each user registers a new account. 2. Each user logs in. 3. Each user uploads a material. 4. Record the total time per user.		
Expected result	Each user completes registration, login and upload in under 5 minutes without instructions.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-AN-05 - Invalid input and unsupported files rejected without crash			
Requirements	NFR-QUAL-03		
Use case	-	**SAD reference**	Components: API Layer | API: n/a
Test level	System (environment)	**Test type**	Non-functional – Robustness (Negative)
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Jagadeesh (M1)
Required role	Faculty, Student	**Readiness**	Ready - executable once a build is deployed (build

			status unverified)
Preconditions	F1 is uploading; S1 has an active session on another browser.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Upload a 0-byte file. 2. Upload an .exe file renamed to .pdf. 3. Send malformed JSON to a registration/login and a material endpoint (Postman). 4. After each step, check the server and let S1 continue using the platform.		
Expected result	All three inputs are rejected with an error message. The server stays up (no crash or restart). S1's session and requests are unaffected.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-AN-06 - Browser and device compatibility			
Requirements	NFR-QUAL-04, DC-05		
Use case	-	**SAD reference**	Components: Web Client | API: n/a
Test level	Compatibility (manual)	**Test type**	Non-functional – Portability
Priority	Medium - Verifies the stated quality or inspection requirement.	**Assigned to**	Jagadeesh (M1)
Required role	Faculty, Student	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	Current versions of Chrome, Firefox, Edge and Safari on desktop and on mobile devices; F1 and S1 exist.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. On each browser/device combination, register a new user. 2. Log in. 3. Upload a material. 4. Download the material.		
Expected result	Registration, login, upload and download work on all listed browsers on desktop and mobile.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		
Dependencies	Tooling/environment per s6 (JMeter, load generator, devices, backup or TLS host as applicable). All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: TOD-10.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-AN-07 - Availability measured over the test period			
Requirements	NFR-QUAL-02		
Use case	-	**SAD reference**	Components: Nginx; Docker deployment; monitoring | API: n/a
Test level	System (environment)	**Test type**	Non-functional – Availability
Priority	Medium - Verifies the stated quality or inspection requirement.	**Assigned to**	Pushkar (M3)
Required role	Public / any	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	Platform is deployed on the test server.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Monitor the platform during the defined test period. 2. Record service downtime. 3. Exclude announced maintenance periods. 4. Calculate system availability.		
Expected result	Platform availability is at least 99% per month excluding announced maintenance. Rev 1.1 clarification: Limitation (TOD-07): a monthly 99% figure cannot be demonstrated inside the sprint window. The test records measured availability over the observed period and reports it as measured, not as proof of the monthly target.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		
Dependencies	Tooling/environment per s6 (JMeter, load generator, devices, backup or TLS host as applicable). All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: TOD-07, OD-06.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-AN-08 - Feature areas independently testable	
Requirements	NFR-QUAL-05

Use case	-	**SAD reference**	Components: Layered modular monolith | API: n/a
Test level	Review / inspection	**Test type**	Non-functional – Modularity
Priority	Low - Verifies the stated quality or inspection requirement.	**Assigned to**	Pushkar (M3)
Required role	Public / any	**Readiness**	Manual / inspection
Preconditions	All four major feature areas are deployed.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Identify Account Management. 2. Identify Material Sharing. 3. Identify Question Bank & Quizzes. 4. Identify Performance & Collaboration. 5. Test the modules independently through their service/API boundaries.		
Expected result	The major feature areas are separately identifiable and independently testable without requiring unrelated module internals. Rev 1.1 clarification: "Four major feature areas" means the four feature areas of the SAD (Account Management, Material Sharing, Question Bank & Quizzes, Performance & Collaboration), not team members.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-AN-09 - Every requirement traceable to a test case			
Requirements	NFR-QUAL-06, DC-10		
Use case	-	**SAD reference**	Components: SAD s3.8; SRS RTM; this STP | API: n/a
Test level	Review / inspection	**Test type**	Non-functional – Traceability
Priority	High - Core use-case flow or integrity control; failure blocks the use case.	**Assigned to**	Pushkar (M3)
Required role	Public / any	**Readiness**	Manual / inspection
Preconditions	Final SRS, SAD, Test Plan and RTM are available.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Review all SRS requirements. 2. Verify each has a unique ID. 3. Verify each requirement appears in the RTM. 4. Verify each requirement maps to at least one test case. 5. Verify Member 1, Member 2, Member 3 and Member 4 requirements are included.		
Expected result	Every requirement has a unique ID and is traceable through the RTM to at least one test case.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-AN-10 - Technology stack conformance			
Requirements	DC-01, DC-04		
Use case	-	**SAD reference**	Components: Whole architecture; Auth & Session Service | API: n/a
Test level	Review / inspection	**Test type**	Non-functional - Inspection
Priority	Low - Verifies the stated quality or inspection requirement.	**Assigned to**	Jagadeesh (M1)
Required role	Team	**Readiness**	Manual / inspection
Preconditions	Repository and deployment files available.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Inspect package manifests, Docker and Nginx configuration. 2. Confirm MongoDB, Express.js, React and Node.js are used and that bcrypt is the password-hashing library. 3. Confirm no other database or server framework is used.		
Expected result	The stack matches DC-01 and DC-04; deviations are listed in the deviation log (Deliverables Part-2, item 2i).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Source repository access.. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-AN-11 - API contract conformance			
Requirements	DC-02, NFR-QUAL-03		
Use case	-	**SAD reference**	Components: API Layer | API: n/a
Test level	Integration (API)	**Test type**	Non-functional - Contract
Priority	Medium - Verifies the stated quality or inspection requirement.	**Assigned to**	Alifiya (M2)
Required role	All roles	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	All endpoints of SAD s4.3 deployed (35 endpoints, base path /api).		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. For each of the 35 endpoints: confirm method, path, required authentication and role. 2. Send one valid and one invalid request; compare status codes with the "Errors" line of SAD s4.3. 3. Verify every error body has the form {error:{code,message}}.		
Expected result	All endpoints conform; every difference is logged as a defect or deviation. Operations that SAD s4.3 does not define (OD-08) are listed as gaps and not counted as failures.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Postman/Supertest collection generated from SAD s4.3.. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-AN-12 - Server-side enforcement independent of the client			
Requirements	DC-07, DC-08, NFR-QUAL-03		
Use case	-	**SAD reference**	Components: API Layer middleware; all services; Auth & Session Service | API: n/a
Test level	Security (API)	**Test type**	Security - Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Jagadeesh (M1)
Required role	Student, Faculty	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Test accounts available.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. For each rule class - required fields, length limits, file type/size, role, ownership, quiz window - bypass the UI and send the violating request directly. 2. Edit client-side stored data (for example a role value in browser storage) and repeat a privileged action.		
Expected result	The server rejects every violating request with the specified status; no privilege is gained by changing client data.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-AN-13 - Central error handling hides internals			
Requirements	NFR-QUAL-03		
Use case	-	**SAD reference**	Components: API Layer | API: n/a
Test level	Security (API)	**Test type**	Non-functional - Robustness
Priority	Medium - Verifies the stated quality or inspection requirement.	**Assigned to**	Jagadeesh (M1)
Required role	Any	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	Fault injection possible in the test environment (stop MongoDB; send malformed JSON; oversized body).		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Send malformed JSON to a login and a material endpoint. 2. Send a body larger than the configured limit. 3. Stop MongoDB and call a read endpoint; restart MongoDB.		
Expected result	Client errors: HTTP 400/413 with {error:{code,message}}. Server fault: HTTP 500 INTERNAL_ERROR with a generic message; no stack trace, file path or query text; detail only in the server log; service resumes after MongoDB restarts.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		
Dependencies	Tooling/environment per s6 (JMeter, load generator, devices, backup or TLS host as applicable). All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-AN-14 - Domain requirement: English user messages			
Requirements	DOM-NOID		
Use case	-	**SAD reference**	Components: Web Client; API Layer (messages) | API: n/a
Test level	Review / inspection	**Test type**	Non-functional - Inspection
Priority	Low - Verifies the stated quality or inspection requirement.	**Assigned to**	Pushkar (M3)
Required role	All roles	**Readiness**	Manual / inspection
Preconditions	All screens reachable.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Walk all screens and trigger each error/confirmation message listed in SAD s4.5. 2. Check that messages are in English.		
Expected result	All user-facing messages are in English and match the SAD s4.5 wording.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-AN-15 - UI inventory and role-based navigation			
Requirements	UI-01..18		
Use case	-	**SAD reference**	Components: Web Client | API: n/a
Test level	System (UI)	**Test type**	Functional - Inspection
Priority	Medium - Verifies the stated functional requirement under normal conditions.	**Assigned to**	Pushkar (M3; provisional - UI items not allocated in SRS App. D)
Required role	Student, Faculty, Admin	**Readiness**	Manual / inspection
Preconditions	One account per role.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. For each role, record the navigation entries. 2. Open every screen reachable by that role and record its identifier (UI-nn). 3. Request, by URL, a screen not permitted for the role.		
Expected result	Only permitted functions are shown for each role; non-permitted URLs are refused (HTTP 403/redirect); every UI-01 to UI-18 screen is reached by at least one role. Item-by-item content (SRS 3.1.1) is not asserted by this test; item-level UI traceability is open (TOD-12).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-AN-16 - Active attempt survives a server restart			
Requirements	NFR-QUAL-03, SEC-QZ-04		
Use case	-	**SAD reference**	Components: API Layer; Quiz Engine (execution) | API: POST /quizzes/:id/attempts; PUT /attempts/:id/answers
Test level	System (environment)	**Test type**	Non-functional - Recovery
Priority	Medium - Verifies the stated quality or inspection requirement.	**Assigned to**	Jagadeesh (M1)
Required role	Student (S1)	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	S1 has an in-progress attempt with 2 saved answers.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Restart the API container. 2. During the restart, call the answer endpoint. 3. After restart, continue the attempt and submit.		
Expected result	Step 2 fails cleanly with no partial write; after restart the 2 saved answers and the original deadline are intact; submission is graded correctly.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		
Dependencies	Docker deployment of the test server.. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-AN-17 - Concurrent update and removal of a material	
Requirements	NFR-SAFE-02, REQ-MAT-07, REQ-MAT-08

Use case	UC-MAT-05/06 (update, remove; order not verified)	**SAD reference**	Components: Material Service; File Storage; Material Service, AccessGrant, File Storage | API: PUT /materials/:id; DELETE /materials/:id
Test level	Integration (API)	**Test type**	Non-functional - Concurrency
Priority	Medium - Race conditions can break one-attempt/one-membership/immutability rules.	**Assigned to**	Vedh (M4)
Required role	Faculty (F1)	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	F1 owns M1 (file A).		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Send PUT /api/materials/M1 (replace with file B) and DELETE /api/materials/M1 simultaneously (repeat 20 times with fresh materials). 2. Inspect the database and file storage after each run.		
Expected result	In every run the final state is consistent: either M1 is removed (404, no file) or updated (file B); no orphan file, no partial record, no success reported for a lost operation.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		
Dependencies	Tooling/environment per s6 (JMeter, load generator, devices, backup or TLS host as applicable). All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

A.6 Question Bank

TC-QB-01 - Add a valid question			
Requirements	REQ-QB-01, REQ-QB-03		
Use case	UC-QB-01 to 04 (pairing not verified)	**SAD reference**	Components: Question Bank Service; API Layer | API: POST /questions
Test level	System (UI + API)	**Test type**	Functional – Positive
Priority	High - Core use-case flow or integrity control; failure blocks the use case.	**Assigned to**	Alifiya (M2)
Required role	Faculty	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Faculty F1 is logged in.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Open Question Bank → Add Question. 2. Enter text "What is 2 + 2?", options 3 / 4 / 5 / 6, correct option B, topic "Arithmetic", difficulty Easy, explanation "2 + 2 = 4". 3. Submit.		
Expected result	Question is saved and a confirmation is shown. Question appears in the bank with owner F1. Rev 1.1 clarification: Response HTTP 201 {id, owner, createdAt}.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-QB-02 - Question validation: empty text, topic, correct option			
Requirements	REQ-QB-02		
Use case	UC-QB-01 to 04 (pairing not verified)	**SAD reference**	Components: Question Bank Service; API Layer | API: POST, PUT /questions
Test level	System (UI + API)	**Test type**	Functional – Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Alifiya (M2)
Required role	Faculty	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Faculty F1 is logged in.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Open Add Question and leave the question text empty; fill all other fields. 2. Submit. 3. Repeat leaving the topic empty, then with no correct option selected.		
Expected result	Each submission is rejected and the message names the failing field ("Question text", "Topic", "Correct option"). No question is saved. Rev 1.1 clarification: HTTP 400 VALIDATION_ERROR naming the failing field.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any		

	difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-QB-03 - Question field-length boundaries			
Requirements	REQ-QB-01, REQ-QB-02		
Use case	UC-QB-01 to 04 (pairing not verified)	**SAD reference**	Components: Question Bank Service; API Layer | API: POST /questions; POST, PUT /questions
Test level	System (UI + API)	**Test type**	Functional – Boundary
Priority	Medium - Boundary values defined in the specification are a common defect location.	**Assigned to**	Alifiya (M2)
Required role	Faculty	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Faculty F1 is logged in.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Add a question whose text is exactly 500 characters, options of exactly 200 characters, topic of 50 characters, explanation of 300 characters. 2. Add a question whose text is 501 characters. 3. Repeat with an option of 201 characters.		
Expected result	The question at the limits is accepted. The 501-character text and the 201-character option are rejected with the field and the length rule named.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-QB-04 - Four distinct options required			
Requirements	REQ-QB-02		
Use case	UC-QB-01 to 04 (pairing not verified)	**SAD reference**	Components: Question Bank Service; API Layer | API: POST, PUT /questions
Test level	System (UI + API)	**Test type**	Functional – Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Alifiya (M2)
Required role	Faculty	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Faculty F1 is logged in.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Add a question with options A = "4" and B = "4" (identical). 2. Add a question with only three options filled.		
Expected result	Both submissions are rejected; the message names the options field and the rule (four distinct options required).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-QB-05 - Search and filter question bank			
Requirements	REQ-QB-04		
Use case	UC-QB-01 to 04 (pairing not verified)	**SAD reference**	Components: Question Bank Service; API Layer | API: GET /questions
Test level	System (UI + API)	**Test type**	Functional – Positive
Priority	Medium - Verifies the stated functional requirement under normal conditions.	**Assigned to**	Alifiya (M2)
Required role	Faculty	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Bank holds 20 questions with mixed topics and difficulties.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Open the Question Bank as Faculty F1. 2. Search keyword "photosynthesis". 3. Filter topic = Biology and difficulty = Medium. 4. Search keyword "zzzz".		
Expected result	Only matching questions are listed with text, topic, difficulty and owner. Step 4 shows "No questions found".		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		

Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-QB-06 - Edit own question (valid and invalid)			
Requirements	REQ-QB-05, REQ-QB-02		
Use case	UC-QB-01 to 04 (pairing not verified)	**SAD reference**	Components: Question Bank Service; API Layer | API: PUT /questions/:id; POST, PUT /questions
Test level	System (UI + API)	**Test type**	Functional – Positive / Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Alifiya (M2)
Required role	Faculty	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 owns question Q1.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Edit Q1: change the text and the correct option; save. 2. Edit Q1 again: clear the topic; save.		
Expected result	Step 1: changes saved and confirmed. Step 2: rejected, "Topic" named; stored question keeps the previous valid values.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-QB-07 - Remove question; published quiz keeps its copy			
Requirements	REQ-QB-06, REQ-QZ-04		
Use case	UC-QB-01 to 04 (pairing not verified)	**SAD reference**	Components: Question Bank Service; API Layer; Quiz Engine (authoring) | API: DELETE /questions/:id; POST /quizzes/:id/publish
Test level	System (UI + API)	**Test type**	Functional – Positive
Priority	High - Core use-case flow or integrity control; failure blocks the use case.	**Assigned to**	Alifiya (M2)
Required role	Faculty, Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 owns question Q2; Q2 is part of a published quiz Z1.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Select Remove on Q2 and cancel the prompt. 2. Select Remove on Q2 again and confirm. 3. Open quiz Z1 as a Student and start it.		
Expected result	Step 1: Q2 is kept. Step 2: Q2 disappears from the bank. Step 3: Z1 still shows Q2 unchanged (copy stored at publication).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-QB-08 - Student denied question-bank access			
Requirements	REQ-QB-07, SEC-QZ-01		
Use case	UC-QB-01 to 04 (pairing not verified)	**SAD reference**	Components: Question Bank Service; API Layer; Quiz Engine (authoring) | API: All /questions endpoints; /questions; /quizzes
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Alifiya (M2)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Student S1 is logged in.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Try to open the Question Bank page by URL. 2. Call GET /api/questions and POST /api/questions with S1's token (Postman).		
Expected result	Both are refused with "Access denied" (HTTP 403); no question data is returned; nothing is created. Rev 1.1 clarification: Role failure is HTTP 403 ACCESS_DENIED; add: the same calls without a token return HTTP 401 (TC-QB-12).		

Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-QB-09 - Non-owner Faculty cannot edit or remove question			
Requirements	REQ-QB-07, SEC-QZ-01		
Use case	UC-QB-01 to 04 (pairing not verified)	**SAD reference**	Components: Question Bank Service; API Layer; Quiz Engine (authoring) | API: All /questions endpoints; /questions; /quizzes
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Alifiya (M2)
Required role	Faculty	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Faculty F1 owns Q1; Faculty F2 is logged in.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. As F2 call PUT /api/questions/Q1 and DELETE /api/questions/Q1. 2. In the UI, check whether Edit/Remove are shown for Q1.		
Expected result	Both API calls return "Access denied" (403); Q1 is unchanged. Edit/Remove controls are not shown to F2 for Q1; F2 can still view Q1.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-QB-10 - Question validation: difficulty, topic, explanation, option count, correct option			
Requirements	REQ-QB-02		
Use case	UC-QB-01	**SAD reference**	Components: Question Bank Service; API Layer | API: POST, PUT /questions
Test level	Integration (API)	**Test type**	Functional - Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Alifiya (M2)
Required role	Faculty (F1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 is logged in.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. POST /api/questions with difficulty "Extreme", then with difficulty empty. 2. With a topic of 51 characters. 3. With an explanation of 301 characters. 4. With five options. 5. With a correctOption that is not one of the four options.		
Expected result	Each request: HTTP 400 VALIDATION_ERROR naming the failing field and rule; no question stored.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-QB-11 - Search and filter combinations and paging			
Requirements	REQ-QB-04		
Use case	UC-QB-02 (not verified)	**SAD reference**	Components: Question Bank Service; API Layer | API: GET /questions
Test level	Integration (API)	**Test type**	Functional - Positive
Priority	Medium - Verifies the stated functional requirement under normal conditions.	**Assigned to**	Alifiya (M2)
Required role	Faculty (F1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Bank of 20 questions in three topics and mixed difficulty (fixture FX-QB20).		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. GET /api/questions?topic=Biology. 2. GET /api/questions?difficulty=Hard. 3. GET /api/questions?q=<keyword>&topic=<t>&difficulty=<d>. 4. GET /api/questions?page=1&limit=5 and ?page=2&limit=5.		

Expected result	Each result contains only questions matching every applied criterion; total matches the fixture. Pages do not overlap; union of all pages equals the unpaged list.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-QB-12 - Unauthenticated access to question endpoints			
Requirements	REQ-QB-07, REQ-ACC-10, SEC-QZ-01		
Use case	UC-QB-01 to 04	**SAD reference**	Components: Question Bank Service; API Layer; Auth & Session Service | API: All /questions endpoints; All protected endpoints; /questions
Test level	Security (API)	**Test type**	Security - Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Alifiya (M2)
Required role	Unauthenticated	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Question Q1 exists.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Call GET /api/questions and POST /api/questions without a token. 2. Call PUT and DELETE /api/questions/Q1 without a token. 3. Repeat step 1 with a token invalidated by logout.		
Expected result	HTTP 401 NOT_AUTHENTICATED in every case; no data returned; Q1 unchanged.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

A.7 Quizzes: authoring, execution, grading

TC-QZ-01 - Create quiz as Draft			
Requirements	REQ-QZ-01, REQ-QZ-03		
Use case	UC-QZ-01, UC-QZ-02	**SAD reference**	Components: Quiz Engine (authoring) | API: POST /quizzes; POST /quizzes/:id/publish
Test level	System (UI + API)	**Test type**	Functional – Positive
Priority	High - Core use-case flow or integrity control; failure blocks the use case.	**Assigned to**	Alifiya (M2)
Required role	Faculty, Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 logged in; bank has ≥ 5 questions.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Create Quiz: title "Unit 1 Quiz", time limit 30, start tomorrow 10:00, end tomorrow 18:00, select 5 questions. 2. Save. 3. Log in as Student S1 and open the quiz list.		
Expected result	Quiz is saved with status Draft and owner F1. The quiz is not visible to S1.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-QZ-02 - Quiz validation failures			
Requirements	REQ-QZ-02		
Use case	UC-QZ-01, UC-QZ-02	**SAD reference**	Components: Quiz Engine (authoring) | API: POST, PUT /quizzes
Test level	System (UI + API)	**Test type**	Functional – Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Alifiya (M2)
Required role	Faculty	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 logged in.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Try to save a quiz with: (a) empty title, (b) no questions selected, (c) 51 questions, (d) time limit 4, (e) time limit 181, (f) end		

	date-time before start date-time.		
Expected result	Each case is rejected and the message names the failing field and rule; no quiz is created.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-QZ-03 - Quiz boundary values (1/50 questions, 5/180 minutes, title 100, description 300)			
Requirements	REQ-QZ-01, REQ-QZ-02		
Use case	UC-QZ-01, UC-QZ-02	**SAD reference**	Components: Quiz Engine (authoring) | API: POST /quizzes; POST, PUT /quizzes
Test level	System (UI + API)	**Test type**	Functional – Boundary
Priority	Medium - Boundary values defined in the specification are a common defect location.	**Assigned to**	Alifiya (M2)
Required role	Faculty	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 logged in; bank has ≥ 50 questions.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Save quizzes with: (a) 1 question, (b) 50 questions, (c) time limit 5, (d) time limit 180, (e) title of 100 characters, (f) description of 300 characters.		
Expected result	All six quizzes are accepted and saved as Draft.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-QZ-04 - Publish quiz; question copy independent of bank edits			
Requirements	REQ-QZ-03, REQ-QZ-04		
Use case	UC-QZ-01, UC-QZ-02	**SAD reference**	Components: Quiz Engine (authoring); Question Bank Service | API: POST /quizzes; POST /quizzes/:id/publish
Test level	System (UI + API)	**Test type**	Functional – Positive
Priority	High - Core use-case flow or integrity control; failure blocks the use case.	**Assigned to**	Alifiya (M2)
Required role	Faculty, Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Draft quiz Z1 owned by F1.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Publish Z1. 2. As S1, open the quiz list. 3. As F1, edit a bank question used in Z1 (change the correct option). 4. As S1 attempt Z1 and view the result.		
Expected result	Z1 becomes Published and is visible to S1. Grading uses the original correct option stored in Z1, not the edited bank question.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-QZ-05 - Edit/delete Draft; edit/delete rejected after an attempt has started			
Requirements	REQ-QZ-05		
Use case	UC-QZ-01, UC-QZ-02	**SAD reference**	Components: Quiz Engine (authoring) | API: PUT, DELETE /quizzes/:id
Test level	System (UI + API)	**Test type**	Functional – Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Alifiya (M2)
Required role	Faculty, Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Published quiz Z1 owned by F1; Draft quiz Z2 owned by F1.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Edit Z2 and save; then delete Z2 and confirm. 2. S1 starts an attempt on Z1. 3. F1 tries to edit and delete Z1.		
Expected result	Step 1: edit saved; Z2 deleted after confirmation.		

	Step 3: both rejected (HTTP 409) with a message that an attempt has started; Z1 is unchanged. Rev 1.1 clarification: Rejections are HTTP 409 (SAD s4.3: quiz already attempted or no longer Draft). Whether a Published quiz without attempts may be edited is unresolved (TOD-03; see TC-QZ-22).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: TOD-03.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-QZ-06 - Student quiz list contains no question data			
Requirements	REQ-QZ-06, SEC-QZ-02		
Use case	UC-QZ-03 (not verified)	**SAD reference**	Components: Quiz Engine (execution); API Layer (projection) | API: GET /quizzes; POST /quizzes/:id/attempts
Test level	System (UI + API)	**Test type**	Functional – Positive
Priority	Medium - Verifies the stated functional requirement under normal conditions.	**Assigned to**	Vedh (M4)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Published quizzes: Z1 (open), Z3 (window already closed).		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. As S1 open the Quizzes page. 2. Inspect the response of GET /api/quizzes in the browser network tab.		
Expected result	Z1 is listed with title, number of questions, time limit, availability window and attempted status; Z3 is not listed. The response contains no question text, options, answers or explanations.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-QZ-07 - Attempt start rules: window and one attempt per Student			
Requirements	REQ-QZ-07		
Use case	UC-QZ-04, UC-QZ-05	**SAD reference**	Components: Quiz Engine (execution) | API: POST /quizzes/:id/attempts
Test level	System (UI + API)	**Test type**	Functional – Negative / Boundary
Priority	High - Boundary values defined in the specification are a common defect location.	**Assigned to**	Vedh (M4)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Quiz Z1 window 10:00–18:00; S1 has not attempted it.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. At 09:59 press Start Quiz. 2. At 10:00 press Start Quiz. 3. After finishing, press Start Quiz again. 4. At 18:01 log in as S2 and press Start Quiz.		
Expected result	Step 1 refused: "not yet open". Step 2 allowed. Step 3 refused: "already attempted". Step 4 refused: "closed". Rev 1.1 clarification: Refusal codes (SAD s4.4): before window HTTP 403 QUIZ_NOT_OPEN "Quiz not yet open"; after window HTTP 403 QUIZ_CLOSED "Quiz closed"; second attempt HTTP 409 ALREADY_ATTEMPTED "Quiz already attempted" (code naming: OD-12). Behaviour at exactly the end instant: TOD-04.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: OD-12, TOD-04.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-QZ-08 - Attempt start shows questions with timer and no answer key			
Requirements	REQ-QZ-08, SEC-QZ-02		
Use case	UC-QZ-04, UC-QZ-05	**SAD reference**	Components: Quiz Engine (execution); API Layer (projection) | API: POST /quizzes/:id/attempts; GET /quizzes
Test level	Security (API / system)	**Test type**	Functional / Security

Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Vedh (M4)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Published open quiz Z1 with 10 questions.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. S1 starts Z1. 2. Inspect the response of the start request in the network tab.		
Expected result	Questions and options are shown with a countdown timer. The response contains no correct option and no explanation.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-QZ-09 - Answer selection, unanswered warning and submission			
Requirements	REQ-QZ-09		
Use case	UC-QZ-04, UC-QZ-05	**SAD reference**	Components: Quiz Engine (execution) | API: PUT /attempts/:id/answers; POST /attempts/:id/submit
Test level	System (UI + API)	**Test type**	Functional – Positive
Priority	Medium - Verifies the stated functional requirement under normal conditions.	**Assigned to**	Vedh (M4)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 is attempting Z1 (10 questions).		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Select option B for question 1, then change it to C. 2. Leave question 2 unanswered. 3. Press Submit. 4. Confirm the warning.		
Expected result	Only one option per question can be selected; the latest choice (C) is kept. A warning about unanswered questions is shown before submission. After confirmation the attempt is submitted with the recorded answers.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-QZ-10 - Auto-submit at time limit; late answer rejected			
Requirements	REQ-QZ-10, SEC-QZ-04		
Use case	UC-QZ-04, UC-QZ-05	**SAD reference**	Components: Quiz Engine (execution) | API: PUT /attempts/:id/answers (410); POST /quizzes/:id/attempts; PUT /attempts/:id/answers
Test level	System (UI + API)	**Test type**	Functional – Boundary
Priority	High - Boundary values defined in the specification are a common defect location.	**Assigned to**	Vedh (M4)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Quiz Z4 with 5 questions and a 5-minute limit.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. S1 starts Z4 and answers 2 questions. 2. Wait until the time limit expires. 3. Send PUT /api/attempts/:id/answers for question 3 after expiry (Postman).		
Expected result	The attempt is auto-submitted with the 2 saved answers and graded (score based on those 2). The late answer is rejected (HTTP 410 TIME_EXPIRED) and does not change the result. Rev 1.1 clarification: Late answer: HTTP 410 TIME_EXPIRED, message "Time limit expired". Auto-submission trigger mechanism: TOD-04.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: TOD-04.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-QZ-11 - Grading correctness (6/10, 10/10, 0/10, 1/1)			
Requirements	REQ-QZ-11		
Use case	UC-QZ-04, UC-QZ-05	**SAD reference**	Components: Grading Service; Quiz Engine (execution) | API: POST /attempts/:id/submit
Test level	System (UI + API)	**Test type**	Functional – Positive / Boundary
Priority	High - Boundary values defined in the specification are a common defect location.	**Assigned to**	Vedh (M4)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	10-question quiz Z5, known answer key.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. S1 answers 6 correctly, 3 incorrectly, 1 unanswered; submit. 2. S2 answers all 10 correctly; submit. 3. S3 submits with no answers. 4. Repeat with a 1-question quiz (answer correct).		
Expected result	S1: score 6, 60.0 %. S2: score 10, 100.0 %. S3: score 0, 0.0 %. No negative marking; 1-question quiz gives 1/1 = 100.0 %.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-QZ-12 - Result details after submission			
Requirements	REQ-QZ-13		
Use case	UC-QZ-06 (not verified)	**SAD reference**	Components: Quiz Engine (execution) | API: GET /attempts/:id/result
Test level	System (UI + API)	**Test type**	Functional – Positive
Priority	High - Core use-case flow or integrity control; failure blocks the use case.	**Assigned to**	Vedh (M4)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 has submitted Z5.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Open the result page of the attempt.		
Expected result	Score, percentage and, for every question, S1's answer, the correct answer, correct/incorrect status and the explanation (when present) are shown.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-QZ-13 - Submitted attempt is immutable			
Requirements	REQ-QZ-14, SEC-QZ-05		
Use case	UC-QZ-04, UC-QZ-05	**SAD reference**	Components: Quiz Engine (execution); Grading Service | API: PUT /attempts/:id/answers; POST /attempts/:id/submit
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Vedh (M4)
Required role	Student, Admin	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 has submitted attempt A1.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Press the browser Back button and try to change an answer and submit again. 2. Call PUT /api/attempts/A1/answers and POST /api/attempts/A1/submit again (Postman).		
Expected result	Both attempts are rejected (HTTP 409 already submitted); the stored answers and score are unchanged. Rev 1.1 clarification: HTTP 409 ALREADY_SUBMITTED for both calls.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-QZ-14 - Student views own attempts			
Requirements	REQ-QZ-15		
Use case	UC-QZ-06 (not verified)	**SAD reference**	Components: Quiz Engine (execution) | API: GET /attempts/mine
Test level	System (UI + API)	**Test type**	Functional – Positive
Priority	Medium - Verifies the stated functional requirement under normal conditions.	**Assigned to**	Vedh (M4)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 has submitted 2 attempts on different quizzes.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Open My Results. 2. Open the detailed result of one attempt.		
Expected result	Both attempts are listed with quiz title, score, percentage and submission date; the detailed result opens. Only S1's own attempts are shown.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-QZ-15 - Faculty views quiz submissions			
Requirements	REQ-QZ-16		
Use case	UC-QZ-07 (not verified)	**SAD reference**	Components: Quiz Engine (execution) | API: GET /quizzes/:id/submissions
Test level	System (UI + API)	**Test type**	Functional – Positive / Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Alifiya (M2)
Required role	Faculty	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 owns Z1 (2 submissions) and Z6 (no submissions).		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. F1 opens Submissions for Z1. 2. F1 opens Submissions for Z6.		
Expected result	Z1 lists each Student's name, score, percentage and submission time. Z6 shows "No submissions yet".		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: OD-08.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-QZ-16 - Result and submission privacy; Faculty cannot attempt			
Requirements	REQ-QZ-17, SEC-QZ-03		
Use case	UC-QZ-01 to 07	**SAD reference**	Components: API Layer (RBAC/ownership); Quiz Engine; Quiz Engine (execution) | API: All /quizzes and /attempts endpoints; GET /attempts/:id/result; GET /quizzes/:id/submissions
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Alifiya (M2)
Required role	Faculty, Student, Admin	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Attempt A1 belongs to S1; quiz Z1 owned by F1; F2 and S2 exist.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. S2 requests the result of A1 (change the id in the URL / API). 2. F2 requests the submissions of Z1. 3. F2 (Faculty) calls POST /api/quizzes/Z1/attempts.		
Expected result	Steps 1 and 2 return "Access denied" (403) with no data. Step 3 is refused (403): only Students can attempt quizzes. Rev 1.1 clarification: Step 1 and 2: HTTP 403 ACCESS_DENIED. Step 3: HTTP 403 ACCESS_DENIED (role).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-QZ-17 - Unauthenticated and Student access to quiz management denied			
Requirements	REQ-QZ-17, SEC-QZ-01		
Use case	UC-QZ-01 to 07	**SAD reference**	Components: API Layer (RBAC/ownership); Quiz Engine; API Layer | API: All /quizzes and /attempts endpoints; /questions; /quizzes
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Alifiya (M2)
Required role	Student, Unauthenticated	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	No valid session / Student session.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Call GET /api/quizzes without a token. 2. As S1 call POST /api/quizzes and POST /api/quizzes/Z1/publish.		
Expected result	Step 1: HTTP 401, authentication required. Step 2: HTTP 403 "Access denied"; no quiz is created or published. Rev 1.1 clarification: Unauthenticated: HTTP 401; Student on Faculty endpoints: HTTP 403 "Access denied".		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-QZ-18 - Unavailable quiz, attempt or question handled without details			
Requirements	REQ-QZ-18		
Use case	UC-QZ-01 to 07	**SAD reference**	Components: API Layer (central error handler); Quiz Engine; Question Bank Service | API: 404 branches of /quizzes, /attempts, /questions
Test level	System (UI + API)	**Test type**	Functional – Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Alifiya (M2)
Required role	Faculty, Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	A quiz, question and attempt that have been deleted/never existed.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Open the URL of a deleted quiz as S1. 2. Open the URL of a non-existent attempt. 3. F1 edits a question deleted in another session.		
Expected result	In each case the system says the item is unavailable (HTTP 404) and shows none of its details. Rev 1.1 clarification: HTTP 404 NOT_FOUND with messages "Quiz unavailable" / "Question unavailable" / "Attempt unavailable"; no details of the item.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-QZ-19 - Publish after the availability end has passed			
Requirements	REQ-QZ-04, REQ-QZ-02		
Use case	UC-QZ-01, UC-QZ-02	**SAD reference**	Components: Quiz Engine (authoring); Question Bank Service | API: POST /quizzes/:id/publish; POST, PUT /quizzes
Test level	Integration (API)	**Test type**	Functional - Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Alifiya (M2)
Required role	Faculty (F1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Draft quiz Z7 owned by F1 whose availableUntil is in the past (create with a future window, then advance the server test clock).		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. POST /api/quizzes/Z7/publish. 2. GET the quiz list as S1.		
Expected result	Step 1: HTTP 400 VALIDATION_ERROR (availability end already passed); Z7 remains Draft; no question copies stored. Step 2: Z7 not listed.		

Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Injectable server clock.. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-QZ-20 - Publish or create with a question removed from the bank			
Requirements	REQ-QZ-18, REQ-QZ-04		
Use case	UC-QZ-01 (alternate flow A2)	**SAD reference**	Components: API Layer (central error handler); Quiz Engine; Question Bank Service | API: 404 branches of /quizzes, /attempts, /questions; POST /quizzes/:id/publish
Test level	Integration (API)	**Test type**	Functional - Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Alifiya (M2)
Required role	Faculty (F1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 owns bank questions Q10 and Q11; Draft quiz Z8 contains both.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. F1 removes Q11 from the bank. 2. POST /api/quizzes/Z8/publish. 3. POST /api/quizzes with questionIds containing Q11.		
Expected result	Step 2: publication refused with an "unavailable" response (HTTP 404 expected, REQ-QZ-18; the SAD lists only "quiz unavailable" for publish, TOD-11); Z8 stays Draft; no partial copy stored. Step 3: HTTP 404 (question unavailable); no quiz created.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: TOD-11.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-QZ-21 - Availability window boundary			
Requirements	REQ-QZ-02		
Use case	UC-QZ-01	**SAD reference**	Components: Quiz Engine (authoring) | API: POST, PUT /quizzes
Test level	Integration (API)	**Test type**	Functional - Boundary
Priority	Medium - Boundary values defined in the specification are a common defect location.	**Assigned to**	Alifiya (M2)
Required role	Faculty (F1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 is logged in; bank has at least 5 questions.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Create a quiz with availableUntil exactly 1 minute after availableFrom. 2. Create a quiz with availableUntil equal to availableFrom. 3. Create a quiz with availableUntil 1 minute before availableFrom.		
Expected result	Step 1: HTTP 201 Draft. Steps 2-3: HTTP 400 VALIDATION_ERROR (end must be later than start); no quiz created.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-QZ-22 - Editing or deleting a Published quiz that has no attempts			
Requirements	REQ-QZ-05		
Use case	UC-QZ-02	**SAD reference**	Components: Quiz Engine (authoring) | API: PUT, DELETE /quizzes/:id
Test level	Integration (API)	**Test type**	Functional - Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Alifiya (M2)
Required role	Faculty (F1)	**Readiness**	Blocked - expected result not agreed
Preconditions	F1 owns Published quiz Z9 with no attempts.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. PUT /api/quizzes/Z9 with a new title. 2. DELETE /api/quizzes/Z9.		
Expected result	BLOCKED (TOD-03): SAD s4.3 allows PUT on Draft only (409 if no longer Draft) and DELETE unless an attempt exists, whereas		

	STP v1.0 TC-QZ-05 implies edits are refused only after an attempt has started. The expected result must be confirmed against SRS REQ-QZ-05.		
Pass / fail criteria	Cannot pass or fail until the linked decision is recorded (Blocked). When the decision is approved, update the expected result and re-baseline.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: TOD-03.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-QZ-23 - Answer-save validation			
Requirements	REQ-QZ-09		
Use case	UC-QZ-04	**SAD reference**	Components: Quiz Engine (execution) | API: PUT /attempts/:id/answers; POST /attempts/:id/submit
Test level	Integration (API)	**Test type**	Functional - Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Vedh (M4)
Required role	Student (S1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 has an in-progress attempt on a 10-question quiz.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. PUT /api/attempts/<id>/answers with position 0, then 11, then -1. 2. With selectedOption "E", then "". 3. With position or selectedOption missing. 4. With position 10 and selectedOption "D".		
Expected result	Steps 1-3: HTTP 400 VALIDATION_ERROR; saved answers unchanged. Step 4: HTTP 200; the answer for question 10 is D.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-QZ-24 - Attempt ownership and existence on attempt endpoints			
Requirements	REQ-QZ-17, REQ-QZ-18, SEC-QZ-03		
Use case	UC-QZ-04	**SAD reference**	Components: API Layer (RBAC/ownership); Quiz Engine; API Layer (central error handler) | API: All /quizzes and /attempts endpoints; 404 branches of /quizzes, /attempts, /questions; GET /attempts/:id/result
Test level	Security (API)	**Test type**	Security - Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Alifiya (M2)
Required role	Student (S1, S2)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Attempt A1 belongs to S1 (in progress).		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. As S2, PUT /api/attempts/A1/answers. 2. As S2, POST /api/attempts/A1/submit. 3. As S1, PUT answers for a well-formed attempt id that does not exist. 4. As S1, PUT answers for a malformed id.		
Expected result	Steps 1-2: HTTP 403 ACCESS_DENIED; A1 unchanged. Step 3: HTTP 404 "Attempt unavailable". Step 4: HTTP 400 or 404; never HTTP 500.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-QZ-25 - Auto-submission when the Student stops interacting			
Requirements	REQ-QZ-10, SEC-QZ-04		
Use case	UC-QZ-04, UC-QZ-05	**SAD reference**	Components: Quiz Engine (execution) | API: PUT /attempts/:id/answers (410); POST /quizzes/:id/attempts; PUT /attempts/:id/answers
Test level	System (UI + API)	**Test type**	Functional - Boundary
Priority	High - Boundary values defined in the	**Assigned to**	Vedh (M4)

	specification are a common defect location.		
Required role	Student (S1), Faculty (F1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Quiz Z4 (5 questions, 5-minute limit) is open.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. S1 starts Z4, saves 2 answers and closes the browser. 2. Wait until the deadline plus 10 seconds. 3. F1 opens the submissions of Z4. 4. S1 logs in and opens My Results. 5. S1 tries to start Z4 again.		
Expected result	Steps 3-4: the attempt is Submitted by automatic submission with the 2 saved answers and graded from them. Step 5: HTTP 409 ALREADY_ATTEMPTED. Whether closure is scheduled or applied on the next read is unspecified (TOD-04); the pass condition is the correct state at the first read after the deadline.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: TOD-04.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-QZ-26 - Grading failure: pending result and retry			
Requirements	REQ-QZ-11, REQ-QZ-14		
Use case	UC-QZ-05 (alternate flow A2)	**SAD reference**	Components: Grading Service; Quiz Engine (execution) | API: POST /attempts/:id/submit; PUT /attempts/:id/answers
Test level	Integration (API)	**Test type**	Functional - Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Vedh (M4)
Required role	Student (S1)	**Readiness**	Blocked - expected result not agreed
Preconditions	Grading Service can be made to fail in the test environment (fault injection).		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Start an attempt, answer 3 questions, enable the grading fault. 2. POST /api/attempts/<id>/submit. 3. Disable the fault; wait for the retry. 4. GET /api/attempts/<id>/result.		
Expected result	BLOCKED (OD-08): SAD s4.3/s4.4 state that the submission is retained and the result shown as pending, but the response contract for a pending result is undefined. After the contract is agreed: step 2 returns the agreed pending response; the submission is unchanged; step 4 returns the correct score.		
Pass / fail criteria	Cannot pass or fail until the linked decision is recorded (Blocked). When the decision is approved, update the expected result and re-baseline.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: OD-08.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-QZ-27 - Result of an in-progress attempt exposes no answer key			
Requirements	SEC-QZ-02, REQ-QZ-13		
Use case	UC-QZ-06 (not verified)	**SAD reference**	Components: API Layer (projection); Quiz Engine (execution) | API: GET /quizzes; POST /quizzes/:id/attempts; GET /attempts/:id/result
Test level	Security (API)	**Test type**	Security - Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Vedh (M4)
Required role	Student (S1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 has an in-progress attempt A1 on Z1.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. GET /api/attempts/A1/result before submitting. 2. Scan the response body for correctOption, explanation and correct flags.		
Expected result	The request is refused, or any response contains no correct option, explanation or correctness flag. The refusal status is not specified in SAD s4.3 (record it). HTTP 200 containing the key is a Fail.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-QZ-28 - Quiz owner cannot read a Student's attempt result			
Requirements	REQ-QZ-17, SEC-QZ-03		
Use case	UC-QZ-06, UC-QZ-07 (not verified)	**SAD reference**	Components: API Layer (RBAC/ownership); Quiz Engine; Quiz Engine (execution) | API: All /quizzes and /attempts endpoints; GET /attempts/:id/result; GET /quizzes/:id/submissions
Test level	Security (API)	**Test type**	Security - Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Alifiya (M2)
Required role	Faculty (F1), Admin (A1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Attempt A1 (S1) on quiz Z1 owned by F1 is submitted.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. As F1, GET /api/attempts/A1/result. 2. As A1 (Admin), GET /api/attempts/A1/result.		
Expected result	Both: HTTP 403 ACCESS_DENIED; no result data. F1 may see the score only through GET /api/quizzes/Z1/submissions (REQ-QZ-16).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-QZ-29 - Deadline when the time limit extends beyond the availability end			
Requirements	REQ-QZ-10, SEC-QZ-04		
Use case	UC-QZ-04	**SAD reference**	Components: Quiz Engine (execution) | API: PUT /attempts/:id/answers (410); POST /quizzes/:id/attempts; PUT /attempts/:id/answers
Test level	Integration (API)	**Test type**	Functional - Boundary
Priority	Medium - Boundary values defined in the specification are a common defect location.	**Assigned to**	Vedh (M4)
Required role	Student (S1)	**Readiness**	Blocked - expected result not agreed
Preconditions	Quiz Z10: time limit 30 minutes, window ends at 18:00.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. S1 starts Z10 at 17:50. 2. Record the deadline returned. 3. Save an answer at 17:59 and at 18:05.		
Expected result	BLOCKED (TOD-04): it is not specified whether the deadline is start + time limit (18:20) or the window end (18:00). The expected deadline and the outcome of the 18:05 answer must be confirmed against SRS REQ-QZ-07/REQ-QZ-10.		
Pass / fail criteria	Cannot pass or fail until the linked decision is recorded (Blocked). When the decision is approved, update the expected result and re-baseline.		
Dependencies	Server test clock that can be advanced (s6.5). All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: TOD-04.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-QZ-30 - Concurrent start requests allow exactly one attempt			
Requirements	REQ-QZ-07, SEC-QZ-05		
Use case	UC-QZ-04	**SAD reference**	Components: Quiz Engine (execution); Grading Service | API: POST /quizzes/:id/attempts; POST /attempts/:id/submit
Test level	Integration (API)	**Test type**	Functional - Concurrency
Priority	High - Race conditions can break one-attempt/one-membership/immutability rules.	**Assigned to**	Vedh (M4)
Required role	Student (20 generated)	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	Open quiz Z1; 20 generated Students with no attempt.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. For each Student, send two simultaneous POST /api/quizzes/Z1/attempts. 2. Count responses and Attempt records per Student.		
Expected result	For every Student: exactly one HTTP 201 and one HTTP 409 ALREADY_ATTEMPTED; exactly one Attempt record exists.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		
Dependencies	Script able to send parallel requests (Supertest/JMeter).. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-QZ-31 - Concurrent submit requests are graded once			
Requirements	REQ-QZ-14, REQ-QZ-11, SEC-QZ-05		
Use case	UC-QZ-05	**SAD reference**	Components: Quiz Engine (execution); Grading Service | API: PUT /attempts/:id/answers; POST /attempts/:id/submit
Test level	Integration (API)	**Test type**	Functional - Concurrency
Priority	High - Race conditions can break one-attempt/one-membership/immutability rules.	**Assigned to**	Vedh (M4)
Required role	Student (S1)	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	S1 has an in-progress attempt with 5 saved answers.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Send two simultaneous POST /api/attempts/<id>/submit. 2. GET the result twice.		
Expected result	One HTTP 200 and one HTTP 409 ALREADY_SUBMITTED; a single stored score; both result reads identical.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		
Dependencies	Script able to send parallel requests.. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-QZ-32 - Countdown follows the server deadline			
Requirements	REQ-QZ-08, SEC-QZ-04		
Use case	UC-QZ-04	**SAD reference**	Components: Quiz Engine (execution); API Layer (projection) | API: POST /quizzes/:id/attempts; PUT /attempts/:id/answers
Test level	System (UI + API)	**Test type**	Functional - Positive
Priority	Medium - Verifies the stated functional requirement under normal conditions.	**Assigned to**	Vedh (M4)
Required role	Student (S1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Quiz Z4 open.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. S1 starts Z4; note the deadline in the start response. 2. After 1 minute reload the attempt page; then open the attempt in a second browser.		
Expected result	The countdown continues from the original deadline (not restarted) in both views; one-question-per-view navigation and answered/unanswered indicators are shown (SAD s4.5).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

A.8 Quiz security

TC-QS-01 - Server clock governs window and deadline			
Requirements	SEC-QZ-04, REQ-QZ-07		
Use case	UC-QZ-04, UC-QZ-05	**SAD reference**	Components: Quiz Engine (execution) | API: POST /quizzes/:id/attempts; PUT /attempts/:id/answers
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Vedh (M4)
Required role	Public / any	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Quiz Z1 window 10:00–18:00.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Set the device clock back to 09:00 and start Z1 at real time 18:30. 2. During an attempt, set the device clock back by 30 minutes and continue answering past the real deadline.		
Expected result	The server clock decides: start is refused ("closed"); answers received after the real deadline are rejected and the attempt is auto-submitted.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Server test clock that can be advanced (s6.5). All tests: build identified by commit SHA; test accounts per s6.4.		

Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-QS-02 - Client-supplied score ignored			
Requirements	SEC-QZ-05		
Use case	-	**SAD reference**	Components: Quiz Engine (execution); Grading Service | API: POST /attempts/:id/submit
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Vedh (M4)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 is attempting Z1.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Intercept POST /api/attempts/:id/submit and add the field "score": 100 and "isCorrect": true for every question. 2. Forward the request.		
Expected result	The extra fields are ignored; the stored score is computed by the server from the answers.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-QS-03 - NoSQL-operator and script payloads in search and question text			
Requirements	SEC-QZ-01, REQ-QB-02		
Use case	UC-QB-01 to 04 (pairing not verified)	**SAD reference**	Components: API Layer; Question Bank Service; Quiz Engine (authoring) | API: /questions; /quizzes; POST, PUT /questions
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Alifiya (M2)
Required role	Faculty	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 logged in.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Search the question bank with query {"$gt": ""} / q[$ne]=x. 2. Add a question whose text is <script>alert(1)</script>. 3. View that question in the bank and in a quiz.		
Expected result	Step 1: treated as plain text or rejected; no unrelated data is returned. Steps 2–3: the text is shown as plain text; the script does not run.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-QS-04 - Expired or logged-out session refused on quiz endpoints			
Requirements	SEC-QZ-03, SEC-ACC-04		
Use case	-	**SAD reference**	Components: Quiz Engine (execution); API Layer; Auth & Session Service | API: GET /attempts/:id/result; GET /quizzes/:id/submissions; POST /auth/logout
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Alifiya (M2)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 logged in; session idle for 31 minutes.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. After 31 minutes of inactivity press Submit on an open attempt page / call GET /api/attempts/mine. 2. Log out, press Back and request a result.		
Expected result	Both requests are refused (401) and the user must log in again; no result data is shown. Rev 1.1 clarification: HTTP 401 NOT_AUTHENTICATED; idle message "Session expired. Please log in again."		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Server test clock that can be advanced (s6.5). All tests: build identified by commit SHA; test accounts per s6.4.		

Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-QS-05 - Malformed identifiers and operator payloads on quiz and attempt endpoints			
Requirements	NFR-QUAL-03, SEC-QZ-01, SEC-QZ-05		
Use case	-	**SAD reference**	Components: API Layer; Question Bank Service; Quiz Engine (authoring) | API: /questions; /quizzes; POST /attempts/:id/submit
Test level	Security (API)	**Test type**	Security - Negative
Priority	Medium - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Jagadeesh (M1)
Required role	Student (S1), Faculty (F1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Quiz Z1 and attempt A1 exist.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. GET /api/quizzes/not-an-id/submissions as F1. 2. GET /api/attempts/%7B%22%24ne%22%3Anull%7D/result as S1. 3. PUT /api/attempts/A1/answers with {position:{"$gt":0}, selectedOption:{"$ne":""}}.		
Expected result	Each request: HTTP 400 or 404; never HTTP 200 or 500; no data returned; no stack trace; server remains available.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-QS-06 - Answer-key scan across all Student-facing responses			
Requirements	SEC-QZ-02		
Use case	UC-QZ-04	**SAD reference**	Components: API Layer (projection); Quiz Engine (execution) | API: GET /quizzes; POST /quizzes/:id/attempts
Test level	Security (API)	**Test type**	Security - Inspection
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Vedh (M4)
Required role	Student (S1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Open quiz Z1; S1 has not attempted it.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Capture the responses of: quiz list, attempt start, answer save, and a result request before submission. 2. Run an automated scan for correctOption, explanation, isCorrect and correct. 3. Submit; capture the submit response.		
Expected result	Steps 1-2: none of the fields in any pre-submission response. Step 3: the fields appear only in S1's own submit/result response.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

A.9 Quiz performance

TC-QP-01 - Grading within 2 s for 100 simultaneous submissions			
Requirements	NFR-PERF-04, REQ-QZ-12		
Use case	UC-QZ-04, UC-QZ-05	**SAD reference**	Components: Grading Service | API: POST /attempts/:id/submit
Test level	Performance (JMeter / timing)	**Test type**	Performance (NFR)
Priority	High - Verifies a measurable NFR threshold; failure is an NFR breach.	**Assigned to**	Vedh (M4)
Required role	Student	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	Quiz with 50 questions; 100 test Students with started attempts (JMeter).		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. All 100 virtual users submit within the same 5-second window. 2. Record the time from submit to result for each.		
Expected result	Every result is returned within 2 seconds; no submission is lost; scores are correct for all 100.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		

Dependencies	Tooling/environment per s6 (JMeter, load generator, devices, backup or TLS host as applicable). All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: TOD-10.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-QP-02 - Question-bank search within 2 s (1,000 questions)			
Requirements	NFR-PERF-05, REQ-QB-04		
Use case	UC-QB-01 to 04 (pairing not verified)	**SAD reference**	Components: Question Bank Service; Data Access Layer; API Layer | API: GET /questions
Test level	Performance (JMeter / timing)	**Test type**	Performance (NFR)
Priority	Medium - Verifies a measurable NFR threshold; failure is an NFR breach.	**Assigned to**	Alifiya (M2)
Required role	Public / any	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	Bank with 1,000 questions.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Search by keyword, then by topic + difficulty filters (10 runs each). 2. Record response times.		
Expected result	Each search returns within 2 seconds.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		
Dependencies	Tooling/environment per s6 (JMeter, load generator, devices, backup or TLS host as applicable). All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: TOD-10.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-QP-03 - Quiz questions displayed within 3 s			
Requirements	NFR-PERF-06, REQ-QZ-08		
Use case	UC-QZ-04, UC-QZ-05	**SAD reference**	Components: Quiz Engine (execution); API Layer (projection) | API: POST /quizzes/:id/attempts
Test level	Performance (JMeter / timing)	**Test type**	Performance (NFR)
Priority	Medium - Verifies a measurable NFR threshold; failure is an NFR breach.	**Assigned to**	Vedh (M4)
Required role	Student	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	Quiz with 50 questions.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. S1 presses Start Quiz (10 runs). 2. Measure time until the questions are displayed.		
Expected result	Questions are displayed within 3 seconds in every run.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		
Dependencies	Tooling/environment per s6 (JMeter, load generator, devices, backup or TLS host as applicable). All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: TOD-10.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-QP-04 - New question visible within 3 s			
Requirements	REQ-QB-03		
Use case	UC-QB-01 to 04 (pairing not verified)	**SAD reference**	Components: Question Bank Service; API Layer | API: POST /questions
Test level	Performance (JMeter / timing)	**Test type**	Performance (NFR)
Priority	Medium - Verifies a measurable NFR threshold; failure is an NFR breach.	**Assigned to**	Alifiya (M2)
Required role	Public / any	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	Bank with 1,000 questions.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Add a question and measure the time until it appears in the bank list.		
Expected result	The new question is visible within 3 seconds.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		
Dependencies	Tooling/environment per s6 (JMeter, load generator, devices, backup or TLS host as applicable). All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

A.10 Performance Analytics

TC-PA-01 - View completed attempts			
Requirements	REQ-PA-01		
Use case	UC-PA-01	**SAD reference**	Components: Performance Analytics Service; API Layer (RBAC/ownership) | API: GET /performance/personal
Test level	System (UI + API)	**Test type**	Functional – Positive
Priority	High - Core use-case flow or integrity control; failure blocks the use case.	**Assigned to**	Pushkar (M3)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Student S1 is logged in and has completed two quizzes.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Open View Performance. 2. Retrieve completed quiz attempts. 3. Inspect the displayed attempt records.		
Expected result	Each completed attempt displays quiz title, score, percentage and submission date.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-PA-02 - Personal performance calculated; empty state			
Requirements	REQ-PA-02		
Use case	UC-PA-01	**SAD reference**	Components: Performance Analytics Service; API Layer (RBAC/ownership) | API: GET /performance/personal
Test level	System (UI + API)	**Test type**	Functional – Positive / Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Pushkar (M3)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Student S1 has completed quizzes with known scores. Student S3 has no completed quiz attempts.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Log in as S1. 2. Open View Performance. 3. Compare displayed performance with the stored completed attempts. 4. Log in as S3 and open View Performance.		
Expected result	S1's performance is correctly calculated and displayed. S3 receives "No performance data available."		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: TOD-05.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-PA-03 - Topic-wise analysis; empty state			
Requirements	REQ-PA-03		
Use case	UC-PA-03	**SAD reference**	Components: Performance Analytics Service; API Layer (RBAC/ownership) | API: GET /performance/topics
Test level	System (UI + API)	**Test type**	Functional – Positive / Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Pushkar (M3)
Required role	Student	**Readiness**	Blocked - expected result not agreed
Preconditions	S1 has completed quizzes containing questions from multiple topics.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Log in as S1. 2. Open Topic-wise Analysis. 3. Inspect the topic groups. 4. Repeat using a Student with no completed quiz data.		
Expected result	Results are correctly grouped by topic. A Student with no completed data receives "No topic-wise performance data available." Rev 1.1 clarification: BLOCKED in part (OD-03): the source of question topic is undefined (REQ-QZ-04/B.10 omit topic from the question copy). Only the empty-state step can be executed until the		

	team decides how topic is obtained.		
Pass / fail criteria	Empty-state step (4) may be executed now; the grouping steps are Blocked until OD-03 is decided.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: OD-03.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-PA-04 - Faculty views class performance for own quiz			
Requirements	REQ-PA-04		
Use case	UC-PA-02	**SAD reference**	Components: Performance Analytics Service; API Layer (RBAC/ownership) | API: GET /performance/class/:quizId
Test level	System (UI + API)	**Test type**	Functional – Positive / Negative
Priority	High - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Pushkar (M3)
Required role	Faculty, Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Faculty F1 owns quiz Z1. Students S1 and S2 have submitted attempts for Z1. Faculty F2 owns another quiz.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Log in as F1. 2. Open Class Performance for Z1. 3. Inspect the submitted Student attempts. 4. Attempt to access performance for a quiz owned by F2.		
Expected result	F1 can view performance for Students who submitted Z1. Access to F2's quiz performance is denied. Rev 1.1 clarification: Quiz selection through the UI depends on a list-own-quizzes contract that is not defined (OD-08); execute through GET /api/performance/class/:quizId. Denial: HTTP 403 ACCESS_DENIED.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: OD-08.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-PA-05 - Class performance record content; empty state			
Requirements	REQ-PA-05		
Use case	UC-PA-02	**SAD reference**	Components: Performance Analytics Service; API Layer (RBAC/ownership) | API: GET /performance/class/:quizId
Test level	System (UI + API)	**Test type**	Functional – Positive / Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Pushkar (M3)
Required role	Faculty, Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 owns Z1 and multiple Students have submitted attempts.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Log in as F1. 2. Open Class Performance for Z1. 3. Inspect each Student record. 4. Repeat using a quiz with no submitted attempts.		
Expected result	Each Student record contains Student name, score, percentage and submission information. A quiz with no submissions displays "No class performance data available." Rev 1.1 clarification: Empty state message per STP v1.0: "No class performance data available."		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: OD-08.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-PA-06 - Non-owner Faculty denied class performance			
Requirements	REQ-PA-06, REQ-PA-04		
Use case	UC-PA-02	**SAD reference**	Components: Performance Analytics Service; API Layer (RBAC/ownership) | API: GET /performance/class/:quizId
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Pushkar (M3)

Required role	Faculty	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 owns Z1. F2 is not the owner of Z1.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Log in as F2. 2. Request class performance for Z1 through the UI. 3. Call /performance/class/Z1 through Postman.		
Expected result	Both requests are rejected with HTTP 403 and "Access denied". No performance information is exposed. Rev 1.1 clarification: Both requests: HTTP 403 ACCESS_DENIED "Access denied" (path GET /api/performance/class/:quizId).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-PA-07 - Personal performance limited to the authenticated Student			
Requirements	REQ-PA-01, REQ-ACC-10		
Use case	UC-PA-01	**SAD reference**	Components: Performance Analytics Service; API Layer (RBAC/ownership); API Layer | API: GET /performance/personal; All protected endpoints
Test level	Security (API)	**Test type**	Security - Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Pushkar (M3)
Required role	Student (S1), Faculty (F1), Admin (A1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 and S2 each have completed attempts.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. GET /api/performance/personal as S1. 2. As F1. 3. As A1. 4. Without a token.		
Expected result	Step 1: HTTP 200; only S1's attempts (none of S2's). Steps 2-3: HTTP 403 ACCESS_DENIED. Step 4: HTTP 401.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-PA-08 - Displayed values equal stored attempts			
Requirements	REQ-PA-02		
Use case	UC-PA-01	**SAD reference**	Components: Performance Analytics Service; API Layer (RBAC/ownership) | API: GET /performance/personal
Test level	System (UI + API)	**Test type**	Functional - Positive
Priority	High - Core use-case flow or integrity control; failure blocks the use case.	**Assigned to**	Pushkar (M3)
Required role	Student (S1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 has three completed attempts: 6/10, 3/5 and 1/1 (stored percentages 60.0, 60.0, 100.0).		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Query the stored attempts directly (database) and compute percentages independently. 2. GET /api/performance/personal and open View Performance. 3. Compare each attempt's score, percentage and any aggregate value shown.		
Expected result	Each attempt's score and percentage equals the stored/recomputed value (one decimal place, as graded). Any aggregate (for example an average) equals the independent calculation. The aggregate definition is not specified (TOD-05); record the formula observed.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: TOD-05.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-PA-09 - Topic-wise grouping with known topic data			
Requirements	REQ-PA-03		
Use case	UC-PA-03	**SAD reference**	Components: Performance Analytics Service; API Layer (RBAC/ownership) | API: GET /performance/topics
Test level	System (UI + API)	**Test type**	Functional - Positive
Priority	High - Core use-case flow or integrity control; failure blocks the use case.	**Assigned to**	Pushkar (M3)
Required role	Student (S1)	**Readiness**	Blocked - expected result not agreed
Preconditions	S1 has attempts on quizzes whose questions span three topics.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Open Topic-wise Analysis. 2. Compare each topic's attempts and averagePercentage with an independent calculation.		
Expected result	BLOCKED (OD-03): the topic of a question is not stored in the quiz question copy (REQ-QZ-04, Appendix B.10) and bank questions may be deleted (REQ-QB-06), so the source of topic is undefined. Setup and independent calculation can be prepared; the expected grouping needs the agreed topic source.		
Pass / fail criteria	Cannot pass or fail until the linked decision is recorded (Blocked). When the decision is approved, update the expected result and re-baseline.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: OD-03.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-PA-10 - Class performance equals quiz submissions view			
Requirements	REQ-PA-05, REQ-QZ-16		
Use case	UC-PA-02	**SAD reference**	Components: Performance Analytics Service; API Layer (RBAC/ownership); Quiz Engine (execution) | API: GET /performance/class/:quizId; GET /quizzes/:id/submissions
Test level	Integration (API)	**Test type**	Functional - Positive
Priority	Medium - Verifies the stated functional requirement under normal conditions.	**Assigned to**	Pushkar (M3)
Required role	Faculty (F1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 owns Z1 with submissions from S1 and S2.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. GET /api/performance/class/Z1. 2. GET /api/quizzes/Z1/submissions. 3. Compare the two payloads.		
Expected result	Both return the same set of {studentName, score, percentage, submittedAt} records (shared data, SAD interface C).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-PA-11 - Class-performance request validation			
Requirements	REQ-PA-06, REQ-QZ-18		
Use case	UC-PA-02	**SAD reference**	Components: Performance Analytics Service; API Layer (RBAC/ownership); API Layer (central error handler) | API: GET /performance/class/:quizId; 404 branches of /quizzes, /attempts, /questions
Test level	Security (API)	**Test type**	Security - Negative
Priority	Medium - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Pushkar (M3)
Required role	Student (S1), Faculty (F1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	F1 owns Z1.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. GET /api/performance/class/Z1 without a token. 2. As S1. 3. As F1 for a well-formed quiz id that does not exist. 4. As F1 for a malformed id.		
Expected result	Step 1: HTTP 401. Step 2: HTTP 403 ACCESS_DENIED. Step 3: HTTP 404 "Quiz unavailable" without details. Step 4: HTTP 400 or 404; never 500.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any		

	difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-PA-12 - Faculty selects an own quiz for class performance			
Requirements	REQ-PA-04		
Use case	UC-PA-02	**SAD reference**	Components: Performance Analytics Service; API Layer (RBAC/ownership) | API: GET /performance/class/:quizId
Test level	System (UI + API)	**Test type**	Functional - Positive
Priority	Medium - Verifies the stated functional requirement under normal conditions.	**Assigned to**	Pushkar (M3)
Required role	Faculty (F1)	**Readiness**	Blocked - expected result not agreed
Preconditions	F1 owns Z1 and Z6.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Open Class Performance. 2. Choose Z1 from F1's quizzes.		
Expected result	BLOCKED (OD-08): no API contract exists for listing or reading a Faculty user's own quizzes (UC-PA-02, UI-16). Execute REQ-PA-04 through the API (TC-PA-04) until the contract is defined.		
Pass / fail criteria	Cannot pass or fail until the linked decision is recorded (Blocked). When the decision is approved, update the expected result and re-baseline.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: OD-08.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-PA-13 - New submission appears in personal performance			
Requirements	REQ-PA-01, REQ-PA-02, REQ-QZ-15		
Use case	UC-PA-01	**SAD reference**	Components: Performance Analytics Service; API Layer (RBAC/ownership); Quiz Engine (execution) | API: GET /performance/personal; GET /attempts/mine
Test level	System (UI + API)	**Test type**	Functional - Positive
Priority	Medium - Verifies the stated functional requirement under normal conditions.	**Assigned to**	Pushkar (M3)
Required role	Student (S3)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S3 has no completed attempts; quiz Z5 (10 questions, known key) is open.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Open View Performance as S3. 2. Complete and submit Z5 (answer 7 correctly). 3. Reload View Performance and My Results.		
Expected result	Step 1: "No performance data available." Step 3: exactly one attempt, score 7, percentage 70.0, identical in both views.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

A.11 Collaboration

TC-COL-01 - Create study group			
Requirements	REQ-COL-01		
Use case	UC-COL-01, UC-COL-02	**SAD reference**	Components: Study Group & Collaboration Service; API Layer | API: POST /groups
Test level	System (UI + API)	**Test type**	Functional – Positive
Priority	Medium - Verifies the stated functional requirement under normal conditions.	**Assigned to**	Pushkar (M3)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Student S1 is authenticated.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Open Study Groups. 2. Select Create Group.		

	3. Enter valid group name and description. 4. Submit.		
Expected result	Study group is created successfully and S1 is recorded as a member. The group appears in S1's groups.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-COL-02 - Join study group			
Requirements	REQ-COL-02		
Use case	UC-COL-01, UC-COL-02	**SAD reference**	Components: Study Group & Collaboration Service; API Layer | API: POST /groups/:id/join
Test level	System (UI + API)	**Test type**	Functional – Positive
Priority	Medium - Verifies the stated functional requirement under normal conditions.	**Assigned to**	Pushkar (M3)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Group G1 exists and S1 is not a member.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Log in as S1. 2. Open Study Groups. 3. Select G1. 4. Select Join.		
Expected result	S1 is successfully added to G1.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-COL-03 - Duplicate membership prevented			
Requirements	REQ-COL-03		
Use case	UC-COL-01, UC-COL-02	**SAD reference**	Components: Study Group & Collaboration Service; API Layer | API: POST /groups/:id/join
Test level	System (UI + API)	**Test type**	Functional – Negative
Priority	High - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Pushkar (M3)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 is already a member of G1.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Log in as S1. 2. Attempt to join G1 again. 3. Repeat through the API.		
Expected result	Duplicate membership is prevented. The system informs the Student that membership already exists. Only one membership record exists. Rev 1.1 clarification: API response HTTP 409 ALREADY_MEMBER.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-COL-04 - View own study groups			
Requirements	REQ-COL-04		
Use case	UC-COL-01, UC-COL-02	**SAD reference**	Components: Study Group & Collaboration Service; API Layer | API: GET /groups
Test level	System (UI + API)	**Test type**	Functional – Positive
Priority	Medium - Verifies the stated functional requirement under normal conditions.	**Assigned to**	Pushkar (M3)
Required role	Student	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 belongs to G1 and G2.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Log in as S1.		

	2. Open Study Groups. 3. View My Groups.		
Expected result	G1 and G2 are displayed as groups belonging to S1.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0.

TC-COL-05 - Admin moderates content			
Requirements	REQ-COL-05		
Use case	UC-COL-01, UC-COL-02	**SAD reference**	Components: Study Group & Collaboration Service; API Layer | API: DELETE /collaboration/content/:id
Test level	System (UI + API)	**Test type**	Functional – Positive
Priority	High - Core use-case flow or integrity control; failure blocks the use case.	**Assigned to**	Pushkar (M3)
Required role	Admin	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Admin A1 is authenticated and collaborative content C1 exists.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Log in as A1. 2. Open moderation. 3. Select C1. 4. Select a valid moderation action. 5. Confirm.		
Expected result	The moderation action succeeds and the updated content state is stored. Rev 1.1 clarification: The only moderation action in the SAD contract is DELETE /api/collaboration/content/:id (HTTP 204); other actions are unspecified (OD-13) and are not tested. Content listing for moderation is undefined (OD-08).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: OD-08, OD-13.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-COL-06 - Non-Admin denied moderation			
Requirements	REQ-COL-06		
Use case	UC-COL-01, UC-COL-02	**SAD reference**	Components: Study Group & Collaboration Service; API Layer | API: DELETE /collaboration/content/:id
Test level	Security (API / system)	**Test type**	Security – Negative
Priority	High - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Pushkar (M3)
Required role	Faculty, Student, Admin	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 or F1 is authenticated but is not an Admin.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Attempt to access moderation through the UI. 2. Call the moderation endpoint directly. 3. Attempt a moderation operation.		
Expected result	Requests are rejected with HTTP 403 and "Access denied". Collaborative content remains unchanged. Rev 1.1 clarification: Add the unauthenticated case (HTTP 401). Non-Admin: HTTP 403 ACCESS_DENIED "Access denied".		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-COL-07 - Unavailable group/content reported without details			
Requirements	REQ-COL-07		
Use case	UC-COL-01, UC-COL-02	**SAD reference**	Components: Study Group & Collaboration Service; API Layer | API: POST /groups/:id/join;

			DELETE /collaboration/content/:id
Test level	System (UI + API)	**Test type**	Functional – Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Pushkar (M3)
Required role	Public / any	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	G3 or C2 has been removed or is unavailable.		
Test data	As stated in the preconditions and steps (v1.0 column "Test Steps and Test Data"); fixtures per s6.4.		
Steps	1. Request the unavailable group/content through the UI. 2. Repeat through the API.		
Expected result	The system reports that the requested item is unavailable and does not expose its details. Team-Level Quality Test Cases Rev 1.1 clarification: HTTP 404 with messages "Group unavailable" / "Content unavailable".		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: Retained from STP v1.0 (amended).

TC-COL-08 - Group creation requires a name			
Requirements	REQ-COL-01		
Use case	UC-COL-01	**SAD reference**	Components: Study Group & Collaboration Service; API Layer | API: POST /groups
Test level	Integration (API)	**Test type**	Functional - Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Pushkar (M3)
Required role	Student (S1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 is logged in.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. POST /api/groups with name "" ; then with name omitted; then with only a description. 2. POST /api/groups with a valid name and no description.		
Expected result	Step 1: HTTP 400 VALIDATION_ERROR naming the name field; no group created. Step 2: HTTP 201 {id, name, description, createdBy}; description is optional.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-COL-09 - Group field length limits			
Requirements	REQ-COL-01		
Use case	UC-COL-01	**SAD reference**	Components: Study Group & Collaboration Service; API Layer | API: POST /groups
Test level	Integration (API)	**Test type**	Functional - Boundary
Priority	Low - Boundary values defined in the specification are a common defect location.	**Assigned to**	Pushkar (M3)
Required role	Student (S1)	**Readiness**	Blocked - expected result not agreed
Preconditions	S1 is logged in.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Create groups with names and descriptions of increasing length up to a large size.		
Expected result	BLOCKED (OD-13): the SRS/SAD do not specify group-name or description length limits. Limits must be defined before boundary values can be tested.		
Pass / fail criteria	Cannot pass or fail until the linked decision is recorded (Blocked). When the decision is approved, update the expected result and re-baseline.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: OD-13.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-COL-10 - Only Students can create and join groups			
Requirements	REQ-COL-01, REQ-COL-02		
Use case	UC-COL-01	**SAD reference**	Components: Study Group & Collaboration Service; API Layer | API: POST /groups; POST /groups/:id/join
Test level	Security (API)	**Test type**	Security - Negative

Priority	Medium - Protects a security requirement; a defect exposes data or privileges.	**Assigned to**	Pushkar (M3)
Required role	Faculty (F1), Admin (A1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Group G1 exists.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. POST /api/groups as F1 and as A1. 2. POST /api/groups/G1/join as F1. 3. Repeat step 1 without a token.		
Expected result	Steps 1-2: HTTP 403 ACCESS_DENIED; no group or membership created. Step 3: HTTP 401.		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-COL-11 - Concurrent duplicate join			
Requirements	REQ-COL-03		
Use case	UC-COL-01	**SAD reference**	Components: Study Group & Collaboration Service; API Layer | API: POST /groups/:id/join
Test level	Integration (API)	**Test type**	Functional - Concurrency
Priority	High - Race conditions can break one-attempt/one-membership/immutability rules.	**Assigned to**	Pushkar (M3)
Required role	Student (S2)	**Readiness**	Environment-dependent (load/tooling/devices/timing)
Preconditions	S2 is not a member of G1.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Send two simultaneous POST /api/groups/G1/join as S2. 2. Count GroupMember records for S2 and G1.		
Expected result	One HTTP 201 and one HTTP 409 ALREADY_MEMBER; exactly one membership record.		
Pass / fail criteria	Pass only when every stated threshold/condition is met in the recorded run with attached evidence; thresholds not met = Fail; not run = Not executed.		
Dependencies	Script able to send parallel requests.. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-COL-12 - My Groups lists only own groups; empty list			
Requirements	REQ-COL-04		
Use case	UC-COL-01	**SAD reference**	Components: Study Group & Collaboration Service; API Layer | API: GET /groups
Test level	Integration (API)	**Test type**	Functional - Positive / Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Pushkar (M3)
Required role	Student (S1, S3)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	S1 belongs to G1 and G2; S2 alone belongs to G4; S3 belongs to no group.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. GET /api/groups as S1. 2. GET /api/groups as S3.		
Expected result	Step 1: G1 and G2 only (G4 absent). Step 2: HTTP 200 with an empty list; the UI shows an empty state without an error (wording unspecified).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-COL-13 - Moderation: list content and available actions			
Requirements	REQ-COL-05		
Use case	UC-COL-02	**SAD reference**	Components: Study Group & Collaboration Service; API Layer | API: DELETE /collaboration/content/:id
Test level	System (UI + API)	**Test type**	Functional - Positive
Priority	Medium - Verifies the stated functional	**Assigned to**	Pushkar (M3)

	requirement under normal conditions.		
Required role	Admin (A1)	**Readiness**	Blocked - expected result not agreed
Preconditions	Collaborative content C1 exists.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. Open the moderation screen. 2. List the collaborative content. 3. Inspect the available actions.		
Expected result	BLOCKED (OD-08, OD-13): no contract exists for listing collaborative content for moderation, the SRS does not list the moderation actions, and the collaborative-content fields are not defined.		
Pass / fail criteria	Cannot pass or fail until the linked decision is recorded (Blocked). When the decision is approved, update the expected result and re-baseline.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4. Open decisions: OD-08, OD-13.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

TC-COL-14 - Moderating content that is already removed			
Requirements	REQ-COL-05, REQ-COL-07		
Use case	UC-COL-02	**SAD reference**	Components: Study Group & Collaboration Service; API Layer | API: DELETE /collaboration/content/:id; POST /groups/:id/join
Test level	Integration (API)	**Test type**	Functional - Negative
Priority	Medium - Verifies rejection behaviour and error messages specified for invalid use.	**Assigned to**	Pushkar (M3)
Required role	Admin (A1)	**Readiness**	Ready - executable once a build is deployed (build status unverified)
Preconditions	Collaborative content C1 exists.		
Test data	As stated in the preconditions and steps; fixtures per s6.4.		
Steps	1. DELETE /api/collaboration/content/C1 as A1. 2. Repeat the request. 3. Check the server log.		
Expected result	Step 1: HTTP 204. Step 2: HTTP 404 "Content unavailable" with no details. Step 3: one moderation entry for step 1 (SAD s4.4).		
Pass / fail criteria	Pass when every expected-result statement is observed exactly (including status code and message where stated); any difference = Fail; unmet precondition = Blocked.		
Dependencies	Deployed build and fixtures only. All tests: build identified by commit SHA; test accounts per s6.4.		
Execution status	Not executed	**Evidence**	None (not executed). Origin: New in Rev 1.1.

A.12 Unit-test candidates (Jest)

Function names are candidates (OD-10). Status: specified, not written.

ID	Unit	Cases to cover	Requirements	Owner	Readiness	Origin
UT-ACC-01	validateRegistration(input)	Empty fields; invalid email; duplicate email (case-insensitive); password of 7 and 8 characters; missing uppercase/lowercase/digit.	REQ-ACC-02, REQ-ACC-03	Jagadeesh	Ready	Retained from STP v1.0
UT-ACC-02	authenticateUser(email, password)	Valid credentials; wrong password; unknown email; both failures return the identical "Invalid email or password".	REQ-ACC-04, REQ-ACC-05	Jagadeesh	Ready	Retained from STP v1.0
UT-ACC-03	isAccountLocked(user, now)	4 failures; 5th failure; just before 15 minutes; after 15 minutes; counter reset after successful login.	REQ-ACC-06	Jagadeesh	Ready	Retained from STP v1.0
UT-ACC-04	isSessionValid(session, now)	Active session; idle just under 30 minutes; idle over 30 minutes; logged-out session.	REQ-ACC-07	Jagadeesh	Ready	Retained from STP v1.0
UT-ACC-05	isRegistrationRoleAllowed(role)	Student and Faculty accepted; Admin, empty, undefined and unknown values rejected.	REQ-ACC-01, BR-08	Jagadeesh	Ready	New in Rev 1.1
UT-ACC-06	sanitizeInput(body)	Keys beginning with $ or containing . are removed or rejected; plain strings unchanged.	SEC-ACC-01, NFR-QUAL-03, DC-07	Jagadeesh	Ready	New in Rev 1.1
UT-COL-01	validateGroup(input)	Required group information present/missing	REQ-COL-01	Pushkar	Ready	Retained from STP v1.0
UT-COL-02	isAlreadyMember(groupId, userId)	Existing member, new member	REQ-COL-03	Pushkar	Ready	Retained from STP v1.0
UT-COL-03	canModerateContent(user)	Admin allowed; Student/Faculty denied	REQ-COL-05, REQ-COL-06	Pushkar	Ready	Retained from STP v1.0

ID	Unit	Cases to cover	Requirements	Owner	Readiness	Origin
UT-COL-04	canJoinOrCreateGroup(user)	Student allowed; Faculty and Admin denied.	REQ-COL-01, REQ-COL-02	Pushkar	Ready	New in Rev 1.1
UT-MAT-01	validateMaterial(input)	Title 100/101; description 500/501; missing fields; PDF/DOCX/PPTX/PNG; TXT; 25 MB and just above 25 MB.	REQ-MAT-01, REQ-MAT-02	Vedh	Ready	Retained from STP v1.0
UT-MAT-02	canAccessMaterial(user, material, grant)	Owner; granted user; no grant; revoked grant; removed material.	REQ-MAT-04, SEC-MAT-01	Vedh	Ready	Retained from STP v1.0
UT-MAT-03	canManageMaterial(user, material)	Owner allowed; granted non-owner and unrelated user denied for share, update, remove and revoke.	REQ-MAT-05, REQ-MAT-07, REQ-MAT-08, REQ-MAT-09, SEC-MAT-02, SEC-MAT-03	Vedh	Ready	Retained from STP v1.0
UT-MAT-04	validateProfileUpdate(input)	Empty name; picture of exactly 2 MB and above 2 MB; email/role fields in input are ignored or rejected.	REQ-ACC-08, REQ-ACC-09	Jagadeesh	Ready	Retained from STP v1.0
UT-MAT-05	sha256OfFile(path) / verifyChecksum(stored, file)	Identical file; one-byte difference; empty file.	NFR-SAFE-03	Vedh	Ready	New in Rev 1.1
UT-MAT-06	validateShareTarget(userId, users)	Missing; malformed; unknown; registered user.	REQ-MAT-05, BR-05	Vedh	Ready	New in Rev 1.1
UT-PA-01	calculatePerformance(attempts)	Completed attempts, score and percentage calculation, no attempts	REQ-PA-02	Pushkar	Ready	Retained from STP v1.0
UT-PA-02	groupResultsByTopic(results)	Multiple topics, multiple results per topic, no topic data	REQ-PA-03	Pushkar	Blocked (OD-03)	Retained from STP v1.0
UT-PA-03	canViewClassPerformance(user, quiz)	Quiz owner allowed; non-owner Faculty denied	REQ-PA-04, REQ-PA-06	Pushkar	Ready	Retained from STP v1.0
UT-QZ-01	GradingService.grade(attempt, quizCopy)	All correct, all wrong, all blank, mixed; 1 and 50 questions; percentage rounded to one decimal.	REQ-QZ-11	Vedh	Ready	Retained from STP v1.0
UT-QZ-02	validateQuestion(input)	Empty fields, length limits (500/200/50/300), duplicate options, correct option outside A–D.	REQ-QB-02	Alifiya	Ready	Retained from STP v1.0
UT-QZ-03	validateQuiz(input)	Title, 1–50 questions, time limit 5–180, end later than start.	REQ-QZ-02	Alifiya	Ready	Retained from STP v1.0
UT-QZ-04	canStartAttempt(quiz, student, now)	Before window, at start, at end, after end, already attempted.	REQ-QZ-07	Vedh	Ready	Retained from STP v1.0
UT-QZ-05	isAnswerAccepted(attempt, now)	Before deadline, at deadline, after deadline, after submission.	REQ-QZ-10, REQ-QZ-14	Vedh	Ready	Retained from STP v1.0
UT-QZ-06	toStudentQuestionDTO(question)	Output never contains correctOption or explanation.	SEC-QZ-02	Vedh	Ready	Retained from STP v1.0
UT-QZ-07	copyQuestionsAtPublish(quiz, questions)	All fields copied (text, options, correct option, explanation, source id); later edits or removal of the bank question do not change the copy.	REQ-QZ-04, REQ-QB-06	Alifiya	Ready	New in Rev 1.1
UT-QZ-08	canViewResult(user, attempt) / canViewSubmissions(user, quiz)	Attempt owner; other Student; quiz owner; other Faculty; Admin.	SEC-QZ-03, REQ-QZ-17	Alifiya	Ready	New in Rev 1.1
UT-QZ-09	computeDeadline(start, timeLimit, windowEnd)	Time limit inside window; limit extending beyond window end. BLOCKED (TOD-04): expected deadline rule not agreed.	REQ-QZ-10, SEC-QZ-04	Vedh	Blocked (TOD-04)	New in Rev 1.1

Annex B. Test-Case Inventory

Counts: 160 test cases (97 retained, 63 new) + 28 unit candidates. Owner = SRS Appendix D primary owner of the test's first-listed requirement (four members). Status of every test: Not executed.

ID	Title	Requirements	Level / type	Prio	Owner	Ready	Spr.	Automation	Regression	Origin	Open decision
TC-ACC-01	Register a new Student account	REQ-ACC-01	System (UI + API) / Functional – Positive	High	Jagadesh	R1	S1	Cypress or Selenium + Postman	Smoke	Retained (amended)	-
TC-ACC-02	Registration: valid, duplicate (case-insensitive), empty and invalid-email inputs	REQ-ACC-01, REQ-ACC-02	System (UI + API) / Functional – Positive/Negative	High	Jagadesh	R1	S1	Cypress or Selenium + Postman	Smoke	Retained (amended)	-
TC-ACC-03	Registration validation: each required field empty; invalid e-mail formats; duplicate e-mail with different capitalisation	REQ-ACC-02	System (UI + API) / Functional – Boundary	Medium	Jagadesh	R1	S1	Cypress or Selenium + Postman	Full regression	Retained (amended)	-
TC-ACC-04	Password rule boundary (8 vs 7 characters; missing character classes)	REQ-ACC-03	System (UI + API) / Functional – Boundary	High	Jagadesh	R1	S1	Cypress or Selenium + Postman	Smoke	Retained (amended)	-
TC-ACC-05	Valid log-in grants access within 2 seconds	REQ-ACC-04	Performance (JMeter / timing) / Functional / Performance	High	Jagadesh	R2	S1	JMeter / timing script	Full regression	Retained (amended)	-
TC-ACC-06	Invalid credentials return the single generic message	REQ-ACC-05, SEC-ACC-02	System (UI + API) / Functional – Negative	High	Jagadesh	R1	S1	Cypress or Selenium + Postman	Smoke	Retained (amended)	-
TC-ACC-07	Account lock-out after five consecutive failed log-ins (15 minutes)	REQ-ACC-06, SEC-ACC-06, DC-08	Security (API / system) / Security – Negative	High	Jagadesh	R1	S1	Jest + Supertest / Postman	Smoke	Retained (amended)	TOD-04
TC-ACC-08	Session ends on logout and after 30 minutes of inactivity	REQ-ACC-07	Security (API / system) / Security – Negative	High	Jagadesh	R1	S1	Jest + Supertest / Postman	Smoke	Retained (amended)	-
TC-ACC-09	View and update permitted profile fields; e-mail and role protected	REQ-ACC-08, BR-06, DC-06	System (UI + API) / Functional – Positive	Medium	Jagadesh	R1	S1	Cypress or Selenium + Postman	Full regression	Retained (amended)	OD-13
TC-ACC-10	Profile validation: empty name; picture at and above 2 MB	REQ-ACC-09, DC-06	System (UI + API) / Functional – Negative/Boundary	Medium	Jagadesh	R1	S1	Cypress or Selenium + Postman	Full regression	Retained (amended)	TOD-06
TC-ACC-11	Protected endpoints reject missing authentication and unauthorized roles	REQ-ACC-10, BR-01	Security (API / system) / Security – Negative	High	Jagadesh	R1	S1	Jest + Supertest / Postman	Smoke	Retained (amended)	-
TC-ACC-12	Registration rejects the Admin role (BR-08)	REQ-ACC-01, BR-08	Integration (API) / Security - Negative	High	Jagadesh	R1	S1	Jest + Supertest / Postman	Smoke	New (1.1)	OD-11
TC-ACC-13	Registration length boundaries (full name, e-mail)	REQ-ACC-02	Integration (API) / Functional - Boundary	Medium	Jagadesh	R1	S1	Jest + Supertest / Postman	Full regression	New (1.1)	-
TC-ACC-14	View own profile	REQ-ACC-08, REQ-ACC-10, SEC-ACC-03	System (UI + API) / Functional - Positive	Medium	Jagadesh	R1	S1	Cypress or Selenium + Postman	Full regression	New (1.1)	-
TC-ACC-15	Profile text-field boundaries	REQ-ACC-09	Integration (API) / Functional - Boundary	Medium	Jagadesh	R1	S1	Jest + Supertest / Postman	Full regression	New (1.1)	-

ID	Title	Requirements	Level / type	Prio	Owner	Ready	Spr.	Automation	Regression	Origin	Open decision
TC-ACC-16	Profile picture: accepted file types	REQ-ACC-09, REQ-ACC-08	Integration (API) / Functional - Boundary	Medium	Jagadesh	R3	S1	Jest + Supertest / Postman	After decision	New (1.1)	OD-13
TC-ACC-17	Inactivity timer restarts on activity (sliding 30-minute expiry)	REQ-ACC-07, SEC-ACC-04, DC-08	Security (API) / Security - Boundary	High	Jagadesh	R1	S1	Jest + Supertest / Postman	Smoke	New (1.1)	-
TC-ACC-18	Log-in request validation and success contract	REQ-ACC-04, REQ-ACC-05	Integration (API) / Functional - Positive / Negative	Medium	Jagadesh	R1	S1	Jest + Supertest / Postman	Full regression	New (1.1)	-
TC-ACC-19	Seeded Admin can authenticate and is role-restricted	REQ-ACC-10, BR-08	Security (API) / Security - Negative	Medium	Jagadesh	R1	S1	Jest + Supertest / Postman	Full regression	New (1.1)	OD-11
TC-ACC-20	Lock-out is isolated to the failing account	REQ-ACC-06, SEC-ACC-06	Security (API) / Security - Negative	Medium	Jagadesh	R1	S1	Jest + Supertest / Postman	Full regression	New (1.1)	-
TC-MAT-01	Upload a valid material	REQ-MAT-01, BR-02	System (UI + API) / Functional – Positive	High	Vedh	R1	S1	Cypress or Selenium + Postman	Smoke	Retained (amended)	OD-07
TC-MAT-02	Material metadata length boundaries (title 100, description 500)	REQ-MAT-01	System (UI + API) / Functional – Boundary	Medium	Vedh	R1	S1	Cypress or Selenium + Postman	Full regression	Retained (amended)	-
TC-MAT-03	Supported and unsupported file types; oversized file	REQ-MAT-02, DC-06	System (UI + API) / Functional – Negative	High	Vedh	R1	S1	Cypress or Selenium + Postman	Smoke	Retained (amended)	-
TC-MAT-04	25 MB size boundary	REQ-MAT-02, DC-06	System (UI + API) / Functional – Boundary	High	Vedh	R1	S1	Cypress or Selenium + Postman	Smoke	Retained (amended)	TOD-06
TC-MAT-05	Uploaded material visible in owner library within 3 seconds	REQ-MAT-03	Performance (JMeter / timing) / Functional / Performance	Medium	Vedh	R2	S1	JMeter / timing script	Full regression	Retained	-
TC-MAT-06	User without access cannot view material	REQ-MAT-04	Security (API / system) / Security – Negative	High	Vedh	R1	S1	Jest + Supertest / Postman	Smoke	Retained (amended)	-
TC-MAT-07	Owner shares material with a registered user	REQ-MAT-05, BR-05, BR-03	System (UI + API) / Functional – Positive	High	Vedh	R1	S1	Cypress or Selenium + Postman	Smoke	Retained (amended)	-
TC-MAT-08	Authorized user downloads material; download starts within 3 seconds	REQ-MAT-06	Performance (JMeter / timing) / Functional / Performance	High	Vedh	R2	S1	JMeter / timing script	Full regression	Retained	-
TC-MAT-09	Owner updates metadata and replaces the file	REQ-MAT-07	System (UI + API) / Functional – Positive/Negative	Medium	Vedh	R1	S1	Cypress or Selenium + Postman	Full regression	Retained	-
TC-MAT-10	Owner removes material after confirmation	REQ-MAT-08	System (UI + API) / Functional – Positive/Negative	High	Vedh	R1	S1	Cypress or Selenium + Postman	Smoke	Retained (amended)	-
TC-MAT-11	Owner revokes access; next request denied	REQ-MAT-09	Security (API / system) / Security – Negative	High	Vedh	R1	S1	Jest + Supertest / Postman	Smoke	Retained (amended)	-
TC-MAT-12	Non-owner share/update/remove/revoke rejected with "Access denied"	REQ-MAT-10, BR-03, BR-04, SEC-MAT-02, SEC-MAT-03	Security (API / system) / Security – Negative	High	Vedh	R1	S1	Jest + Supertest / Postman	Smoke	Retained (amended)	-
TC-MAT-13	Removed or unretrievable material reported as unavailable	REQ-MAT-11	System (UI + API) / Functional – Negative	Medium	Vedh	R1	S1	Cypress or Selenium + Postman	Full regression	Retained (amended)	-
TC-MAT-14	Admin upload (role scope)	REQ-MAT-01	Integration (API) / Functional - Negative	Medium	Vedh	R3	S1	Jest + Supertest / Postman	After decision	New (1.1)	OD-07
TC-MAT-15	Library lists only owned and shared materials; pagination	REQ-MAT-03, REQ-MAT-04	Integration (API) / Functional - Positive	Medium	Vedh	R1	S1	Jest + Supertest / Postman	Full regression	New (1.1)	-
TC-MAT-16	Upload with missing metadata or file	REQ-MAT-01, REQ-MAT-02, NFR-QUAL-03	Integration (API) / Functional - Negative	Medium	Vedh	R1	S1	Jest + Supertest / Postman	Full regression	New (1.1)	-
TC-MAT-17	Share target validation	REQ-MAT-05, BR-05	Integration (API) / Functional - Negative	Medium	Vedh	R1	S1	Jest + Supertest / Postman	Full regression	New (1.1)	-
TC-MAT-18	Owner lists users who hold access	REQ-MAT-09	System (UI + API) / Functional - Positive	Medium	Vedh	R3	S1	Cypress or Selenium + Postman	After decision	New (1.1)	OD-08
TC-MAT-19	File replacement serves only the new file	REQ-MAT-07, NFR-SAFE-03	Integration (API) / Functional - Positive	High	Vedh	R1	S1	Jest + Supertest / Postman	Smoke	New (1.1)	-
TC-MAT-20	Revocation edge cases	REQ-MAT-09	Integration (API) / Functional - Negative	Medium	Vedh	R1	S1	Jest + Supertest / Postman	Full regression	New (1.1)	-

ID	Title	Requirements	Level / type	Prio	Owner	Ready	Spr.	Automation	Regression	Origin	Open decision
TC-MAT-21	Library shows owner name and access status	REQ-MAT-04	System (UI + API) / Functional - Positive	Low	Vedh	R3	S1	Cypress or Selenium + Postman	After decision	New (1.1)	OD-13
TC-MAT-22	Student can upload and own material	REQ-MAT-01, REQ-MAT-03, BR-02	System (UI + API) / Functional - Positive	Medium	Vedh	R1	S1	Cypress or Selenium + Postman	Full regression	New (1.1)	-
TC-AS-01	Lock-out counter reset, lock-out and release after 15 minutes	SEC-ACC-06, REQ-ACC-06, DC-08	Security (API / system) / Security – Negative	High	Jagadesh	R1	S1	Jest + Supertest / Postman	Smoke	Retained (amended)	-
TC-MS-01	Direct material access without authorization denied	SEC-MAT-01, REQ-MAT-04	Security (API / system) / Security – Negative	High	Vedh	R1	S1	Jest + Supertest / Postman	Smoke	Retained	-
TC-AS-02	Logged-out and idle tokens cannot be reused	SEC-ACC-04, REQ-ACC-07, DC-08	Security (API / system) / Security – Negative	High	Jagadesh	R1	S1	Jest + Supertest / Postman	Smoke	Retained (amended)	-
TC-MS-02	Revoked user denied on the very next request	SEC-MAT-04, REQ-MAT-09	Security (API / system) / Security – Negative	High	Vedh	R1	S1	Jest + Supertest / Postman	Smoke	Retained	-
TC-AS-03	Unauthenticated and malformed-token requests rejected	SEC-ACC-01, REQ-ACC-10, BR-01	Security (API / system) / Security – Negative	High	Jagadesh	R1	S1	Jest + Supertest / Postman	Smoke	Retained (amended)	-
TC-MS-03	Removed material not retrievable by any user or URL	SEC-MAT-04, REQ-MAT-11	Security (API / system) / Security – Negative	High	Vedh	R1	S1	Jest + Supertest / Postman	Smoke	Retained	-
TC-AS-04	Passwords stored only as salted bcrypt hashes	SEC-ACC-03, DC-04	Security (API / system) / Security – Inspection	High	Jagadesh	R1	S1	Jest + Supertest / Postman	Smoke	Retained (amended)	-
TC-MS-04	Non-owner cannot share or revoke	SEC-MAT-02, REQ-MAT-10, BR-03, BR-04	Security (API / system) / Security – Negative	High	Vedh	R1	S1	Jest + Supertest / Postman	Smoke	Retained (amended)	-
TC-AS-05	Cross-user profile update and e-mail/role change rejected	SEC-ACC-05	Security (API / system) / Security – Negative	High	Jagadesh	R1	S1	Jest + Supertest / Postman	Smoke	Retained (amended)	-
TC-MS-05	Non-owner cannot update or remove	SEC-MAT-03, REQ-MAT-07, REQ-MAT-08, REQ-MAT-10, BR-03, BR-04	Security (API / system) / Security – Negative	High	Vedh	R1	S1	Jest + Supertest / Postman	Smoke	Retained (amended)	-
TC-AS-06	TLS 1.2+ enforced; plain HTTP and TLS 1.0/1.1 refused	SEC-COM-01, DC-03, COM-01	Security (API / system) / Security – Negative	High	Jagadesh	R2	S1	Jest + Supertest / Postman	Full regression	Retained (amended)	TOD-10
TC-AS-07	NoSQL-operator injection on authentication endpoints	SEC-ACC-01, SEC-ACC-02, NFR-QUAL-03, DC-07	Security (API) / Security - Negative	High	Jagadesh	R1	S1	Jest + Supertest / Postman	Smoke	New (1.1)	-
TC-AS-08	Security events logged without sensitive data	SEC-ACC-01, SEC-ACC-06, SEC-MAT-01	Security (API) / Security - Inspection	High	Jagadesh	R1	S1	Jest + Supertest / Postman	Smoke	New (1.1)	OD-06
TC-AS-09	Rate limiting returns HTTP 429	NFR-PERF-02, NFR-QUAL-03	Security (API) / Security - Negative	Medium	Jagadesh	R3	S1	Jest + Supertest / Postman	After decision	New (1.1)	TOD-02
TC-AP-01	200 concurrent users: 95% of page loads within 3 s	NFR-PERF-01	Performance (JMeter / timing) / Performance (NFR)	High	Jagadesh	R2	S2	JMeter / timing script	Full regression	Retained	TOD-10
TC-MP-01	Upload of 25 MB file visible within 3 s (10 runs)	REQ-MAT-03	Performance (JMeter / timing) / Performance (NFR)	Medium	Vedh	R2	S2	JMeter / timing script	Full regression	Retained	-
TC-AP-02	200 concurrent users: fewer than 1% failed requests	NFR-PERF-02	Performance (JMeter / timing) / Performance (NFR)	High	Jagadesh	R2	S2	JMeter / timing script	Full regression	Retained	TOD-10
TC-MP-02	Download of 25 MB file starts within 3 s (10 runs)	REQ-MAT-06	Performance (JMeter / timing) / Performance (NFR)	Medium	Vedh	R2	S2	JMeter / timing script	Full regression	Retained	-
TC-MP-03	Share and revocation effective within 2 s	NFR-PERF-03	Performance (JMeter / timing) / Performance (NFR)	High	Vedh	R2	S2	JMeter / timing script	Full regression	Retained	-

ID	Title	Requirements	Level / type	Prio	Owner	Ready	Spr.	Automation	Regression	Origin	Open decision
TC-AP-03	Mixed workload with 200 users including quiz submission	NFR-PERF-01, NFR-PERF-02, NFR-PERF-04	Performance (JMeter) / Performance (NFR)	Medium	Jagadesh	R2	S2	JMeter / timing script	Full regression	New (1.1)	TOD-10
TC-AN-01	Daily backup and restore (backup no older than 24 h)	NFR-SAFE-01	System (environment) / Non-functional – Safety (backup/restore)	High	Jagadesh	R2	S2	Script + environment	Full regression	Retained	OD-06
TC-AN-02	Interrupted upload/update/removal leaves previous state	NFR-SAFE-02	System (environment) / Non-functional – Safety (Negative)	High	Vedh	R2	S2	Script + environment	Full regression	Retained	-
TC-AN-03	Downloaded file identical to uploaded file (SHA-256)	NFR-SAFE-03	System (environment) / Non-functional – Data integrity	High	Vedh	R1	S2	Cypress or Selenium + Postman	Smoke	Retained	-
TC-AN-04	First-time users complete register, log in, upload in under 5 minutes	NFR-QUAL-01	Usability (manual) / Non-functional – Usability	Medium	Jagadesh	R4	S2	Manual / review	Manual	Retained	-
TC-AN-05	Invalid input and unsupported files rejected without crash	NFR-QUAL-03	System (environment) / Non-functional – Robustness (Negative)	Medium	Jagadesh	R1	S2	Cypress or Selenium + Postman	Full regression	Retained (amended)	-
TC-AN-06	Browser and device compatibility	NFR-QUAL-04, DC-05	Compatibility (manual) / Non-functional – Portability	Medium	Jagadesh	R2	S2	Manual / review	Full regression	Retained (amended)	TOD-10
TC-AN-07	Availability measured over the test period	NFR-QUAL-02	System (environment) / Non-functional – Availability	Medium	Pushkar	R2	S2	Script + environment	Full regression	Retained (amended)	TOD-07, OD-06
TC-AN-08	Feature areas independently testable	NFR-QUAL-05	Review / inspection / Non-functional – Modularity	Low	Pushkar	R4	S2	Manual / review	Manual	Retained (amended)	
TC-AN-09	Every requirement traceable to a test case	NFR-QUAL-06, DC-10	Review / inspection / Non-functional – Traceability	High	Pushkar	R4	S2	Manual / review	Manual	Retained (amended)	
TC-AN-10	Technology stack conformance	DC-01, DC-04	Review / inspection / Non-functional - Inspection	Low	Jagadesh	R4	S1	Manual / review	Manual	New (1.1)	-
TC-AN-11	API contract conformance	DC-02, NFR-QUAL-03	Integration (API) / Non-functional - Contract	Medium	Alifiya	R1	S1	Jest + Supertest / Postman	Full regression	New (1.1)	-
TC-AN-12	Server-side enforcement independent of the client	DC-07, DC-08, NFR-QUAL-03	Security (API) / Security - Negative	High	Jagadesh	R1	S1	Jest + Supertest / Postman	Smoke	New (1.1)	-
TC-AN-13	Central error handling hides internals	NFR-QUAL-03	Security (API) / Non-functional - Robustness	Medium	Jagadesh	R2	S2	Jest + Supertest / Postman	Full regression	New (1.1)	-
TC-AN-14	Domain requirement: English user messages	DOM-NOID	Review / inspection / Non-functional - Inspection	Low	Pushkar	R4	S2	Manual / review	Manual	New (1.1)	-
TC-AN-15	UI inventory and role-based navigation	UI-01..18	System (UI) / Functional - Inspection	Medium	Pushkar	R4	S2	Manual / review	Manual	New (1.1)	TOD-12
TC-AN-16	Active attempt survives a server restart	NFR-QUAL-03, SEC-QZ-04	System (environment) / Non-functional - Recovery	Medium	Jagadesh	R2	S2	Script + environment	Full regression	New (1.1)	-
TC-AN-17	Concurrent update and removal of a material	NFR-SAFE-02, REQ-MAT-07, REQ-MAT-08	Integration (API) / Non-functional - Concurrency	Medium	Vedh	R2	S2	Jest + Supertest / Postman	Full regression	New (1.1)	-
TC-QB-01	Add a valid question	REQ-QB-01, REQ-QB-03	System (UI + API) / Functional – Positive	High	Alifiya	R1	S1	Cypress or Selenium + Postman	Smoke	Retained (amended)	-
TC-QB-02	Question validation: empty text, topic, correct option	REQ-QB-02	System (UI + API) / Functional – Negative	Medium	Alifiya	R1	S1	Cypress or Selenium + Postman	Full regression	Retained (amended)	-

ID	Title	Requirements	Level / type	Prio	Owner	Ready	Spr.	Automation	Regression	Origin	Open decision
TC-QB-03	Question field-length boundaries	REQ-QB-01, REQ-QB-02	System (UI + API) / Functional – Boundary	Medium	Alifiya	R1	S1	Cypress or Selenium + Postman	Full regression	Retained	-
TC-QB-04	Four distinct options required	REQ-QB-02	System (UI + API) / Functional – Negative	Medium	Alifiya	R1	S1	Cypress or Selenium + Postman	Full regression	Retained	-
TC-QB-05	Search and filter question bank	REQ-QB-04	System (UI + API) / Functional – Positive	Medium	Alifiya	R1	S1	Cypress or Selenium + Postman	Full regression	Retained	-
TC-QB-06	Edit own question (valid and invalid)	REQ-QB-05, REQ-QB-02	System (UI + API) / Functional – Positive / Negative	Medium	Alifiya	R1	S1	Cypress or Selenium + Postman	Full regression	Retained	-
TC-QB-07	Remove question; published quiz keeps its copy	REQ-QB-06, REQ-QZ-04	System (UI + API) / Functional – Positive	High	Alifiya	R1	S1	Cypress or Selenium + Postman	Smoke	Retained	-
TC-QB-08	Student denied question-bank access	REQ-QB-07, SEC-QZ-01	Security (API / system) / Security – Negative	High	Alifiya	R1	S1	Jest + Supertest / Postman	Smoke	Retained (amended)	-
TC-QB-09	Non-owner Faculty cannot edit or remove question	REQ-QB-07, SEC-QZ-01	Security (API / system) / Security – Negative	High	Alifiya	R1	S1	Jest + Supertest / Postman	Smoke	Retained	-
TC-QB-10	Question validation: difficulty, topic, explanation, option count, correct option	REQ-QB-02	Integration (API) / Functional - Negative	Medium	Alifiya	R1	S1	Jest + Supertest / Postman	Full regression	New (1.1)	-
TC-QB-11	Search and filter combinations and paging	REQ-QB-04	Integration (API) / Functional - Positive	Medium	Alifiya	R1	S1	Jest + Supertest / Postman	Full regression	New (1.1)	-
TC-QB-12	Unauthenticated access to question endpoints	REQ-QB-07, REQ-ACC-10, SEC-QZ-01	Security (API) / Security - Negative	High	Alifiya	R1	S1	Jest + Supertest / Postman	Smoke	New (1.1)	-
TC-QZ-01	Create quiz as Draft	REQ-QZ-01, REQ-QZ-03	System (UI + API) / Functional – Positive	High	Alifiya	R1	S1	Cypress or Selenium + Postman	Smoke	Retained	-
TC-QZ-02	Quiz validation failures	REQ-QZ-02	System (UI + API) / Functional – Negative	Medium	Alifiya	R1	S1	Cypress or Selenium + Postman	Full regression	Retained	-
TC-QZ-03	Quiz boundary values (1/50 questions, 5/180 minutes, title 100, description 300)	REQ-QZ-01, REQ-QZ-02	System (UI + API) / Functional – Boundary	Medium	Alifiya	R1	S1	Cypress or Selenium + Postman	Full regression	Retained	-
TC-QZ-04	Publish quiz; question copy independent of bank edits	REQ-QZ-03, REQ-QZ-04	System (UI + API) / Functional – Positive	High	Alifiya	R1	S1	Cypress or Selenium + Postman	Smoke	Retained	-
TC-QZ-05	Edit/delete Draft; edit/delete rejected after an attempt has started	REQ-QZ-05	System (UI + API) / Functional – Negative	Medium	Alifiya	R1	S1	Cypress or Selenium + Postman	Full regression	Retained (amended)	TOD-03
TC-QZ-06	Student quiz list contains no question data	REQ-QZ-06, SEC-QZ-02	System (UI + API) / Functional – Positive	Medium	Vedh	R1	S2	Cypress or Selenium + Postman	Full regression	Retained	-
TC-QZ-07	Attempt start rules: window and one attempt per Student	REQ-QZ-07	System (UI + API) / Functional – Negative / Boundary	High	Vedh	R1	S2	Cypress or Selenium + Postman	Smoke	Retained (amended)	OD-12, TOD-04
TC-QZ-08	Attempt start shows questions with timer and no answer key	REQ-QZ-08, SEC-QZ-02	Security (API / system) / Functional / Security	High	Vedh	R1	S2	Jest + Supertest / Postman	Smoke	Retained	-
TC-QZ-09	Answer selection, unanswered warning and submission	REQ-QZ-09	System (UI + API) / Functional – Positive	Medium	Vedh	R1	S2	Cypress or Selenium + Postman	Full regression	Retained	-
TC-QZ-10	Auto-submit at time limit; late answer rejected	REQ-QZ-10, SEC-QZ-04	System (UI + API) / Functional – Boundary	High	Vedh	R1	S2	Cypress or Selenium + Postman	Smoke	Retained (amended)	TOD-04
TC-QZ-11	Grading correctness (6/10, 10/10, 0/10, 1/1)	REQ-QZ-11	System (UI + API) / Functional – Positive / Boundary	High	Vedh	R1	S2	Cypress or Selenium + Postman	Smoke	Retained	-
TC-QZ-12	Result details after submission	REQ-QZ-13	System (UI + API) / Functional – Positive	High	Vedh	R1	S2	Cypress or Selenium + Postman	Smoke	Retained	-
TC-QZ-13	Submitted attempt is immutable	REQ-QZ-14, SEC-QZ-05	Security (API / system) / Security – Negative	High	Vedh	R1	S2	Jest + Supertest / Postman	Smoke	Retained (amended)	-
TC-QZ-14	Student views own attempts	REQ-QZ-15	System (UI + API) / Functional – Positive	Medium	Vedh	R1	S2	Cypress or Selenium + Postman	Full regression	Retained	-
TC-QZ-15	Faculty views quiz submissions	REQ-QZ-16	System (UI + API) / Functional – Positive / Negative	Medium	Alifiya	R1	S2	Cypress or Selenium + Postman	Full regression	Retained	OD-08
TC-QZ-16	Result and submission privacy; Faculty cannot attempt	REQ-QZ-17, SEC-QZ-03	Security (API / system) / Security – Negative	High	Alifiya	R1	S2	Jest + Supertest / Postman	Smoke	Retained (amended)	-
TC-QZ-17	Unauthenticated and Student access to quiz management denied	REQ-QZ-17, SEC-QZ-01	Security (API / system) / Security – Negative	High	Alifiya	R1	S2	Jest + Supertest / Postman	Smoke	Retained (amended)	-

ID	Title	Requirements	Level / type	Prio	Owner	Ready	Spr.	Automation	Regression	Origin	Open decision
TC-QZ-18	Unavailable quiz, attempt or question handled without details	REQ-QZ-18	System (UI + API) / Functional – Negative	Medium	Alifiya	R1	S2	Cypress or Selenium + Postman	Full regression	Retained (amended)	-
TC-QZ-19	Publish after the availability end has passed	REQ-QZ-04, REQ-QZ-02	Integration (API) / Functional - Negative	Medium	Alifiya	R1	S1	Jest + Supertest / Postman	Full regression	New (1.1)	-
TC-QZ-20	Publish or create with a question removed from the bank	REQ-QZ-18, REQ-QZ-04	Integration (API) / Functional - Negative	Medium	Alifiya	R1	S1	Jest + Supertest / Postman	Full regression	New (1.1)	TOD-11
TC-QZ-21	Availability window boundary	REQ-QZ-02	Integration (API) / Functional - Boundary	Medium	Alifiya	R1	S1	Jest + Supertest / Postman	Full regression	New (1.1)	-
TC-QZ-22	Editing or deleting a Published quiz that has no attempts	REQ-QZ-05	Integration (API) / Functional - Negative	Medium	Alifiya	R3	S1	Jest + Supertest / Postman	After decision	New (1.1)	TOD-03
TC-QZ-23	Answer-save validation	REQ-QZ-09	Integration (API) / Functional - Negative	Medium	Vedh	R1	S2	Jest + Supertest / Postman	Full regression	New (1.1)	-
TC-QZ-24	Attempt ownership and existence on attempt endpoints	REQ-QZ-17, REQ-QZ-18, SEC-QZ-03	Security (API) / Security - Negative	High	Alifiya	R1	S2	Jest + Supertest / Postman	Smoke	New (1.1)	-
TC-QZ-25	Auto-submission when the Student stops interacting	REQ-QZ-10, SEC-QZ-04	System (UI + API) / Functional - Boundary	High	Vedh	R1	S2	Cypress or Selenium + Postman	Smoke	New (1.1)	TOD-04
TC-QZ-26	Grading failure: pending result and retry	REQ-QZ-11, REQ-QZ-14	Integration (API) / Functional - Negative	Medium	Vedh	R3	S2	Jest + Supertest / Postman	After decision	New (1.1)	OD-08
TC-QZ-27	Result of an in-progress attempt exposes no answer key	SEC-QZ-02, REQ-QZ-13	Security (API) / Security - Negative	High	Vedh	R1	S2	Jest + Supertest / Postman	Smoke	New (1.1)	-
TC-QZ-28	Quiz owner cannot read a Student's attempt result	REQ-QZ-17, SEC-QZ-03	Security (API) / Security - Negative	High	Alifiya	R1	S2	Jest + Supertest / Postman	Smoke	New (1.1)	-
TC-QZ-29	Deadline when the time limit extends beyond the availability end	REQ-QZ-10, SEC-QZ-04	Integration (API) / Functional - Boundary	Medium	Vedh	R3	S2	Jest + Supertest / Postman	After decision	New (1.1)	TOD-04
TC-QZ-30	Concurrent start requests allow exactly one attempt	REQ-QZ-07, SEC-QZ-05	Integration (API) / Functional - Concurrency	High	Vedh	R2	S2	Jest + Supertest / Postman	Full regression	New (1.1)	-
TC-QZ-31	Concurrent submit requests are graded once	REQ-QZ-14, REQ-QZ-11, SEC-QZ-05	Integration (API) / Functional - Concurrency	High	Vedh	R2	S2	Jest + Supertest / Postman	Full regression	New (1.1)	-
TC-QZ-32	Countdown follows the server deadline	REQ-QZ-08, SEC-QZ-04	System (UI + API) / Functional - Positive	Medium	Vedh	R1	S2	Cypress or Selenium + Postman	Full regression	New (1.1)	-
TC-QS-01	Server clock governs window and deadline	SEC-QZ-04, REQ-QZ-07	Security (API / system) / Security – Negative	High	Vedh	R1	S2	Jest + Supertest / Postman	Smoke	Retained	-
TC-QS-02	Client-supplied score ignored	SEC-QZ-05	Security (API / system) / Security – Negative	High	Vedh	R1	S2	Jest + Supertest / Postman	Smoke	Retained	-
TC-QS-03	NoSQL-operator and script payloads in search and question text	SEC-QZ-01, REQ-QB-02	Security (API / system) / Security – Negative	High	Alifiya	R1	S1	Jest + Supertest / Postman	Smoke	Retained	-
TC-QS-04	Expired or logged-out session refused on quiz endpoints	SEC-QZ-03, SEC-ACC-04	Security (API / system) / Security – Negative	High	Alifiya	R1	S2	Jest + Supertest / Postman	Smoke	Retained (amended)	-
TC-QS-05	Malformed identifiers and operator payloads on quiz and attempt endpoints	NFR-QUAL-03, SEC-QZ-01, SEC-QZ-05	Security (API) / Security - Negative	Medium	Jagadeesh	R1	S1	Jest + Supertest / Postman	Full regression	New (1.1)	-
TC-QS-06	Answer-key scan across all Student-facing responses	SEC-QZ-02	Security (API) / Security - Inspection	High	Vedh	R1	S2	Jest + Supertest / Postman	Smoke	New (1.1)	-
TC-QP-01	Grading within 2 s for 100 simultaneous submissions	NFR-PERF-04, REQ-QZ-12	Performance (JMeter / timing) / Performance (NFR)	High	Vedh	R2	S2	JMeter / timing script	Full regression	Retained	TOD-10
TC-QP-02	Question-bank search within 2 s (1,000 questions)	NFR-PERF-05, REQ-QB-04	Performance (JMeter / timing) / Performance (NFR)	Medium	Alifiya	R2	S2	JMeter / timing script	Full regression	Retained	TOD-10
TC-QP-03	Quiz questions displayed within 3 s	NFR-PERF-06, REQ-QZ-08	Performance (JMeter / timing) / Performance (NFR)	Medium	Vedh	R2	S2	JMeter / timing script	Full regression	Retained	TOD-10
TC-QP-04	New question visible within 3 s	REQ-QB-03	Performance (JMeter / timing) / Performance (NFR)	Medium	Alifiya	R2	S2	JMeter / timing script	Full regression	Retained	-
TC-PA-01	View completed attempts	REQ-PA-01	System (UI + API) / Functional – Positive	High	Pushkar	R1	S2	Cypress or Selenium + Postman	Smoke	Retained	-

ID	Title	Requirements	Level / type	Prio	Owner	Ready	Spr.	Automation	Regression	Origin	Open decision
TC-PA-02	Personal performance calculated; empty state	REQ-PA-02	System (UI + API) / Functional – Positive / Negative	Medium	Pushkar	R1	S2	Cypress or Selenium + Postman	Full regression	Retained	TOD-05
TC-PA-03	Topic-wise analysis; empty state	REQ-PA-03	System (UI + API) / Functional – Positive / Negative	Medium	Pushkar	R3	S2	Cypress or Selenium + Postman	After decision	Retained (amended)	OD-03
TC-PA-04	Faculty views class performance for own quiz	REQ-PA-04	System (UI + API) / Functional – Positive / Negative	High	Pushkar	R1	S2	Cypress or Selenium + Postman	Smoke	Retained (amended)	OD-08
TC-PA-05	Class performance record content; empty state	REQ-PA-05	System (UI + API) / Functional – Positive / Negative	Medium	Pushkar	R1	S2	Cypress or Selenium + Postman	Full regression	Retained (amended)	OD-08
TC-PA-06	Non-owner Faculty denied class performance	REQ-PA-06, REQ-PA-04	Security (API / system) / Security – Negative	High	Pushkar	R1	S2	Jest + Supertest / Postman	Smoke	Retained (amended)	-
TC-PA-07	Personal performance limited to the authenticated Student	REQ-PA-01, REQ-ACC-10	Security (API) / Security - Negative	High	Pushkar	R1	S2	Jest + Supertest / Postman	Smoke	New (1.1)	-
TC-PA-08	Displayed values equal stored attempts	REQ-PA-02	System (UI + API) / Functional - Positive	High	Pushkar	R1	S2	Cypress or Selenium + Postman	Smoke	New (1.1)	TOD-05
TC-PA-09	Topic-wise grouping with known topic data	REQ-PA-03	System (UI + API) / Functional - Positive	High	Pushkar	R3	S2	Cypress or Selenium + Postman	After decision	New (1.1)	OD-03
TC-PA-10	Class performance equals quiz submissions view	REQ-PA-05, REQ-QZ-16	Integration (API) / Functional - Positive	Medium	Pushkar	R1	S2	Jest + Supertest / Postman	Full regression	New (1.1)	-
TC-PA-11	Class-performance request validation	REQ-PA-06, REQ-QZ-18	Security (API) / Security - Negative	Medium	Pushkar	R1	S2	Jest + Supertest / Postman	Full regression	New (1.1)	-
TC-PA-12	Faculty selects an own quiz for class performance	REQ-PA-04	System (UI + API) / Functional - Positive	Medium	Pushkar	R3	S2	Cypress or Selenium + Postman	After decision	New (1.1)	OD-08
TC-PA-13	New submission appears in personal performance	REQ-PA-01, REQ-PA-02, REQ-QZ-15	System (UI + API) / Functional - Positive	Medium	Pushkar	R1	S2	Cypress or Selenium + Postman	Full regression	New (1.1)	-
TC-COL-01	Create study group	REQ-COL-01	System (UI + API) / Functional – Positive	Medium	Pushkar	R1	S1	Cypress or Selenium + Postman	Full regression	Retained	-
TC-COL-02	Join study group	REQ-COL-02	System (UI + API) / Functional – Positive	Medium	Pushkar	R1	S1	Cypress or Selenium + Postman	Full regression	Retained	-
TC-COL-03	Duplicate membership prevented	REQ-COL-03	System (UI + API) / Functional – Negative	High	Pushkar	R1	S1	Cypress or Selenium + Postman	Smoke	Retained (amended)	-
TC-COL-04	View own study groups	REQ-COL-04	System (UI + API) / Functional – Positive	Medium	Pushkar	R1	S1	Cypress or Selenium + Postman	Full regression	Retained	-
TC-COL-05	Admin moderates content	REQ-COL-05	System (UI + API) / Functional – Positive	High	Pushkar	R1	S2	Cypress or Selenium + Postman	Smoke	Retained (amended)	OD-08, OD-13
TC-COL-06	Non-Admin denied moderation	REQ-COL-06	Security (API / system) / Security – Negative	High	Pushkar	R1	S2	Jest + Supertest / Postman	Smoke	Retained (amended)	-

ID	Title	Requirements	Level / type	Prio	Owner	Ready	Spr.	Automation	Regression	Origin	Open decision
TC-COL-07	Unavailable group/content reported without details	REQ-COL-07	System (UI + API) / Functional – Negative	Medium	Pushkar	R1	S2	Cypress or Selenium + Postman	Full regression	Retained (amended)	-
TC-COL-08	Group creation requires a name	REQ-COL-01	Integration (API) / Functional - Negative	Medium	Pushkar	R1	S1	Jest + Supertest / Postman	Full regression	New (1.1)	-
TC-COL-09	Group field length limits	REQ-COL-01	Integration (API) / Functional - Boundary	Low	Pushkar	R3	S2	Jest + Supertest / Postman	After decision	New (1.1)	OD-13
TC-COL-10	Only Students can create and join groups	REQ-COL-01, REQ-COL-02	Security (API) / Security - Negative	Medium	Pushkar	R1	S1	Jest + Supertest / Postman	Full regression	New (1.1)	-
TC-COL-11	Concurrent duplicate join	REQ-COL-03	Integration (API) / Functional - Concurrency	High	Pushkar	R2	S1	Jest + Supertest / Postman	Full regression	New (1.1)	-
TC-COL-12	My Groups lists only own groups; empty list	REQ-COL-04	Integration (API) / Functional - Positive / Negative	Medium	Pushkar	R1	S1	Jest + Supertest / Postman	Full regression	New (1.1)	-
TC-COL-13	Moderation: list content and available actions	REQ-COL-05	System (UI + API) / Functional - Positive	Medium	Pushkar	R3	S2	Cypress or Selenium + Postman	After decision	New (1.1)	OD-08, OD-13
TC-COL-14	Moderating content that is already removed	REQ-COL-05, REQ-COL-07	Integration (API) / Functional - Negative	Medium	Pushkar	R1	S2	Jest + Supertest / Postman	Full regression	New (1.1)	-

Unit-test candidates

ID	Unit	Requirements	Owner	Ready	Origin
UT-ACC-01	validateRegistration(input)	REQ-ACC-02, REQ-ACC-03	Jagadeesh	R1	Retained from STP v1.0
UT-ACC-02	authenticateUser(email, password)	REQ-ACC-04, REQ-ACC-05	Jagadeesh	R1	Retained from STP v1.0
UT-ACC-03	isAccountLocked(user, now)	REQ-ACC-06	Jagadeesh	R1	Retained from STP v1.0
UT-ACC-04	isSessionValid(session, now)	REQ-ACC-07	Jagadeesh	R1	Retained from STP v1.0
UT-ACC-05	isRegistrationRoleAllowed(role)	REQ-ACC-01, BR-08	Jagadeesh	R1	New in Rev 1.1
UT-ACC-06	sanitizeInput(body)	SEC-ACC-01, NFR-QUAL-03, DC-07	Jagadeesh	R1	New in Rev 1.1
UT-COL-01	validateGroup(input)	REQ-COL-01	Pushkar	R1	Retained from STP v1.0
UT-COL-02	isAlreadyMember(groupId, userId)	REQ-COL-03	Pushkar	R1	Retained from STP v1.0
UT-COL-03	canModerateContent(user)	REQ-COL-05, REQ-COL-06	Pushkar	R1	Retained from STP v1.0
UT-COL-04	canJoinOrCreateGroup(user)	REQ-COL-01, REQ-COL-02	Pushkar	R1	New in Rev 1.1
UT-MAT-01	validateMaterial(input)	REQ-MAT-01, REQ-MAT-02	Vedh	R1	Retained from STP v1.0
UT-MAT-02	canAccessMaterial(user, material, grant)	REQ-MAT-04, SEC-MAT-01	Vedh	R1	Retained from STP v1.0
UT-MAT-03	canManageMaterial(user, material)	REQ-MAT-05, REQ-MAT-07, REQ-MAT-08, REQ-MAT-09, SEC-MAT-02, SEC-MAT-03	Vedh	R1	Retained from STP v1.0
UT-MAT-04	validateProfileUpdate(input)	REQ-ACC-08, REQ-ACC-09	Jagadeesh	R1	Retained from STP v1.0
UT-MAT-05	sha256OfFile(path) / verifyChecksum(stored, file)	NFR-SAFE-03	Vedh	R1	New in Rev 1.1
UT-MAT-06	validateShareTarget(userId, users)	REQ-MAT-05, BR-05	Vedh	R1	New in Rev 1.1

ID	Unit	Requirements	Owner	Ready	Origin
UT-PA-01	calculatePerformance(attempts)	REQ-PA-02	Pushkar	R1	Retained from STP v1.0
UT-PA-02	groupResultsByTopic(results)	REQ-PA-03	Pushkar	R3	Retained from STP v1.0
UT-PA-03	canViewClassPerformance(user, quiz)	REQ-PA-04, REQ-PA-06	Pushkar	R1	Retained from STP v1.0
UT-QZ-01	GradingService.grade(attempt, quizCopy)	REQ-QZ-11	Vedh	R1	Retained from STP v1.0
UT-QZ-02	validateQuestion(input)	REQ-QB-02	Alifiya	R1	Retained from STP v1.0
UT-QZ-03	validateQuiz(input)	REQ-QZ-02	Alifiya	R1	Retained from STP v1.0
UT-QZ-04	canStartAttempt(quiz, student, now)	REQ-QZ-07	Vedh	R1	Retained from STP v1.0
UT-QZ-05	isAnswerAccepted(attempt, now)	REQ-QZ-10, REQ-QZ-14	Vedh	R1	Retained from STP v1.0
UT-QZ-06	toStudentQuestionDTO(question)	SEC-QZ-02	Vedh	R1	Retained from STP v1.0
UT-QZ-07	copyQuestionsAtPublish(quiz, questions)	REQ-QZ-04, REQ-QB-06	Alifiya	R1	New in Rev 1.1
UT-QZ-08	canViewResult(user, attempt) / canViewSubmissions(user, quiz)	SEC-QZ-03, REQ-QZ-17	Alifiya	R1	New in Rev 1.1
UT-QZ-09	computeDeadline(start, timeLimit, windowEnd)	REQ-QZ-10, SEC-QZ-04	Vedh	R3	New in Rev 1.1

Annex C. Requirement-to-Test Traceability Matrix

One row per requirement. "Test owner" = owners of the tests listed in the row (Annex B); "SRS App. D" = primary owner of the requirement in SRS Appendix D (UI-01 to UI-18 and COM-01 are not allocated there: TOD-12). Evidence: all "Not executed". Coverage status is design coverage only.

Req. ID	Description	Use case	SAD component / API	Tests (TC, UT)	Type	Test owner	SRS App. D	Coverage	Evidence	Open decision / note
REQ-ACC-01	Register with full name, e-mail, password and permitted role (Student or Faculty).	UC-ACC-01	Auth & Session Service; API Layer (validation) | POST /auth/register	TC-ACC-01, TC-ACC-02, TC-ACC-12, UT-ACC-05	Functional, Security, Unit	Jagadeesh	M1	Covered	Not executed	OD-11
REQ-ACC-02	Registration validation: empty fields, invalid e-mail, duplicate e-mail (case-insensitive).	UC-ACC-01	Auth & Session Service; API Layer (validation) | POST /auth/register	TC-ACC-02, TC-ACC-03, TC-ACC-13, UT-ACC-01	Functional, Unit	Jagadeesh	M1	Covered	Not executed	
REQ-ACC-03	Password rule: minimum length (8) with uppercase, lowercase and digit.	UC-ACC-01	Auth & Session Service | POST /auth/register	TC-ACC-04, UT-ACC-01	Functional, Unit	Jagadeesh	M1	Covered	Not executed	
REQ-ACC-04	Authenticate with valid credentials; access granted within 2 seconds.	UC-ACC-02	Auth & Session Service | POST /auth/login	TC-ACC-05, TC-ACC-18, UT-ACC-02	Functional, Performance, Unit	Jagadeesh	M1	Covered	Not executed	
REQ-ACC-05	Invalid credentials rejected with the single specified message "Invalid email or password".	UC-ACC-02	Auth & Session Service | POST /auth/login	TC-ACC-06, TC-ACC-18, UT-ACC-02	Functional, Unit	Jagadeesh	M1	Covered	Not executed	
REQ-ACC-06	Account locked for 15 minutes after five consecutive failed log-ins.	UC-ACC-02	Auth & Session Service | POST /auth/login	TC-ACC-07, TC-ACC-20, TC-AS-01, UT-ACC-03	Security, Unit	Jagadeesh	M1	Covered	Not executed	TOD-04
REQ-ACC-07	Session ends on logout or after 30 minutes of inactivity; token no longer accepted.	UC-ACC-03 to 05 (UC ID not individually verifiable without SRS)	Auth & Session Service; API Layer (authentication middleware) | POST /auth/logout; all protected endpoints	TC-ACC-08, TC-ACC-17, TC-AS-02, UT-ACC-04	Security, Unit	Jagadeesh	M1	Covered	Not executed	
REQ-ACC-08	View and edit permitted profile information; e-mail and role protected.	UC-ACC-03 to 05 (UC ID not individually verifiable without SRS)	Account Service; File Storage | GET, PUT /account/profile	TC-ACC-09, TC-ACC-14, TC-ACC-16, UT-MAT-04	Functional, Unit	Jagadeesh	M1	Covered	Not executed	OD-13
REQ-ACC-09	Profile validation: name not empty; profile picture not larger than 2 MB.	UC-ACC-03 to 05 (UC ID not individually verifiable without SRS)	Account Service; File Storage | PUT /account/profile	TC-ACC-10, TC-ACC-15, TC-ACC-16, UT-MAT-04	Functional, Unit	Jagadeesh	M1	Partially covered	Not executed	OD-13, TOD-06
REQ-ACC-10	Authentication and role-based access control for protected functionality.	UC-ACC-02	API Layer; Auth & Session Service | All protected endpoints	TC-ACC-11, TC-ACC-14, TC-ACC-19, TC-AS-03, TC-PA-07, TC-QB-12	Functional, Security	Jagadeesh, Pushkar, Alifiya	M1	Covered	Not executed	OD-11
REQ-MAT-01	Upload material with title, subject, description and file.	UC-MAT-01	Material Service, AccessGrant, File Storage; API Layer | POST /materials	TC-MAT-01, TC-MAT-02, TC-MAT-14, TC-MAT-16, TC-MAT-22, UT-MAT-01	Functional, Unit	Vedh	M4	Partially covered	Not executed	OD-07
REQ-MAT-02	File type (PDF, DOCX, PPTX, PNG) and 25 MB size validation.	UC-MAT-01	Material Service, AccessGrant, File Storage; API Layer | POST /materials; PUT /materials/:id	TC-MAT-03, TC-MAT-04, TC-MAT-16, UT-MAT-01	Functional, Unit	Vedh	M4	Covered	Not executed	TOD-06
REQ-MAT-03	Owner recorded; material visible in the owner's library within 3 seconds.	UC-MAT-01	Material Service, AccessGrant, File Storage; API Layer | POST /materials; GET /materials	TC-MAT-05, TC-MAT-15, TC-MAT-22, TC-MP-01	Functional, Performance	Vedh	M4	Covered	Not executed	
REQ-MAT-04	Material visible only to its owner or authorized users.	UC-MAT-02	Material Service, AccessGrant, File Storage; API Layer | GET /materials; GET /materials/:id	TC-MAT-06, TC-MAT-15, TC-MAT-21, TC-MS-01, UT-MAT-02	Functional, Security, Unit	Vedh	M4	Partially covered	Not executed	OD-13

Req. ID	Description	Use case	SAD component / API	Tests (TC, UT)	Type	Test owner	SRS App. D	Coverage	Evidence	Open decision / note
REQ-MAT-05	Owner shares material with another registered user.	UC-MAT-03	Material Service, AccessGrant, File Storage; API Layer | POST /materials/:id/share	TC-MAT-07, TC-MAT-17, UT-MAT-03, UT-MAT-06	Functional, Unit	Vedh	M4	Covered	Not executed	
REQ-MAT-06	Authorized user downloads material; download starts within 3 seconds.	UC-MAT-04	Material Service, AccessGrant, File Storage; API Layer | GET /materials/:id/download	TC-MAT-08, TC-MP-02	Performance	Vedh	M4	Covered	Not executed	
REQ-MAT-07	Owner updates metadata or replaces the file.	UC-MAT-05/06 (update, remove; order not verified)	Material Service, AccessGrant, File Storage; API Layer | PUT /materials/:id	TC-AN-17, TC-MAT-09, TC-MAT-19, TC-MS-05, UT-MAT-03	Functional, Non-functional, Security, Unit	Vedh	M4	Covered	Not executed	
REQ-MAT-08	Owner removes material after confirmation.	UC-MAT-05/06 (update, remove; order not verified)	Material Service, AccessGrant, File Storage; API Layer | DELETE /materials/:id	TC-AN-17, TC-MAT-10, TC-MS-05, UT-MAT-03	Functional, Non-functional, Security, Unit	Vedh	M4	Covered	Not executed	
REQ-MAT-09	Owner revokes a user's access; next request by that user is denied.	UC-MAT-07	Material Service, AccessGrant, File Storage; API Layer | DELETE /materials/:id/access/:userId	TC-MAT-11, TC-MAT-18, TC-MAT-20, TC-MS-02, UT-MAT-03	Functional, Security, Unit	Vedh	M4	Partially covered	Not executed	OD-08
REQ-MAT-10	Unauthorized material operations rejected with "Access denied".	UC-MAT-02 to 07	Material Service, AccessGrant, File Storage; API Layer | All /materials endpoints	TC-MAT-12, TC-MS-04, TC-MS-05	Security	Vedh	M4	Covered	Not executed	
REQ-MAT-11	Removed or unretrievable material handled without exposing its details ("Material unavailable").	UC-MAT-02, 04, 07	Material Service, AccessGrant, File Storage; API Layer | GET /materials/:id; GET /materials/:id/download	TC-MAT-13, TC-MS-03	Functional, Security	Vedh	M4	Covered	Not executed	
REQ-QB-01	Faculty adds a question (text, four options, correct option, topic, difficulty, explanation).	UC-QB-01 to 04 (pairing not verified)	Question Bank Service; API Layer | POST /questions	TC-QB-01, TC-QB-03	Functional	Alifiya	M2	Covered	Not executed	
REQ-QB-02	Question validation (required fields, limits 500/200/50/300, distinct options, correct option among options).	UC-QB-01 to 04 (pairing not verified)	Question Bank Service; API Layer | POST, PUT /questions	TC-QB-02, TC-QB-03, TC-QB-04, TC-QB-06, TC-QB-10, TC-QS-03, UT-QZ-02	Functional, Security, Unit	Alifiya	M2	Covered	Not executed	
REQ-QB-03	Owner recorded; new question visible in the bank within 3 seconds.	UC-QB-01 to 04 (pairing not verified)	Question Bank Service; API Layer | POST /questions	TC-QB-01, TC-QP-04	Functional, Performance	Alifiya	M2	Covered	Not executed	
REQ-QB-04	Search and filter the bank by keyword, topic and difficulty (results within 2 s for 1,000 questions).	UC-QB-01 to 04 (pairing not verified)	Question Bank Service; API Layer | GET /questions	TC-QB-05, TC-QB-11, TC-QP-02	Functional, Performance	Alifiya	M2	Covered	Not executed	TOD-10
REQ-QB-05	Faculty edits own question.	UC-QB-01 to 04 (pairing not verified)	Question Bank Service; API Layer | PUT /questions/:id	TC-QB-06	Functional	Alifiya	M2	Covered	Not executed	
REQ-QB-06	Faculty removes own question; published quizzes are unaffected.	UC-QB-01 to 04 (pairing not verified)	Question Bank Service; API Layer | DELETE /questions/:id	TC-QB-07, UT-QZ-07	Functional, Unit	Alifiya	M2	Covered	Not executed	
REQ-QB-07	Students cannot access the bank; only the owner can edit or remove a question.	UC-QB-01 to 04 (pairing not verified)	Question Bank Service; API Layer | All /questions endpoints	TC-QB-08, TC-QB-09, TC-QB-12	Security	Alifiya	M2	Covered	Not executed	
REQ-QZ-01	Faculty creates a quiz (title, description, time limit, availability window, 1-50 questions).	UC-QZ-01, UC-QZ-02	Quiz Engine (authoring) | POST /quizzes	TC-QZ-01, TC-QZ-03	Functional	Alifiya	M2	Covered	Not executed	
REQ-QZ-02	Quiz validation: title, 1-50 questions, time limit 5-180 minutes, end later than start.	UC-QZ-01, UC-QZ-02	Quiz Engine (authoring) | POST, PUT /quizzes	TC-QZ-02, TC-QZ-03, TC-QZ-19, TC-QZ-21, UT-QZ-03	Functional, Unit	Alifiya	M2	Covered	Not executed	
REQ-QZ-03	Quiz is saved as Draft; Faculty publishes it (Draft/Published life cycle).	UC-QZ-01, UC-QZ-02	Quiz Engine (authoring) | POST /quizzes; POST /quizzes/:id/publish	TC-QZ-01, TC-QZ-04	Functional	Alifiya	M2	Covered	Not executed	
REQ-QZ-04	Selected questions are copied into the quiz at publication.	UC-QZ-01, UC-QZ-02	Quiz Engine (authoring); Question Bank Service | POST /quizzes/:id/publish	TC-QB-07, TC-QZ-04, TC-QZ-19, TC-QZ-20, UT-QZ-07	Functional, Unit	Alifiya	M2	Covered	Not executed	TOD-11
REQ-QZ-05	Edit and delete rules: Draft editable/deletable; rejected once an attempt has started.	UC-QZ-01, UC-QZ-02	Quiz Engine (authoring) | PUT, DELETE /quizzes/:id	TC-QZ-05, TC-QZ-22	Functional	Alifiya	M2	Partially covered	Not executed	TOD-03
REQ-QZ-06	Student lists available quizzes (without questions or answers).	UC-QZ-03 (not verified)	Quiz Engine (execution) | GET /quizzes	TC-QZ-06	Functional	Vedh	M4	Covered	Not executed	

Req. ID	Description	Use case	SAD component / API	Tests (TC, UT)	Type	Test owner	SRS App. D	Coverage	Evidence	Open decision / note
REQ-QZ-07	Start an attempt only inside the availability window and only once per Student.	UC-QZ-04, UC-QZ-05	Quiz Engine (execution) | POST /quizzes/:id/attempts	TC-QS-01, TC-QZ-07, TC-QZ-30, UT-QZ-04	Functional, Security, Unit	Vedh	M4	Covered	Not executed	OD-12, TOD-04
REQ-QZ-08	Questions displayed with a server-based countdown timer.	UC-QZ-04, UC-QZ-05	Quiz Engine (execution); API Layer (projection) | POST /quizzes/:id/attempts	TC-QP-03, TC-QZ-08, TC-QZ-32	Functional, Performance, Security	Vedh	M4	Covered	Not executed	TOD-10
REQ-QZ-09	Student selects one answer per question and submits.	UC-QZ-04, UC-QZ-05	Quiz Engine (execution) | PUT /attempts/:id/answers; POST /attempts/:id/submit	TC-QZ-09, TC-QZ-23	Functional	Vedh	M4	Covered	Not executed	
REQ-QZ-10	Attempt auto-submitted when the time limit is reached.	UC-QZ-04, UC-QZ-05	Quiz Engine (execution) | PUT /attempts/:id/answers (410)	TC-QZ-10, TC-QZ-25, TC-QZ-29, UT-QZ-05, UT-QZ-09	Functional, Unit	Vedh	M4	Partially covered	Not executed	TOD-04
REQ-QZ-11	Automatic grading of the submitted attempt.	UC-QZ-04, UC-QZ-05	Grading Service; Quiz Engine (execution) | POST /attempts/:id/submit	TC-QZ-11, TC-QZ-26, TC-QZ-31, UT-QZ-01	Functional, Unit	Vedh	M4	Partially covered	Not executed	OD-08
REQ-QZ-12	Result available within 2 seconds of submission.	UC-QZ-04, UC-QZ-05	Grading Service | POST /attempts/:id/submit	TC-QP-01	Performance	Vedh	M4	Covered	Not executed	TOD-10
REQ-QZ-13	Result details: score, percentage, per-question answers, correct answers, explanations.	UC-QZ-06 (not verified)	Quiz Engine (execution) | GET /attempts/:id/result	TC-QZ-12, TC-QZ-27	Functional, Security	Vedh	M4	Covered	Not executed	
REQ-QZ-14	Submitted attempt is immutable.	UC-QZ-04, UC-QZ-05	Quiz Engine (execution); Grading Service | PUT /attempts/:id/answers; POST /attempts/:id/submit	TC-QZ-13, TC-QZ-26, TC-QZ-31, UT-QZ-05	Functional, Security, Unit	Vedh	M4	Covered	Not executed	OD-08
REQ-QZ-15	Student views own attempts (title, score, percentage, date).	UC-QZ-06 (not verified)	Quiz Engine (execution) | GET /attempts/mine	TC-PA-13, TC-QZ-14	Functional	Pushkar, Vedh	M4	Covered	Not executed	
REQ-QZ-16	Faculty views submissions for own quizzes.	UC-QZ-07 (not verified)	Quiz Engine (execution) | GET /quizzes/:id/submissions	TC-PA-10, TC-QZ-15	Functional	Pushkar, Alifiya	M2	Covered	Not executed	OD-08
REQ-QZ-17	Role and ownership access control for quiz functions.	UC-QZ-01 to 07	API Layer (RBAC/ownership); Quiz Engine | All /quizzes and /attempts endpoints	TC-QZ-16, TC-QZ-17, TC-QZ-24, TC-QZ-28, UT-QZ-08	Security, Unit	Alifiya	M2	Covered	Not executed	
REQ-QZ-18	Unavailable (removed or non-existent) quiz, question or attempt handled without exposing details.	UC-QZ-01 to 07	API Layer (central error handler); Quiz Engine; Question Bank Service | 404 branches of /quizzes, /attempts, /questions	TC-PA-11, TC-QZ-18, TC-QZ-20, TC-QZ-24	Functional, Security	Pushkar, Alifiya	M2	Covered	Not executed	TOD-11
REQ-PA-01	Student views completed quiz attempts with quiz title, score, percentage and submission date.	UC-PA-01	Performance Analytics Service; API Layer (RBAC/ownership) | GET /performance/personal	TC-PA-01, TC-PA-07, TC-PA-13	Functional, Security	Pushkar	M3	Covered	Not executed	
REQ-PA-02	Student performance calculated and displayed from completed attempts.	UC-PA-01	Performance Analytics Service; API Layer (RBAC/ownership) | GET /performance/personal	TC-PA-02, TC-PA-08, TC-PA-13, UT-PA-01	Functional, Unit	Pushkar	M3	Covered	Not executed	TOD-05
REQ-PA-03	Student views quiz performance grouped by topic.	UC-PA-03	Performance Analytics Service; API Layer (RBAC/ownership) | GET /performance/topics	TC-PA-03, TC-PA-09, UT-PA-02	Functional, Unit	Pushkar	M3	Blocked pending clarification	Not executed	OD-03. Topic source undefined (OD-03)
REQ-PA-04	Faculty views performance of Students who submitted attempts for Faculty-owned quizzes.	UC-PA-02	Performance Analytics Service; API Layer (RBAC/ownership) | GET /performance/class/:quizId	TC-PA-04, TC-PA-06, TC-PA-12, UT-PA-03	Functional, Security, Unit	Pushkar	M3	Partially covered	Not executed	OD-08
REQ-PA-05	Class performance shows Student name, score, percentage and submission information.	UC-PA-02	Performance Analytics Service; API Layer (RBAC/ownership) | GET /performance/class/:quizId	TC-PA-05, TC-PA-10	Functional	Pushkar	M3	Covered	Not executed	OD-08
REQ-PA-06	Faculty access to performance of quizzes they do not own is denied.	UC-PA-02	Performance Analytics Service; API Layer (RBAC/ownership) | GET /performance/class/:quizId	TC-PA-06, TC-PA-11, UT-PA-03	Security, Unit	Pushkar	M3	Covered	Not executed	
REQ-COL-01	Student creates a study group with required group information.	UC-COL-01, UC-COL-02	Study Group & Collaboration Service; API Layer | POST /groups	TC-COL-01, TC-COL-08, TC-COL-09, TC-COL-10, UT-COL-01, UT-COL-04	Functional, Security, Unit	Pushkar	M3	Partially covered	Not executed	OD-13
REQ-COL-02	Student joins an existing study group.	UC-COL-01, UC-COL-02	Study Group & Collaboration Service; API Layer | POST /groups/:id/join	TC-COL-02, TC-COL-10, UT-COL-04	Functional, Security, Unit	Pushkar	M3	Covered	Not executed	

Req. ID	Description	Use case	SAD component / API	Tests (TC, UT)	Type	Test owner	SRS App. D	Coverage	Evidence	Open decision / note
REQ-COL-03	Duplicate study-group membership is prevented.	UC-COL-01, UC-COL-02	Study Group & Collaboration Service; API Layer | POST /groups/:id/join	TC-COL-03, TC-COL-11, UT-COL-02	Functional, Unit	Pushkar	M3	Covered	Not executed	
REQ-COL-04	Student views the study groups to which they belong.	UC-COL-01, UC-COL-02	Study Group & Collaboration Service; API Layer | GET /groups	TC-COL-04, TC-COL-12	Functional	Pushkar	M3	Covered	Not executed	
REQ-COL-05	Admin moderates collaborative content.	UC-COL-01, UC-COL-02	Study Group & Collaboration Service; API Layer | DELETE /collaboration/content/:id	TC-COL-05, TC-COL-13, TC-COL-14, UT-COL-03	Functional, Unit	Pushkar	M3	Partially covered	Not executed	OD-08, OD-13
REQ-COL-06	Non-Admin users are denied collaborative-content moderation.	UC-COL-01, UC-COL-02	Study Group & Collaboration Service; API Layer | DELETE /collaboration/content/:id	TC-COL-06, UT-COL-03	Security, Unit	Pushkar	M3	Covered	Not executed	
REQ-COL-07	Removed or unavailable groups/content handled without exposing their details.	UC-COL-01, UC-COL-02	Study Group & Collaboration Service; API Layer | POST /groups/:id/join; DELETE /collaboration/content/:id	TC-COL-07, TC-COL-14	Functional	Pushkar	M3	Covered	Not executed	
SEC-ACC-01	Authentication enforced for protected functionality.	-	API Layer; Auth & Session Service | All protected endpoints	TC-AS-03, TC-AS-07, TC-AS-08, UT-ACC-06	Security, Unit	Jagadeesh	M1	Covered	Not executed	OD-06
SEC-ACC-02	Generic credential error (no disclosure of whether the e-mail exists).	-	Auth & Session Service | POST /auth/login	TC-ACC-06, TC-AS-07	Functional, Security	Jagadeesh	M1	Covered	Not executed	
SEC-ACC-03	Passwords stored only as salted bcrypt hashes.	-	Auth & Session Service | POST /auth/register	TC-ACC-14, TC-AS-04	Functional, Security	Jagadeesh	M1	Covered	Not executed	
SEC-ACC-04	Session termination on logout and inactivity (server-side session record).	-	API Layer; Auth & Session Service | POST /auth/logout	TC-ACC-17, TC-AS-02, TC-QS-04	Security	Jagadeesh, Alifiya	M1	Covered	Not executed	
SEC-ACC-05	A user can change only their own profile; e-mail and role not changeable.	-	Account Service | PUT /account/profile	TC-AS-05	Security	Jagadeesh	M1	Covered	Not executed	
SEC-ACC-06	Account lock-out after repeated failed log-ins.	-	Auth & Session Service | POST /auth/login	TC-ACC-07, TC-ACC-20, TC-AS-01, TC-AS-08	Security	Jagadeesh	M1	Covered	Not executed	OD-06, TOD-04
SEC-MAT-01	Material access control: only owner and authorized users can view/download.	-	Material Service, AccessGrant, File Storage; API Layer | GET /materials/:id; GET /materials/:id/download	TC-AS-08, TC-MS-01, UT-MAT-02	Security, Unit	Jagadeesh, Vedh	M4	Covered	Not executed	OD-06
SEC-MAT-02	Only the owner can share material and revoke access.	-	Material Service, AccessGrant, File Storage | POST /materials/:id/share; DELETE /materials/:id/access/:userId	TC-MAT-12, TC-MS-04, UT-MAT-03	Security, Unit	Vedh	M4	Covered	Not executed	
SEC-MAT-03	Only the owner can update or remove material.	-	Material Service, AccessGrant, File Storage | PUT, DELETE /materials/:id	TC-MAT-12, TC-MS-05, UT-MAT-03	Security, Unit	Vedh	M4	Covered	Not executed	
SEC-MAT-04	Access denied after revocation or removal.	-	Material Service, AccessGrant, File Storage | GET /materials/:id; DELETE /materials/:id/access/:userId	TC-MS-02, TC-MS-03	Security	Vedh	M4	Covered	Not executed	
SEC-COM-01	TLS 1.2 or higher for all client-server communication; unencrypted connections rejected.	-	Nginx reverse proxy (TLS) | All endpoints	TC-AS-06	Security	Jagadeesh	M1	Covered	Not executed	TOD-10
SEC-QZ-01	Question-bank and quiz management restricted to Faculty (owner).	-	API Layer; Question Bank Service; Quiz Engine (authoring) | /questions; /quizzes	TC-QB-08, TC-QB-09, TC-QB-12, TC-QS-03, TC-QS-05, TC-QZ-17	Security	Alifiya, Jagadeesh	M2	Covered	Not executed	
SEC-QZ-02	No answer keys or explanations sent before submission.	-	API Layer (projection); Quiz Engine (execution) | GET /quizzes; POST /quizzes/:id/attempts	TC-QS-06, TC-QZ-06, TC-QZ-08, TC-QZ-27, UT-QZ-06	Functional, Security, Unit	Vedh	M4	Covered	Not executed	
SEC-QZ-03	Results visible only to the attempt owner; submissions only to the quiz owner.	-	Quiz Engine (execution); API Layer | GET /attempts/:id/result; GET /quizzes/:id/submissions	TC-QS-04, TC-QZ-16, TC-QZ-24, TC-QZ-28, UT-QZ-08	Security, Unit	Alifiya	M2	Covered	Not executed	

Req. ID	Description	Use case	SAD component / API	Tests (TC, UT)	Type	Test owner	SRS App. D	Coverage	Evidence	Open decision / note
SEC-QZ-04	Quiz timer and availability enforced on the server clock.	-	Quiz Engine (execution) | POST /quizzes/:id/attempts; PUT /attempts/:id/answers	TC-AN-16, TC-QS-01, TC-QZ-10, TC-QZ-25, TC-QZ-29, TC-QZ-32, UT-QZ-09	Functional, Non-functional, Security, Unit	Jagadeesh, Vedh	M4	Partially covered	Not executed	TOD-04
SEC-QZ-05	Scores computed on the server; submitted attempts immutable.	-	Quiz Engine (execution); Grading Service | POST /attempts/:id/submit	TC-QS-02, TC-QS-05, TC-QZ-13, TC-QZ-30, TC-QZ-31	Functional, Security	Vedh, Jagadeesh	M4	Covered	Not executed	
NFR-PERF-01	At least 95% of page loads within 3 seconds with 200 concurrent users.	-	Web Client; API Layer; Nginx; Data Access Layer	TC-AP-01, TC-AP-03	Performance	Jagadeesh	M1	Covered	Not executed	TOD-10
NFR-PERF-02	200 concurrent users supported with fewer than 1% failed requests.	-	API Layer (rate limiting); Nginx; MongoDB	TC-AP-02, TC-AP-03, TC-AS-09	Performance, Security	Jagadeesh	M1	Covered	Not executed	TOD-02, TOD-10
NFR-PERF-03	Sharing and access revocation take effect within 2 seconds.	-	Material Service; AccessGrant	TC-MP-03	Performance	Vedh	M4	Covered	Not executed	
NFR-PERF-04	Grading within 2 seconds with 100 concurrent submissions.	-	Grading Service	TC-AP-03, TC-QP-01	Performance	Jagadeesh, Vedh	M4	Covered	Not executed	TOD-10
NFR-PERF-05	Question-bank search within 2 seconds.	-	Question Bank Service; Data Access Layer	TC-QP-02	Performance	Alifiya	M2	Covered	Not executed	TOD-10
NFR-PERF-06	Quiz questions available within 3 seconds of starting.	-	Quiz Engine (execution)	TC-QP-03	Performance	Vedh	M4	Covered	Not executed	TOD-10
NFR-SAFE-01	Daily backups; recovery from a backup no more than 24 hours old.	-	MongoDB; File Storage; Docker deployment	TC-AN-01	Non-functional	Jagadeesh	M1	Covered	Not executed	OD-06
NFR-SAFE-02	Interrupted/failed material operation leaves previous state unchanged and is not reported as success.	-	Material Service; File Storage	TC-AN-02, TC-AN-17	Non-functional	Vedh	M4	Covered	Not executed	
NFR-SAFE-03	Downloaded file identical to uploaded file (SHA-256).	-	Material Service; File Storage	TC-AN-03, TC-MAT-19, UT-MAT-05	Functional, Non-functional, Unit	Vedh	M4	Covered	Not executed	
NFR-QUAL-01	First-time user completes registration, login and upload in under five minutes without instructions.	-	Web Client	TC-AN-04	Non-functional	Jagadeesh	M1	Covered	Not executed	
NFR-QUAL-02	At least 99% monthly availability, excluding announced maintenance.	-	Nginx; Docker deployment; monitoring	TC-AN-07	Non-functional	Pushkar	M3	Partially covered	Not executed	OD-06, TOD-07. A monthly 99% availability cannot be demonstrated in the sprint window; the test records measured availability over the observed period (TOD-07, OD-06)
NFR-QUAL-03	Invalid input and unsupported files rejected without a crash or effect on other users.	-	API Layer	TC-AN-05, TC-AN-11, TC-AN-12, TC-AN-13, TC-AN-16, TC-AS-07, TC-AS-09, TC-MAT-16, TC-QS-05, UT-ACC-06	Functional, Non-functional, Security, Unit	Jagadeesh, Alifiya, Vedh	M1	Covered	Not executed	TOD-02

Req. ID	Description	Use case	SAD component / API	Tests (TC, UT)	Type	Test owner	SRS App. D	Coverage	Evidence	Open decision / note
NFR-QUAL-04	Current Chrome, Firefox, Edge, Safari on desktop and mobile.	-	Web Client	TC-AN-06	Non-functional	Jagadeesh	M1	Covered	Not executed	TOD-10
NFR-QUAL-05	Modular, independently testable feature areas.	-	Layered modular monolith	TC-AN-08	Non-functional	Pushkar	M3	Covered	Not executed	
NFR-QUAL-06	Requirements traceable to test cases.	-	SAD s3.8; SRS RTM; this STP	TC-AN-09	Non-functional	Pushkar	M3	Covered	Not executed	
SO-01	Security objective Account confidentiality: Protect user credentials and account data from unauthorized disclosure.	-	Security architecture (SAD s3.9); linked SEC requirements: SEC-ACC-02, SEC-ACC-03, SEC-ACC-06	TC-ACC-06, TC-AS-07, TC-ACC-14, TC-AS-04, TC-ACC-07, TC-ACC-20, TC-AS-01, TC-AS-08	Security (via linked SEC tests)	Jagadeesh	M1	Partially covered	Not executed	Mapped through SRS 3.3.3.4 to SEC-ACC-02, SEC-ACC-03, SEC-ACC-06. Linked tests not yet reviewed against the objective wording; no objective-level test
SO-02	Security objective Material access control: Ensure study materials are accessible only to their owner and users the owner has authorized.	-	Security architecture (SAD s3.9); linked SEC requirements: SEC-MAT-01, SEC-MAT-02, SEC-MAT-04	TC-AS-08, TC-MS-01, UT-MAT-02, TC-MAT-12, TC-MS-04, UT-MAT-03, TC-MS-02, TC-MS-03	Security (via linked SEC tests)	Jagadeesh, Vedh	M1	Partially covered	Not executed	Mapped through SRS 3.3.3.4 to SEC-MAT-01, SEC-MAT-02, SEC-MAT-04. Linked tests not yet reviewed against the objective wording; no objective-level test
SO-03	Security objective Integrity: Prevent unauthorized modification of accounts, materials and material permissions.	-	Security architecture (SAD s3.9); linked SEC requirements: SEC-ACC-05, SEC-MAT-02, SEC-MAT-03	TC-AS-05, TC-MAT-12, TC-MS-04, UT-MAT-03, TC-MS-05	Security (via linked SEC tests)	Jagadeesh, Vedh	M1	Partially covered	Not executed	Mapped through SRS 3.3.3.4 to SEC-ACC-05, SEC-MAT-02, SEC-MAT-03. Linked tests not yet reviewed against the objective wording; no objective-level test

Req. ID	Description	Use case	SAD component / API	Tests (TC, UT)	Type	Test owner	SRS App. D	Coverage	Evidence	Open decision / note
SO-04	Security objective Secure authentication and sessions: Ensure protected functionality is reachable only through a valid session and that ended sessions cannot be reused.	-	Security architecture (SAD s3.9); linked SEC requirements: SEC-ACC-01, SEC-ACC-04	TC-AS-03, TC-AS-07, TC-AS-08, UT-ACC-06, TC-ACC-17, TC-AS-02, TC-QS-04	Security (via linked SEC tests)	Jagadeesh, Alifiya	M1	Partially covered	Not executed	Mapped through SRS 3.3.3.4 to SEC-ACC-01, SEC-ACC-04. Linked tests not yet reviewed against the objective wording; no objective-level test
SO-05	Security objective Secure communication: Protect data in transit between client and server.	-	Security architecture (SAD s3.9); linked SEC requirements: SEC-COM-01	TC-AS-06	Security (via linked SEC tests)	Jagadeesh	M1	Partially covered	Not executed	Mapped through SRS 3.3.3.4 to SEC-COM-01. Linked tests not yet reviewed against the objective wording; no objective-level test
SO-06	Security objective Assessment integrity: Ensure that quiz answers and results are disclosed only to authorized users and that attempts and scores cannot be tampered with.	-	Security architecture (SAD s3.9); linked SEC requirements: SEC-QZ-01, SEC-QZ-02, SEC-QZ-03, SEC-QZ-04, SEC-QZ-05	TC-QB-08, TC-QB-09, TC-QB-12, TC-QS-03, TC-QS-05, TC-QZ-17, TC-QS-06, TC-QZ-06, TC-QZ-08, TC-QZ-27, UT-QZ-06, TC-QS-04, TC-QZ-16, TC-QZ-24, TC-QZ-28, UT-QZ-08, TC-AN-16, TC-QS-01, TC-QZ-10, TC-QZ-25, TC-QZ-29, TC-QZ-32, UT-QZ-09, TC-QS-02, TC-QZ-13, TC-QZ-30, TC-QZ-31	Security (via linked SEC tests)	Alifiya, Jagadeesh, Vedh	M1	Partially covered	Not executed	Mapped through SRS 3.3.3.4 to SEC-QZ-01, SEC-QZ-02, SEC-QZ-03, SEC-QZ-04, SEC-QZ-05. Linked tests not yet reviewed against the objective wording; no objective-level test
BR-01	Business rule BR-01: Protected functions are available only to authenticated users.	-	API Layer (authentication middleware); Auth & Session Service	TC-ACC-11, TC-AS-03	Security	Jagadeesh	M1	Partially covered	Not executed	SRS 3.3.5 wording now available; mapped tests have not yet been re-reviewed against it, so the status is retained

Req. ID	Description	Use case	SAD component / API	Tests (TC, UT)	Type	Test owner	SRS App. D	Coverage	Evidence	Open decision / note
BR-02	Business rule BR-02: Each study material has exactly one owner: the user who uploaded it.	-	Material Service, AccessGrant, File Storage	TC-MAT-01, TC-MAT-22	Functional	Vedh	M4	Covered	Not executed	OD-07
BR-03	Business rule BR-03: Only the owner may share, update, remove or revoke access to a material.	-	Material Service, AccessGrant, File Storage	TC-MAT-07, TC-MAT-12, TC-MS-04, TC-MS-05	Functional, Security	Vedh	M4	Partially covered	Not executed	SRS 3.3.5 wording now available; mapped tests have not yet been re-reviewed against it, so the status is retained
BR-04	Business rule BR-04: A user may view or download a material only if they are its owner or have been granted access.	-	Material Service, AccessGrant, File Storage	TC-MAT-12, TC-MS-04, TC-MS-05	Security	Vedh	M4	Partially covered	Not executed	SRS 3.3.5 wording now available; mapped tests have not yet been re-reviewed against it, so the status is retained
BR-05	Business rule BR-05: Materials may be shared only with registered users.	-	Material Service, AccessGrant, File Storage	TC-MAT-07, TC-MAT-17, UT-MAT-06	Functional, Unit	Vedh	M4	Covered	Not executed	
BR-06	Business rule BR-06: Users may change only their own profile; email address and role cannot be changed.	-	Account Service	TC-ACC-09	Functional	Jagadeesh	M1	Partially covered	Not executed	OD-13. SRS 3.3.5 wording now available; mapped tests have not yet been re-reviewed against it, so the status is retained
BR-07	Business rule BR-07: Users may upload only material they have the right to share. The uploader is responsible for copyright compliance.	-	Policy rule	-	-	Jagadeesh	M4	Not directly testable	Not executed	Policy/project constraint, not a system behaviour (copyright responsibility; no mechanism, OD-11)

Req. ID	Description	Use case	SAD component / API	Tests (TC, UT)	Type	Test owner	SRS App. D	Coverage	Evidence	Open decision / note
BR-08	Business rule BR-08: Admin accounts are created by platform administrators, not through public registration.	-	Auth & Session Service	TC-ACC-12, TC-ACC-19, UT-ACC-05	Security, Unit	Jagadeesh	M1	Covered	Not executed	OD-11
DC-01	MERN stack (MongoDB, Express.js, React, Node.js).	-	Whole architecture	TC-AN-10	Non-functional	Jagadeesh	M1	Covered	Not executed	
DC-02	Defined integrated API.	-	API Layer	TC-AN-11	Non-functional	Alifiya	M2	Covered	Not executed	
DC-03	TLS 1.2 or higher.	-	Nginx reverse proxy	TC-AS-06	Security	Jagadeesh	M3	Covered	Not executed	TOD-10
DC-04	bcrypt password hashing.	-	Auth & Session Service	TC-AN-10, TC-AS-04	Non-functional, Security	Jagadeesh	M1	Covered	Not executed	
DC-05	Supported browsers.	-	Web Client	TC-AN-06	Non-functional	Jagadeesh	M1	Covered	Not executed	TOD-10
DC-06	File types, size and field limits.	-	API Layer; Material Service; Account Service; File Storage	TC-ACC-09, TC-ACC-10, TC-MAT-03, TC-MAT-04	Functional	Jagadeesh, Vedh	M4	Covered	Not executed	OD-13, TOD-06
DC-07	Server-side validation and authorization checks.	-	API Layer middleware; all services	TC-AN-12, TC-AS-07, UT-ACC-06	Security, Unit	Jagadeesh	M1	Covered	Not executed	
DC-08	Server-side session expiry and lock-out.	-	Auth & Session Service	TC-ACC-07, TC-ACC-17, TC-AN-12, TC-AS-01, TC-AS-02	Security	Jagadeesh	M1	Covered	Not executed	TOD-04
DC-09	Scope constraint (features outside the SRS are excluded).	-	Whole architecture (SAD s1.2)	-	-	Pushkar	M3	Not directly testable	Not executed	Policy/project constraint, not a system behaviour (scope statement; checked by review of s3/s4 of this STP)
DC-10	Requirement IDs unchanged across documents.	-	SAD s3.8	TC-AN-09	Non-functional	Pushkar	M2	Covered	Not executed	
DC-11	The system shall be developed by the four-member project team within the course schedule (SRS DC-11).	-	Not an architecture element	-	-	Unassigned	-	Not directly testable	Not executed	Policy/project constraint, not a system behaviour (four-member project team; SRS DC-11)
DOM-NOID	Domain requirement without SRS ID: user-facing messages are in English (SAD s3.8). Label DOM-NOID is a test-plan placeholder, not an SRS ID.	-	Web Client; API Layer (messages)	TC-AN-14	Non-functional	Pushkar	M3	Covered	Not executed	

Req. ID	Description	Use case	SAD component / API	Tests (TC, UT)	Type	Test owner	SRS App. D	Coverage	Evidence	Open decision / note
UI-01..18	User-interface requirements UI-01 to UI-18 (SRS 3.1.1).	-	Web Client	TC-AN-15	Functional	Pushkar (provisional)	not allocated	Partially covered	Not executed	TC-AN-15 verifies navigation and role visibility only; item-level UI traceability is open (TOD-12).
COM-01	Communication-interface requirement COM-01 (SRS 3.1.3): HTTPS with TLS 1.2 or higher; unencrypted connections rejected.	-	Nginx; API Layer | All endpoints	TC-AS-06	Security	Jagadeesh	not allocated	Covered	Not executed	TOD-10, TOD-12 (COM-02 to COM-10 not traced).

Annex D. Test-Case-to-Requirement Index

Test	Requirements traced
TC-ACC-01	REQ-ACC-01
TC-ACC-02	REQ-ACC-01, REQ-ACC-02
TC-ACC-03	REQ-ACC-02
TC-ACC-04	REQ-ACC-03
TC-ACC-05	REQ-ACC-04
TC-ACC-06	REQ-ACC-05, SEC-ACC-02
TC-ACC-07	REQ-ACC-06, SEC-ACC-06, DC-08
TC-ACC-08	REQ-ACC-07
TC-ACC-09	REQ-ACC-08, BR-06, DC-06
TC-ACC-10	REQ-ACC-09, DC-06
TC-ACC-11	REQ-ACC-10, BR-01
TC-ACC-12	REQ-ACC-01, BR-08
TC-ACC-13	REQ-ACC-02
TC-ACC-14	REQ-ACC-08, REQ-ACC-10, SEC-ACC-03
TC-ACC-15	REQ-ACC-09
TC-ACC-16	REQ-ACC-09, REQ-ACC-08
TC-ACC-17	REQ-ACC-07, SEC-ACC-04, DC-08
TC-ACC-18	REQ-ACC-04, REQ-ACC-05
TC-ACC-19	REQ-ACC-10, BR-08
TC-ACC-20	REQ-ACC-06, SEC-ACC-06
TC-MAT-01	REQ-MAT-01, BR-02
TC-MAT-02	REQ-MAT-01
TC-MAT-03	REQ-MAT-02, DC-06
TC-MAT-04	REQ-MAT-02, DC-06
TC-MAT-05	REQ-MAT-03
TC-MAT-06	REQ-MAT-04
TC-MAT-07	REQ-MAT-05, BR-05, BR-03
TC-MAT-08	REQ-MAT-06
TC-MAT-09	REQ-MAT-07
TC-MAT-10	REQ-MAT-08
TC-MAT-11	REQ-MAT-09
TC-MAT-12	REQ-MAT-10, BR-03, BR-04, SEC-MAT-02, SEC-MAT-03
TC-MAT-13	REQ-MAT-11
TC-MAT-14	REQ-MAT-01
TC-MAT-15	REQ-MAT-03, REQ-MAT-04
TC-MAT-16	REQ-MAT-01, REQ-MAT-02, NFR-QUAL-03
TC-MAT-17	REQ-MAT-05, BR-05
TC-MAT-18	REQ-MAT-09
TC-MAT-19	REQ-MAT-07, NFR-SAFE-03
TC-MAT-20	REQ-MAT-09
TC-MAT-21	REQ-MAT-04
TC-MAT-22	REQ-MAT-01, REQ-MAT-03, BR-02
TC-AS-01	SEC-ACC-06, REQ-ACC-06, DC-08
TC-MS-01	SEC-MAT-01, REQ-MAT-04
TC-AS-02	SEC-ACC-04, REQ-ACC-07, DC-08
TC-MS-02	SEC-MAT-04, REQ-MAT-09
TC-AS-03	SEC-ACC-01, REQ-ACC-10, BR-01
TC-MS-03	SEC-MAT-04, REQ-MAT-11
TC-AS-04	SEC-ACC-03, DC-04
TC-MS-04	SEC-MAT-02, REQ-MAT-10, BR-03, BR-04
TC-AS-05	SEC-ACC-05
TC-MS-05	SEC-MAT-03, REQ-MAT-07, REQ-MAT-08, REQ-MAT-10, BR-03, BR-04
TC-AS-06	SEC-COM-01, DC-03, COM-01
TC-AS-07	SEC-ACC-01, SEC-ACC-02, NFR-QUAL-03, DC-07
TC-AS-08	SEC-ACC-01, SEC-ACC-06, SEC-MAT-01
TC-AS-09	NFR-PERF-02, NFR-QUAL-03
TC-AP-01	NFR-PERF-01
TC-MP-01	REQ-MAT-03
TC-AP-02	NFR-PERF-02
TC-MP-02	REQ-MAT-06
TC-MP-03	NFR-PERF-03
TC-AP-03	NFR-PERF-01, NFR-PERF-02, NFR-PERF-04
TC-AN-01	NFR-SAFE-01
TC-AN-02	NFR-SAFE-02

Test	Requirements traced
TC-AN-03	NFR-SAFE-03
TC-AN-04	NFR-QUAL-01
TC-AN-05	NFR-QUAL-03
TC-AN-06	NFR-QUAL-04, DC-05
TC-AN-07	NFR-QUAL-02
TC-AN-08	NFR-QUAL-05
TC-AN-09	NFR-QUAL-06, DC-10
TC-AN-10	DC-01, DC-04
TC-AN-11	DC-02, NFR-QUAL-03
TC-AN-12	DC-07, DC-08, NFR-QUAL-03
TC-AN-13	NFR-QUAL-03
TC-AN-14	DOM-NOID
TC-AN-15	UI-01..18
TC-AN-16	NFR-QUAL-03, SEC-QZ-04
TC-AN-17	NFR-SAFE-02, REQ-MAT-07, REQ-MAT-08
TC-QB-01	REQ-QB-01, REQ-QB-03
TC-QB-02	REQ-QB-02
TC-QB-03	REQ-QB-01, REQ-QB-02
TC-QB-04	REQ-QB-02
TC-QB-05	REQ-QB-04
TC-QB-06	REQ-QB-05, REQ-QB-02
TC-QB-07	REQ-QB-06, REQ-QZ-04
TC-QB-08	REQ-QB-07, SEC-QZ-01
TC-QB-09	REQ-QB-07, SEC-QZ-01
TC-QB-10	REQ-QB-02
TC-QB-11	REQ-QB-04
TC-QB-12	REQ-QB-07, REQ-ACC-10, SEC-QZ-01
TC-QZ-01	REQ-QZ-01, REQ-QZ-03
TC-QZ-02	REQ-QZ-02
TC-QZ-03	REQ-QZ-01, REQ-QZ-02
TC-QZ-04	REQ-QZ-03, REQ-QZ-04
TC-QZ-05	REQ-QZ-05
TC-QZ-06	REQ-QZ-06, SEC-QZ-02
TC-QZ-07	REQ-QZ-07
TC-QZ-08	REQ-QZ-08, SEC-QZ-02
TC-QZ-09	REQ-QZ-09
TC-QZ-10	REQ-QZ-10, SEC-QZ-04
TC-QZ-11	REQ-QZ-11
TC-QZ-12	REQ-QZ-13
TC-QZ-13	REQ-QZ-14, SEC-QZ-05
TC-QZ-14	REQ-QZ-15
TC-QZ-15	REQ-QZ-16
TC-QZ-16	REQ-QZ-17, SEC-QZ-03
TC-QZ-17	REQ-QZ-17, SEC-QZ-01
TC-QZ-18	REQ-QZ-18
TC-QZ-19	REQ-QZ-04, REQ-QZ-02
TC-QZ-20	REQ-QZ-18, REQ-QZ-04
TC-QZ-21	REQ-QZ-02
TC-QZ-22	REQ-QZ-05
TC-QZ-23	REQ-QZ-09
TC-QZ-24	REQ-QZ-17, REQ-QZ-18, SEC-QZ-03
TC-QZ-25	REQ-QZ-10, SEC-QZ-04
TC-QZ-26	REQ-QZ-11, REQ-QZ-14
TC-QZ-27	SEC-QZ-02, REQ-QZ-13
TC-QZ-28	REQ-QZ-17, SEC-QZ-03
TC-QZ-29	REQ-QZ-10, SEC-QZ-04
TC-QZ-30	REQ-QZ-07, SEC-QZ-05
TC-QZ-31	REQ-QZ-14, REQ-QZ-11, SEC-QZ-05
TC-QZ-32	REQ-QZ-08, SEC-QZ-04
TC-QS-01	SEC-QZ-04, REQ-QZ-07
TC-QS-02	SEC-QZ-05
TC-QS-03	SEC-QZ-01, REQ-QB-02
TC-QS-04	SEC-QZ-03, SEC-ACC-04
TC-QS-05	NFR-QUAL-03, SEC-QZ-01, SEC-QZ-05
TC-QS-06	SEC-QZ-02
TC-QP-01	NFR-PERF-04, REQ-QZ-12
TC-QP-02	NFR-PERF-05, REQ-QB-04
TC-QP-03	NFR-PERF-06, REQ-QZ-08
TC-QP-04	REQ-QB-03

Test	Requirements traced
TC-PA-01	REQ-PA-01
TC-PA-02	REQ-PA-02
TC-PA-03	REQ-PA-03
TC-PA-04	REQ-PA-04
TC-PA-05	REQ-PA-05
TC-PA-06	REQ-PA-06, REQ-PA-04
TC-PA-07	REQ-PA-01, REQ-ACC-10
TC-PA-08	REQ-PA-02
TC-PA-09	REQ-PA-03
TC-PA-10	REQ-PA-05, REQ-QZ-16
TC-PA-11	REQ-PA-06, REQ-QZ-18
TC-PA-12	REQ-PA-04
TC-PA-13	REQ-PA-01, REQ-PA-02, REQ-QZ-15
TC-COL-01	REQ-COL-01
TC-COL-02	REQ-COL-02
TC-COL-03	REQ-COL-03
TC-COL-04	REQ-COL-04
TC-COL-05	REQ-COL-05
TC-COL-06	REQ-COL-06
TC-COL-07	REQ-COL-07
TC-COL-08	REQ-COL-01
TC-COL-09	REQ-COL-01
TC-COL-10	REQ-COL-01, REQ-COL-02
TC-COL-11	REQ-COL-03
TC-COL-12	REQ-COL-04
TC-COL-13	REQ-COL-05
TC-COL-14	REQ-COL-05, REQ-COL-07
UT-ACC-01	REQ-ACC-02, REQ-ACC-03
UT-ACC-02	REQ-ACC-04, REQ-ACC-05
UT-ACC-03	REQ-ACC-06
UT-ACC-04	REQ-ACC-07
UT-ACC-05	REQ-ACC-01, BR-08
UT-ACC-06	SEC-ACC-01, NFR-QUAL-03, DC-07
UT-COL-01	REQ-COL-01
UT-COL-02	REQ-COL-03
UT-COL-03	REQ-COL-05, REQ-COL-06
UT-COL-04	REQ-COL-01, REQ-COL-02
UT-MAT-01	REQ-MAT-01, REQ-MAT-02
UT-MAT-02	REQ-MAT-04, SEC-MAT-01
UT-MAT-03	REQ-MAT-05, REQ-MAT-07, REQ-MAT-08, REQ-MAT-09, SEC-MAT-02, SEC-MAT-03
UT-MAT-04	REQ-ACC-08, REQ-ACC-09
UT-MAT-05	NFR-SAFE-03
UT-MAT-06	REQ-MAT-05, BR-05
UT-PA-01	REQ-PA-02
UT-PA-02	REQ-PA-03
UT-PA-03	REQ-PA-04, REQ-PA-06
UT-QZ-01	REQ-QZ-11
UT-QZ-02	REQ-QB-02
UT-QZ-03	REQ-QZ-02
UT-QZ-04	REQ-QZ-07
UT-QZ-05	REQ-QZ-10, REQ-QZ-14
UT-QZ-06	SEC-QZ-02
UT-QZ-07	REQ-QZ-04, REQ-QB-06
UT-QZ-08	SEC-QZ-03, REQ-QZ-17
UT-QZ-09	REQ-QZ-10, SEC-QZ-04

Annex E. Coverage Summary

E.1 Requirement coverage by family

Family	Requirements	Covered	Partially	Blocked	Not directly testable	Not covered	Unique tests mapped
REQ-ACC	10	9	1	0	0	0	25
REQ-MAT	11	8	3	0	0	0	30
REQ-QB	7	7	0	0	0	0	15

Family	Requirements	Covered	Partially	Blocked	Not directly testable	Not covered	Unique tests mapped
REQ-QZ	18	15	3	0	0	0	39
REQ-PA	6	4	1	1	0	0	13
REQ-COL	7	5	2	0	0	0	14
SEC (ACC, MAT, COM, QZ)	16	15	1	0	0	0	43
NFR	15	14	1	0	0	0	26
SO	6	0	6	0	0	0	43
BR	8	3	4	0	1	0	12
DC	11	9	0	0	2	0	16
Domain / UI / COM	3	2	1	0	0	0	3
Total	118	91	23	1	3	0	160

"Unique tests mapped" counts each test once within its family row; a test tracing to several families appears in each, so the column does not sum to the total (the bottom cell is the number of distinct tests that trace to at least one requirement: all 160).

E.2 Test cases by feature area (each test counted once; total 160)

Feature area	Tests	Ready (R1)	Environment (R2)	Blocked (R3)	Manual (R4)	New in 1.1
Account Management	20	18	1	1	0	9
Material Sharing	22	17	2	3	0	9
Security: Account and Material	14	12	1	1	0	3
Performance: Account and Material	6	0	6	0	0	1
Safety, Quality and Team-level	17	4	7	0	6	8
Question Bank	12	12	0	0	0	3
Quizzes: authoring, execution, grading	32	27	2	3	0	14
Quiz security	6	6	0	0	0	2
Quiz performance	4	0	4	0	0	0
Performance Analytics	13	10	0	3	0	7
Collaboration	14	11	1	2	0	7
Total	160	117	24	13	6	63

E.3 Tests by owner and priority

Owner	Tests	High	Medium	Low	Unit candidates
Jagadeesh (M1)	41	22	18	1	7
Alifiya (M2)	32	13	19	0	4
Pushkar (M3)	32	11	18	3	7
Vedh (M4)	55	32	22	1	10
Total	160	78	77	5	28

E.4 Requirements not Covered, blocked or only partly covered

Requirement	Status	Open decision	Reason
REQ-ACC-09	Partially covered	OD-13, TOD-06	-
REQ-MAT-01	Partially covered	OD-07	-
REQ-MAT-04	Partially covered	OD-13	-
REQ-MAT-09	Partially covered	OD-08	-
REQ-QZ-05	Partially covered	TOD-03	-
REQ-QZ-10	Partially covered	TOD-04	-
REQ-QZ-11	Partially covered	OD-08	-
REQ-PA-03	Blocked pending clarification	OD-03	Topic source undefined (OD-03).
REQ-PA-04	Partially covered	OD-08	-
REQ-COL-01	Partially covered	OD-13	-
REQ-COL-05	Partially covered	OD-08, OD-13	-
SEC-QZ-04	Partially covered	TOD-04	-
NFR-QUAL-02	Partially covered	OD-06, TOD-07	A monthly 99% availability cannot be demonstrated in the sprint window; the test records measured availability over the observed period (TOD-07, OD-06).
SO-01	Partially covered	-	Objective realised by SEC-ACC-02, SEC-ACC-03, SEC-ACC-06 (SRS 3.3.3.4); evidence is the tests of those requirements, not yet reviewed against the objective wording.

Requirement	Status	Open decision	Reason
SO-02	Partially covered	-	Objective realised by SEC-MAT-01, SEC-MAT-02, SEC-MAT-04 (SRS 3.3.3.4); evidence is the tests of those requirements, not yet reviewed against the objective wording.
SO-03	Partially covered	-	Objective realised by SEC-ACC-05, SEC-MAT-02, SEC-MAT-03 (SRS 3.3.3.4); evidence is the tests of those requirements, not yet reviewed against the objective wording.
SO-04	Partially covered	-	Objective realised by SEC-ACC-01, SEC-ACC-04 (SRS 3.3.3.4); evidence is the tests of those requirements, not yet reviewed against the objective wording.
SO-05	Partially covered	-	Objective realised by SEC-COM-01 (SRS 3.3.3.4); evidence is the tests of those requirements, not yet reviewed against the objective wording.
SO-06	Partially covered	-	Objective realised by SEC-QZ-01, SEC-QZ-02, SEC-QZ-03, SEC-QZ-04, SEC-QZ-05 (SRS 3.3.3.4); evidence is the tests of those requirements, not yet reviewed against the objective wording.
BR-01	Partially covered	-	SRS 3.3.5 wording now available; mapped tests have not yet been re-reviewed against it, so the status is retained.
BR-03	Partially covered	-	SRS 3.3.5 wording now available; mapped tests have not yet been re-reviewed against it, so the status is retained.
BR-04	Partially covered	-	SRS 3.3.5 wording now available; mapped tests have not yet been re-reviewed against it, so the status is retained.
BR-06	Partially covered	OD-13	SRS 3.3.5 wording now available; mapped tests have not yet been re-reviewed against it, so the status is retained.
BR-07	Not directly testable	-	Policy/project constraint, not a system behaviour (copyright responsibility; no mechanism, OD-11)
DC-09	Not directly testable	-	Policy/project constraint, not a system behaviour (scope statement; checked by review of s3/s4 of this STP)
DC-11	Not directly testable	-	Policy/project constraint, not a system behaviour (four-member project team; SRS DC-11)
UI-01..18	Partially covered	TOD-12	TC-AN-15 verifies navigation and role visibility only; individual UI items (SRS 3.1.1) are not traced (TOD-12).

Annex F. Open-Decision and Blocked-Test Register

F.1 SAD open decisions (OD-01 to OD-14) that affect testing

ID	Priority	Issue (SAD Appendix 5.4)	Impact on this STP	Linked tests	Decision / action needed
-	High	Team size/ownership: SRS Rev 1.2 (title page, DC-11, Appendix D) states four members; earlier SAD/STP drafts used a three-member division. DECIDED by the team (instruction of 18-10-2026): four members (Jagadeesh, Alifiya, Pushkar, Vedh) with the baseline workload allocation of SRS Appendix D. Residual: Member 4 SRN not supplied.	Test and unit-test ownership reassigned by SRS Appendix D (s9, Annex B/C). TC-AN-08 and TC-AN-09 test modularity and traceability, not ownership.	TC-AN-08, TC-AN-09	Closed for team size and allocation. Supply the Member 4 SRN. The team may still revisit workload (ownership workbook, Option B) under change control.
OD-02	High	Deliverables Part-1 and the official SRS/SAD/STP templates not supplied (Project Guidelines and Part-2 supplied).	STP follows the STP v1.0 structure; compliance with the official STP template and Part-1 is unverified.	None directly	Supply Part-1 and the templates for a compliance check.
OD-03	High	Source of question topic for REQ-PA-03 undefined; the quiz question copy omits topic and bank questions can be deleted.	TC-PA-03, TC-PA-09, UT-PA-02 blocked; REQ-PA-03 Blocked.	TC-PA-03, TC-PA-09, UT-PA-02	Decide how topic is obtained; SRS clarification likely.
OD-04	Medium	SRS Rev 1.2 is a draft; STP baseline cited as v1.0 (revised in 1.1 and 1.2).	Requirement wording may change; IDs treated as stable (DC-10).	None directly	Re-baseline after SRS approval.
OD-05	Medium	No owner for Data Access Layer conventions.	Index/pool-related performance tests have no named fixer; defects are routed to the module owner.	None directly	Team agreement.
OD-06	Medium	No owner for Nginx, Docker, CI, logging/monitoring operation, backup operation.	TC-AN-01, TC-AN-07, TC-AS-08 need an environment owner; assumed to be the test-execution owner of the TC.	TC-AN-01, TC-AN-07, TC-AS-08	Team agreement.
OD-07	Medium	Upload role scope: SAD allows Student and Faculty; SRS REQ-MAT-01/UC-MAT-01 name any authenticated user incl. Admin.	TC-MAT-14 blocked; REQ-MAT-01 Partially covered.	TC-MAT-01, TC-MAT-14	Decide and align SRS/SAD.
OD-08	High	Missing API contracts: Faculty list/read own quizzes; list users with access to a material; list content for moderation; pending-grading-result response.	TC-MAT-18, TC-PA-12, TC-COL-13, TC-QZ-26 blocked; TC-PA-04/05, TC-QZ-15 executed via API only.	TC-COL-05, TC-COL-13, TC-MAT-18, TC-PA-04, TC-PA-05, TC-PA-12, TC-QZ-15, TC-QZ-26	Specify the contracts (not invented here).
OD-09	Low	Notification records not required by SRS; DC-09 scope.	No tests for notifications (Section 4).	None directly	Keep or remove.
OD-10	Medium	req.user shape, roles, route index and middleware naming not verifiable (CONTRIBUTING.md not supplied).	Unit-test candidate names are provisional.	None directly	Verify against CONTRIBUTING.md.
OD-11	Low	BR-07 and BR-08 have no mechanism; Admin account creation unspecified.	Admin test accounts are seeded; BR-07 Not directly testable; TC-ACC-12/19 depend on seeding.	TC-ACC-12, TC-ACC-19	Decide mechanism.
OD-12	Low	ALREADY_ATTEMPTED vs QUIZ_ALREADY_ATTEMPTED code naming.	TC-QZ-07/25/30 assert the message and HTTP 409; code string compared once agreed.	TC-QZ-07	Confirm code and wording.
OD-13	Low	Field-contract gaps: material list owner name/access status, picture types, group field limits, moderation actions, collaborative-content fields.	TC-ACC-16, TC-MAT-21, TC-COL-09, TC-COL-13 blocked; REQ-COL-05 uses DELETE only.	TC-ACC-09, TC-ACC-16, TC-COL-05, TC-COL-09, TC-COL-13, TC-MAT-21	Specify.
OD-14	Low	SRS RTM architecture references to correct (TLS terminates at Nginx, etc.).	None on test design; SEC-COM-01 is tested at the TLS endpoint.	None directly	Controlled SRS change.

F.2 Decisions introduced by this STP (TOD-01 to TOD-12)

ID	Priority	Issue	Impact	Linked tests	Action needed
-	Medium	SAD Revision 1.1 file not supplied to this revision: SAD-derived status codes, error-code strings and messages in the test cards were retained from STP Rev 1.1 and could not be re-verified. (SRS Rev 1.2 was supplied: its limits (App. B), messages (B.7) and SO/BR/UI/COM/DC wording were checked.)	Expected results taken from SAD s4.3-4.5 may differ from the SAD; no test is blocked by this.	-	Supply the SAD; re-verify status codes and error strings before Sprint 1 execution.
TOD-02	Medium	Rate-limit threshold and window are not specified (SAD names rate limiting and HTTP 429 only).	TC-AS-09 blocked.	TC-AS-09	Specify values.
TOD-03	Medium	REQ-QZ-05: SAD allows PUT on Draft only; STP v1.0 implies edit/delete refused only after an attempt has started. SRS REQ-QZ-05: the owner may edit a Draft quiz and delete a quiz after confirmation; editing or deletion of a published quiz is rejected once any Student has started an attempt. The SRS therefore allows deletion of a published quiz without attempts, but does not say whether such a quiz may be edited.	TC-QZ-22 blocked; TC-QZ-05 asserts HTTP 409 for the case both documents agree on.	TC-QZ-05, TC-QZ-22	Confirm the rule for editing a Published quiz without attempts (and align SAD s4.3).
TOD-04	Medium	Time semantics unspecified: deadline when the limit extends beyond the window end; inclusivity at end instants; how auto-submission is triggered; lock-out release instant.	TC-QZ-29, UT-QZ-09 blocked; TC-QZ-07, TC-QZ-10, TC-QZ-25, TC-ACC-07 carry notes.	TC-ACC-07, TC-QZ-07, TC-QZ-10, TC-QZ-25, TC-QZ-29, UT-QZ-09	Define rules.
TOD-05	Low	REQ-PA-02 does not define the aggregate performance metrics (SAD returns attempt rows only).	TC-PA-08 records the formula observed; TC-PA-02 compares to stored attempts.	TC-PA-02, TC-PA-08	Define metrics.
TOD-06	Low	Unit of "MB" (1,048,576 vs 1,000,000 bytes) for the 25 MB and 2 MB limits.	Boundary tests use 1 MB = 1,048,576 bytes and also test the adjacent byte values.	TC-ACC-10, TC-MAT-04	Confirm.
TOD-07	Medium	NFR-QUAL-02 (99% monthly availability) cannot be demonstrated in the sprint window.	TC-AN-07 reports availability over the observed period only; requirement Partially covered.	TC-AN-07	Accept proxy or provide longer monitoring.
TOD-08	Low	SAD s3.1 lists Admin "manage users" but no requirement defines it.	Not tested; Section 4.	-	Add a requirement or remove.
TOD-09	Low	Re-sharing an already shared material has no specified behaviour.	Not tested (TC-MAT-17 note).	-	Specify.
TOD-10	Medium	Capacity of the load generator/test server and availability of Safari/iOS devices unverified.	R2 tests may be Not executed; reported as such.	TC-AN-06, TC-AP-01, TC-AP-02, TC-AP-03, TC-AS-06, TC-QP-01, TC-QP-02, TC-QP-03	Confirm environment before Sprint 2.
TOD-11	Low	Status/message for publishing a quiz whose bank question was removed is not specified for the publish endpoint.	TC-QZ-20 expects HTTP 404 "unavailable" (REQ-QZ-18).	TC-QZ-20	Confirm.
TOD-12	Low	UI-01 to UI-18 and COM-01 have no owner in SRS Appendix D; COM-02 to COM-10 (SRS 3.1.3) and the individual UI items have no trace row of their own in this STP.	TC-AN-15 verifies navigation and role visibility only; COM-02 to COM-10 are not traced to tests.	TC-AN-15, TC-AS-06	Team to allocate an owner and decide whether to add trace rows (no IDs are invented here).

F.3 Blocked tests (R3)

Test	Title	Requirements	Blocking decision	Note
TC-ACC-16	Profile picture: accepted file types	REQ-ACC-09, REQ-ACC-08	OD-13	Setup and purpose defined; expected result not approved - see test card
TC-AS-09	Rate limiting returns HTTP 429	NFR-PERF-02, NFR-QUAL-03	TOD-02	Setup and purpose defined; expected result not approved - see test card
TC-COL-09	Group field length limits	REQ-COL-01	OD-13	Setup and purpose defined; expected result not approved - see test card
TC-COL-13	Moderation: list content and available actions	REQ-COL-05	OD-08, OD-13	Setup and purpose defined; expected result not approved - see test card
TC-MAT-14	Admin upload (role scope)	REQ-MAT-01	OD-07	Setup and purpose defined; expected result not approved - see test card
TC-MAT-18	Owner lists users who hold access	REQ-MAT-09	OD-08	Setup and purpose defined; expected result not approved - see test card
TC-MAT-21	Library shows owner name and access status	REQ-MAT-04	OD-13	Setup and purpose defined; expected result not approved - see test card
TC-PA-03	Topic-wise analysis; empty state	REQ-PA-03	OD-03	Setup and purpose defined; expected result not approved - see test card
TC-PA-09	Topic-wise grouping with known topic data	REQ-PA-03	OD-03	Setup and purpose defined; expected result not approved - see test card
TC-PA-12	Faculty selects an own quiz for class performance	REQ-PA-04	OD-08	Setup and purpose defined; expected result not approved - see test card
TC-QZ-22	Editing or deleting a Published quiz that has no attempts	REQ-QZ-05	TOD-03	Setup and purpose defined; expected result not approved - see test card
TC-QZ-26	Grading failure: pending result and retry	REQ-QZ-11, REQ-QZ-14	OD-08	Setup and purpose defined; expected result not approved - see test card
TC-QZ-29	Deadline when the time limit extends beyond the availability end	REQ-QZ-10, SEC-QZ-04	TOD-04	Setup and purpose defined; expected result not approved - see test card
UT-PA-02	groupResultsByTopic(results)	REQ-PA-03	OD-03	Specified; cases not final
UT-QZ-09	computeDeadline(start, timeLimit, windowEnd)	REQ-QZ-10, SEC-QZ-04	TOD-04	Specified; cases not final

Annex G. Old-to-New ID Mapping

Result: 97 test cases and 20 unit candidates of v1.0 retained with the same IDs. None replaced, split or renumbered. Uniqueness check: 160 TC IDs and 28 UT IDs, no duplicates (script-verified at build). Changes to retained tests are limited to clarifications and requirement-column additions below; unchanged tests are listed as "retained as is".

Old ID (v1.0)	Current ID (Rev 1.2)	Change
TC-QB-01	TC-QB-01	Expected result clarified (status code/message/note)
TC-QB-02	TC-QB-02	Expected result clarified (status code/message/note)
TC-QB-03	TC-QB-03	Retained as is
TC-QB-04	TC-QB-04	Retained as is
TC-QB-05	TC-QB-05	Retained as is
TC-QB-06	TC-QB-06	Retained as is
TC-QB-07	TC-QB-07	Retained as is
TC-QB-08	TC-QB-08	Expected result clarified (status code/message/note)
TC-QB-09	TC-QB-09	Retained as is
TC-QZ-01	TC-QZ-01	Retained as is
TC-QZ-02	TC-QZ-02	Retained as is
TC-QZ-03	TC-QZ-03	Retained as is
TC-QZ-04	TC-QZ-04	Retained as is
TC-QZ-05	TC-QZ-05	Expected result clarified (status code/message/note)
TC-QZ-06	TC-QZ-06	Retained as is
TC-QZ-07	TC-QZ-07	Expected result clarified (status code/message/note)
TC-QZ-08	TC-QZ-08	Retained as is
TC-QZ-09	TC-QZ-09	Retained as is
TC-QZ-10	TC-QZ-10	Expected result clarified (status code/message/note)
TC-QZ-11	TC-QZ-11	Retained as is

Old ID (v1.0)	Current ID (Rev 1.2)	Change
TC-QZ-12	TC-QZ-12	Retained as is
TC-QZ-13	TC-QZ-13	Expected result clarified (status code/message/note)
TC-QZ-14	TC-QZ-14	Retained as is
TC-QZ-15	TC-QZ-15	Retained as is
TC-QZ-16	TC-QZ-16	Expected result clarified (status code/message/note)
TC-QZ-17	TC-QZ-17	Expected result clarified (status code/message/note)
TC-QZ-18	TC-QZ-18	Expected result clarified (status code/message/note)
TC-QS-01	TC-QS-01	Retained as is
TC-QS-02	TC-QS-02	Retained as is
TC-QS-03	TC-QS-03	Retained as is
TC-QS-04	TC-QS-04	Expected result clarified (status code/message/note)
TC-QP-01	TC-QP-01	Retained as is
TC-QP-02	TC-QP-02	Retained as is
TC-QP-03	TC-QP-03	Retained as is
TC-QP-04	TC-QP-04	Retained as is
TC-ACC-01	TC-ACC-01	Expected result clarified (status code/message/note)
TC-ACC-02	TC-ACC-02	Expected result clarified (status code/message/note)
TC-ACC-03	TC-ACC-03	Expected result clarified (status code/message/note)
TC-ACC-04	TC-ACC-04	Expected result clarified (status code/message/note)
TC-ACC-05	TC-ACC-05	Expected result clarified (status code/message/note)
TC-ACC-06	TC-ACC-06	Expected result clarified (status code/message/note); Requirement mapping extended: +SEC-ACC-02
TC-ACC-07	TC-ACC-07	Expected result clarified (status code/message/note); Requirement mapping extended: +SEC-ACC-06, DC-08
TC-ACC-08	TC-ACC-08	Expected result clarified (status code/message/note)
TC-ACC-09	TC-ACC-09	Expected result clarified (status code/message/note); Requirement mapping extended: +BR-06, DC-06
TC-ACC-10	TC-ACC-10	Expected result clarified (status code/message/note); Requirement mapping extended: +DC-06
TC-ACC-11	TC-ACC-11	Expected result clarified (status code/message/note); Requirement mapping extended: +BR-01
TC-MAT-01	TC-MAT-01	Expected result clarified (status code/message/note); Requirement mapping extended: +BR-02
TC-MAT-02	TC-MAT-02	Expected result clarified (status code/message/note)
TC-MAT-03	TC-MAT-03	Expected result clarified (status code/message/note); Requirement mapping extended: +DC-06
TC-MAT-04	TC-MAT-04	Expected result clarified (status code/message/note); Requirement mapping extended: +DC-06
TC-MAT-05	TC-MAT-05	Retained as is
TC-MAT-06	TC-MAT-06	Expected result clarified (status code/message/note)
TC-MAT-07	TC-MAT-07	Requirement mapping extended: +BR-05, BR-03
TC-MAT-08	TC-MAT-08	Retained as is
TC-MAT-09	TC-MAT-09	Retained as is
TC-MAT-10	TC-MAT-10	Expected result clarified (status code/message/note)
TC-MAT-11	TC-MAT-11	Expected result clarified (status code/message/note)
TC-MAT-12	TC-MAT-12	Expected result clarified (status code/message/note); Requirement mapping extended: +BR-03, BR-04, SEC-MAT-02, SEC-MAT-03
TC-MAT-13	TC-MAT-13	Expected result clarified (status code/message/note)
TC-AS-01	TC-AS-01	Expected result clarified (status code/message/note); Requirement mapping extended: +DC-08
TC-AS-02	TC-AS-02	Expected result clarified (status code/message/note); Requirement mapping extended: +DC-08
TC-AS-03	TC-AS-03	Expected result clarified (status code/message/note); Requirement mapping extended: +BR-01
TC-AS-04	TC-AS-04	Requirement mapping extended: +DC-04
TC-AS-05	TC-AS-05	Expected result clarified (status code/message/note)
TC-AS-06	TC-AS-06	Requirement mapping extended: +DC-03, COM-01
TC-MS-01	TC-MS-01	Retained as is
TC-MS-02	TC-MS-02	Retained as is

Old ID (v1.0)	Current ID (Rev 1.2)	Change
TC-MS-03	TC-MS-03	Retained as is
TC-MS-04	TC-MS-04	Requirement mapping extended: +BR-03, BR-04
TC-MS-05	TC-MS-05	Requirement mapping extended: +BR-03, BR-04
TC-AP-01	TC-AP-01	Retained as is
TC-AP-02	TC-AP-02	Retained as is
TC-MP-01	TC-MP-01	Retained as is
TC-MP-02	TC-MP-02	Retained as is
TC-MP-03	TC-MP-03	Retained as is
TC-AN-01	TC-AN-01	Retained as is
TC-AN-02	TC-AN-02	Retained as is
TC-AN-03	TC-AN-03	Retained as is
TC-AN-04	TC-AN-04	Retained as is
TC-AN-05	TC-AN-05	Retained as is
TC-AN-06	TC-AN-06	Requirement mapping extended: +DC-05
TC-PA-01	TC-PA-01	Retained as is
TC-PA-02	TC-PA-02	Retained as is
TC-PA-03	TC-PA-03	Expected result clarified (status code/message/note)
TC-PA-04	TC-PA-04	Expected result clarified (status code/message/note)
TC-PA-05	TC-PA-05	Expected result clarified (status code/message/note)
TC-PA-06	TC-PA-06	Expected result clarified (status code/message/note); Requirement mapping extended: +REQ-PA-04
TC-COL-01	TC-COL-01	Retained as is
TC-COL-02	TC-COL-02	Retained as is
TC-COL-03	TC-COL-03	Expected result clarified (status code/message/note)
TC-COL-04	TC-COL-04	Retained as is
TC-COL-05	TC-COL-05	Expected result clarified (status code/message/note)
TC-COL-06	TC-COL-06	Expected result clarified (status code/message/note)
TC-COL-07	TC-COL-07	Expected result clarified (status code/message/note)
TC-AN-07	TC-AN-07	Expected result clarified (status code/message/note)
TC-AN-08	TC-AN-08	Expected result clarified (status code/message/note)
TC-AN-09	TC-AN-09	Requirement mapping extended: +DC-10
UT-QZ-01	UT-QZ-01	Retained as is
UT-QZ-02	UT-QZ-02	Retained as is
UT-QZ-03	UT-QZ-03	Retained as is
UT-QZ-04	UT-QZ-04	Retained as is
UT-QZ-05	UT-QZ-05	Retained as is
UT-QZ-06	UT-QZ-06	Retained as is
UT-PA-01	UT-PA-01	Retained as is
UT-PA-02	UT-PA-02	Retained as is
UT-PA-03	UT-PA-03	Retained as is
UT-COL-01	UT-COL-01	Retained as is
UT-COL-02	UT-COL-02	Retained as is
UT-COL-03	UT-COL-03	Retained as is
UT-ACC-01	UT-ACC-01	Retained as is
UT-ACC-02	UT-ACC-02	Retained as is
UT-ACC-03	UT-ACC-03	Retained as is
UT-ACC-04	UT-ACC-04	Retained as is
UT-MAT-01	UT-MAT-01	Retained as is

Old ID (v1.0)	Current ID (Rev 1.2)	Change
UT-MAT-02	UT-MAT-02	Retained as is
UT-MAT-03	UT-MAT-03	Retained as is
UT-MAT-04	UT-MAT-04	Retained as is (prefix MAT validates profile REQ-ACC-08/09; kept to preserve the ID)

Annex H. Acceptance Scenarios, Regression Suite and Deviation Log

H.1 Acceptance scenarios (compositions of existing tests, not extra test cases)

ID	Scenario	Tests composed
AT-01	Account lifecycle: register, log in, edit profile, log out, session expiry	TC-ACC-01, 05, 09, 08, 17
AT-02	Material lifecycle: upload, share, download, update, revoke, remove	TC-MAT-01, 07, 08, 09, 11, 10
AT-03	Faculty authoring: add question, create quiz, publish	TC-QB-01, TC-QZ-01, TC-QZ-04
AT-04	Student attempt: start, answer, submit, graded result, immutability	TC-QZ-07, 08, 09, 11, 12, 13
AT-05	Analytics: personal performance and class performance after AT-04	TC-PA-01, 04, 05, TC-PA-13
AT-06	Groups and moderation: create, join, view, Admin moderates	TC-COL-01, 02, 04, 05
AT-07	Security sampler: lock-out, cross-user denial, answer-key, role denial	TC-ACC-07, TC-MS-01, TC-QZ-08, TC-COL-06
AT-08	NFR demonstration: performance run, backup/restore, browser smoke	TC-AP-03, TC-QP-01, TC-AN-01, TC-AN-06

H.2 Regression suite

Smoke set (High priority, Ready): 64 tests - run on every build and pull request where automated. Full regression set: 77 further tests (Ready and Environment-dependent) - run on the release candidate and after defect fixes. Manual/blocked tests: 19. The "Regression" column of Annex B lists the membership of each test.

H.3 Deviation log (Deliverables Part-2 2i)

No.	Document / section	Deviation	Reason	Date	Raised by
-	Team17_Deviation_Log.docx	Document-level differences between the submitted v1.0 SRS/SAD/STP and the current drafts are listed in the deviation log (DEV-01 onward). No implementation deviation is recorded: no implementation evidence exists.	See log	18-10-2026	Project team

Annex I. Final Compliance Checklist

Check	Result	Note
Part-2 s1: backlog from FR/NFR, story points, assignment, two sprints	Partially met	Import CSV prepared (Team17_Part2_Backlog.csv); GitHub Project, repository link and the 12-Oct ETA are not verified
Part-2 s2b/3: GitHub Actions for commits and PRs	Planned	Test workflow planned (s7); not verified
Part-2 s2c-d: PR by assigned member; owners review	Planned	Evidence links in PRs (s7, s13.3)
Part-2 s2f: follow submitted SRS, SAD and STP	Partially met	STP aligned to SRS Rev 1.2 (read in full) and SAD Rev 1.1 (via audit report); SAD file not supplied (TOD-01)
Part-2 s2g-h, 3b: 1-minute and 2-minute sprint videos in repo	Planned	Dates in s7; none claimed
Part-2 s2i: deviations documented	Met (mechanism)	Annex H.3 and Team17_Deviation_Log.docx; no implementation deviation recorded
Part-2 s3c-d: documentation complete; development frozen	Planned	30-Oct
Master prompt: existing STP audited before revision	Met	Audit report
Existing valid content and IDs preserved	Met	Annex G: 97 TC and 20 UT retained
All 115 SRS IDs (REQ, SEC, NFR, SO, BR, DC) present in the RTM	Met	Annex C; SO Blocked, BR/UI Partially, DC-09/11 and BR-07 Not directly testable
IDs validated against the latest SRS	Met	All 115 REQ/SEC/NFR/SO/BR/DC IDs checked programmatically against SRS Rev 1.2 (see final report)

Check	Result	Note
Every applicable requirement has coverage or a documented reason	Met	No row "Not covered"; blocked rows explained
Unique test IDs; consistent references	Met	Script check at build
Status codes/messages from SAD s4.3-4.5 used; none invented	Met	Clarifications in cards; gaps flagged OD/TOD
Open decisions OD-01/03/07/08/13 and others recorded and linked	Met	Annex F
Test ownership: four members, SRS Appendix D allocation	Met	Ownership by SRS Appendix D; s9, Annex B/C; OD-01 decided; Member 4 SRN pending
Planned vs executable vs blocked distinguished	Met	Readiness R1-R4
NFR thresholds specified with workload, measurement, evidence	Met	s5.2
Security tests mapped to SEC-* IDs; SO-* mapped	Partially met	SO-01 to SO-06 linked to the tests of their SEC requirements (SRS 3.3.3.4); SO rows Partially covered
No fabricated execution evidence, approvals or implementation claims	Met	All tests Not executed; approvals blank
Project Guidelines / official STP template compliance	Partially verified	Project Guidelines checked (final report); official STP template and Part-1 not supplied (OD-02)
Traceability totals agree (RTM, inventory, coverage)	Met	Generated from one dataset; checked at build

Annex J. Glossary

Term	Meaning
AccessGrant	Record giving a registered user access to a material
BR / DC / SO	Business rule / design constraint / security objective of the SRS
COM / UI	Communication / user-interface requirement items of the SRS
OD	Open decision of the SAD (Appendix 5.4)
TOD	Open decision introduced by this STP (Annex F.2)
R1-R4	Readiness classes (s How to read)
RTM	Requirements Traceability Matrix
TC / UT	Test case / unit-test candidate
Quiz copy	Snapshot of a bank question stored in a published quiz
Projection	Response shape omitting fields such as correct options
Student, Faculty, Admin	Platform roles (not team members)
Member 1-4	Jagadeesh (PES1UG24CS347), Alifiya (PES1UG25CD841), Pushkar (PES1UG24CS350), Vedh (SRN not yet supplied); ownership per SRS Appendix D
Test clock	Controllable server time used by timer tests